# An independent double-difference tomography implementation: synthetic reconstruction and limits of manuscript reproduction

Independent computational reproduction

October 2026

**Abstract**

Double-difference tomography uses paired seismic arrivals to reduce shared travel-time errors while estimating source locations and velocity jointly. Here we implement this inverse problem independently in C++ and test it against the written model of Zhang and Thurber (2003). The code combines variational ray tracing, trilinear velocity interpolation, joint linearization, and damped, smoothed least squares. We reconstruct the specified synthetic velocity profile and generate new event and station geometries. Consequently, the true-model panel is independently reproduced, whereas synthetic inversion panels are partial reproductions. The initial project instructions excluded Hayward observations. A later authorized acquisition supplies an independently selected, identified geometry for a partial Figure 4 counterpart. The revised solver also generates partial counterparts to the Hayward relocation, velocity, and conditional resolution panels from catalog observations. All numerical figures in this report use saved output from this implementation. The result is a reproducible method investigation, not a recovery of the published Hayward velocity model. The resumed work permits authoritative observational inputs. A later section records authoritative metadata investigation and ray-kernel corrections. The synthetic inverse solutions retain their original baseline provenance.

## 1 Scientific scope and evidence

The target paper is Zhang and Thurber (2003), *Bulletin of the Seismological Society of America* 93, 1875–1889, doi:10.1785/0120020190. Its written equations, experiment descriptions, and captions are the scientific specification. The publisher page returned HTTP 403; the complete manuscript transcription posted by an author on ResearchGate provided the text. The baseline calculations used no original model code, source observations, synthetic arrival files, plotting scripts, or figure files. No published plot was digitized. No published image appears in this report.

The initial exclusion of source observations had a direct scientific consequence. A physical forward model can generate new arrival times for a specified geometry, but it cannot recover the historical earthquake realization and picking errors from a prose description. The baseline therefore separated a reconstructible velocity specification, an independent synthetic inverse experiment, and unavailable field-data inversions. Later sections document the authorized authoritative-data retrieval and subsequent calculations. The panel audit preserves all eleven manuscript figures and records each remaining experimental difference. No supplement is listed for this article on the original SSA issue contents or its original abstract page. No supplementary figures were identified. The official issue contents marks supplements for other articles explicitly; `reference/specification_notes.md` records those checks.

## 2 Forward problem and joint inversion

Assume an isotropic, stationary medium and high-frequency geometric P rays. Coordinates are in km, velocity in km/s, and time in s; depth is positive downward. The predicted arrival of event $i$ at station

$k$ is

$$T_{ik} = \tau_i + \int_{\Gamma_{ik}} \frac{1}{v(\mathbf{x})}\, ds. \tag{1}$$

Here $\tau_i$ is origin time and $\Gamma_{ik}$ is a stationary travel-time path with fixed endpoints. The calculation includes no reflected or converted phases, attenuation, or wave-amplitude effects. Paths remain inside the rectangular model domain. There are no periodic or reflecting boundary conditions.

Nodal velocities $v_n$ are interpolated trilinearly. Write the update as $q_n = \delta \log(1/v_n)$, so that the nonlinear velocity update is $v_n^{\text{new}} = v_n \exp(-q_n)$. The absolute residual linearization is

$$r_{ik} \simeq \mathbf{g}_{ik} \cdot \delta\mathbf{x}_i + \delta\tau_i + \sum_n K_{ikn} q_n, \qquad K_{ikn} = v_n \int_{\Gamma_{ik}} \frac{w_n(\mathbf{x})}{v(\mathbf{x})^2}\, ds. \tag{2}$$

$w_n$ is the interpolation weight. The source derivative $\mathbf{g}_{ik}$ is evaluated from the differentiated endpoint segment integral. Absolute arrivals constrain structure outside clusters and the common source shift.

For a measured event-pair difference $d_{ijk}$ at a common station, use

$$r_{ijk}^{\text{DD}} = d_{ijk} - (T_{ik} - T_{jk}), \qquad A_{ijk}^{\text{DD}} = A_{ik} - A_{jk}. \tag{3}$$

Velocity sensitivities from both events enter the difference, subject to the 5 m sparsification rule described below. This includes changes in path anomalies with source location. Shared station timing errors cancel in differenced observations, whereas independent pick errors do not. Catalog differences share errors with each other and with absolute picks; our weighting follows the staged formulation without whitening this covariance.

The linear step minimizes a weighted residual norm with first-difference regularization of the final log slowness and damping of the column-scaled increment. A first-difference row divides the difference between adjacent nodes by their separation. Columns are normalized to unit Euclidean norm before damping rows are added. An independently written LSQR implementation solves the sparse system without forming normal equations. Absolute, catalog-difference, and precise-difference rows receive stage weights $(1, 0.1, 0.01)$, $(0.1, 1, 0.01)$, and $(0.001, 0.01, 1)$, respectively. Smoothing and damping choices are explicit in the configuration. After a rejected update, identical repetitions within the same stage are skipped, while every later weight stage is retained.

Each accepted nonlinear step must decrease the current weighted misfit plus roughness. Weights are frozen within that line search. Trial source locations are projected onto the grid domain; velocities remain positive by logarithmic parameterization. No published solution is used to choose an update. Sources at boundaries and failed updates are counted in the results because they can indicate model error or weak constraints.

### 2.1 Ray integration and numerical substitutions

The baseline ray tracer minimizes discretized Fermat time with limited-memory BFGS and Armijo backtracking, starting from a straight path. Each segment is split at grid-plane crossings and integrated with two-point Gaussian quadrature. This independent variational method replaces the paper's pseudo-bending implementation. It searches for a local stationary path, with no guarantee of finding every competing first-arrival branch. Search directions and convergence gradients respect active model boundaries. Near-coincident interior nodes are merged at a numerical tolerance far below ray spacing, with the quasi-Newton history reset after such topology changes. Independent rays are traced in parallel with OpenMP.

The reference configuration uses 1 km initial ray spacing and 180 ray iterations, 18 nonlinear iterations in three equal stages, smoothing weight 5, and scaled damping 0.1. These choices are not all specified by the paper. Distance and residual biweights, a 5 m differential path-coefficient cutoff, regularization scaling, line search, and the residual-scale estimator are fully defined in `METHODS.md` and

the source. A damping or smoothing number cannot be transferred between implementations without its scaling convention. Our equal-direction gradient penalty should therefore not be assumed identical to the original regularization.

# 3 Independent synthetic experiment

The true medium is constant with depth and along strike. Its cross-strike profile is 6 km/s for $x \leq 0$, falls linearly to 4 km/s at $x = 1$ km, remains 4 km/s through $x = 5$ km, rises linearly to 5 km/s at $x = 6$ km, and increases linearly to 6 km/s at $x = 38$ km. The paper describes sharp gradients over $x = 0$–1 and 5–6 km; linear interpolation across these two transitions is our explicit assumption. Figure 1 is generated from saved model sampling and reconstructs the written velocity specification with this assumption, without using published pixel values.

The inversion grid follows the coordinates in manuscript caption 2: $x = (-35, -15, 0, 2, 4, 6, 20, 35)$ km, $y = (-60, -40, -20, 0, 20, 40)$ km, and $z = (0, 3, 7, 11, 16)$ km. The forward grid adds the profile breakpoints $x = 1$ and 5 km. This prevents the data generator from using exactly the same coarse representation as the inverse problem.

We generate 120 sources and 32 surface stations independently with seed 2003. Half the sources sample a 12 km wide region around the trough, and half sample a broader 40 km interval; all sample the model volume in depth and along strike. Stations use a jittered surface array. Figure 2 shows the complete generated geometry. These are not the paper's Loma Prieta observations. The starting velocity is homogeneous at 5.5 km/s, substituting for the paper's referenced layered starting model, whose values are not tabulated in the target article. Independent starting source-coordinate errors have standard deviation 0.6 km, with an initial depth restriction of 0.5–15.5 km. Starting origin-time errors have standard deviation 0.05 s.

We round forward travel times to 0.01 s, add independent 0.04 s Gaussian pick noise and a station-constant uniform bias between −0.3 and 0.3 s. Catalog differences subtract these noisy picks. Precise synthetic differences subtract rounded forward times before added pick noise. This explicitly resolves an ambiguity in the text's noise ordering in favor of precise pair differences. Every event is observed at every station, and pairs within 20 km are selected. The saved arrival table separates exact time, rounding, station bias, pick noise, and final observed time. Subtracting two separately rounded arrivals permits differential rounding errors approaching 0.01 s. The paper quotes 0.005 s for its differential data; this discrepancy remains a limitation of our explicit rounding order.

## 3.1 Starting-model sensitivity

The target paper does not identify a side of the Dietz–Ellsworth model. USGS Open-File Report 93-578, Table 4, tabulates two gradient adaptations attributed to that work. We sample each named profile (LOM and LON) onto the same inversion-depth grid and repeat DD inversion using otherwise identical inputs. The adaptation footnote prevents identifying either as the confirmed target start. This comparison measures sensitivity to two written alternatives without inventing their values.

# 4 Validation and numerical results

All 32 implementation consistency checks passed. Across 32 deterministic source–receiver pairs, halving the ray segment spacing and increasing the iteration limit changed travel times by a median of 0.054 ms and a maximum of 0.892 ms. This check does not establish global ray optimality.

In the small known-model recovery check, the median three-dimensional location error decreased from 0.3035 to 0.0343 km; the final absolute-time RMS was 0.0034 s. Only 0.01 s time quantization remains in this check.

| Start | Median 3D error | Mean $\lvert\Delta v\rvert$ | Interior $\lvert\Delta v\rvert$ | $R_{\mathrm{abs}}$ | $R_{\mathrm{prec}}$ |
|---|---|---|---|---|---|
| Homogeneous | 0.8222 | 0.3050 | 0.2135 | 0.3526 | 0.4390 |
| LOM | 1.4316 | 0.4036 | 0.2589 | 0.3752 | 0.4370 |
| LON | 1.1469 | 0.3606 | 0.2444 | 0.3780 | 0.4348 |

Table 1: Starting-model sensitivity of the independent DD experiment. LOM and LON are written gradient adaptations in USGS OFR 93-578. All runs share geometry, true velocity, and observation noise. Median three-dimensional location errors are in km, mean absolute velocity errors are in km/s, and arrival residual RMS values are in s. The first velocity column uses all saved slices; the interior column restricts $x$ to $[-15, 20]$ km, $y$ to $[-40, 20]$ km, and depths to 3–11 km.

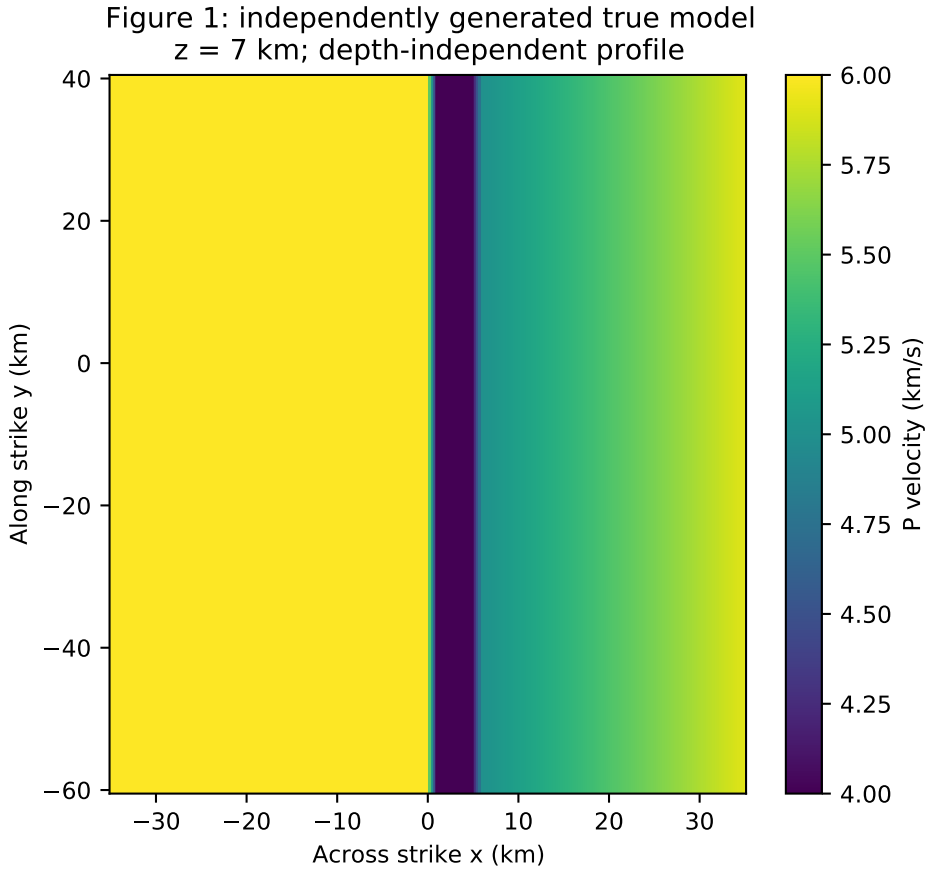



Figure 1: Manuscript Figure 1 reconstruction: true P velocity from the written sandwich profile, sampled at 7 km depth. The model is independent of depth; colors show velocity in km/s. All plotted samples are saved C++ output. The trough occupies $x = 1$–$5$ km and has velocity 4 km/s.

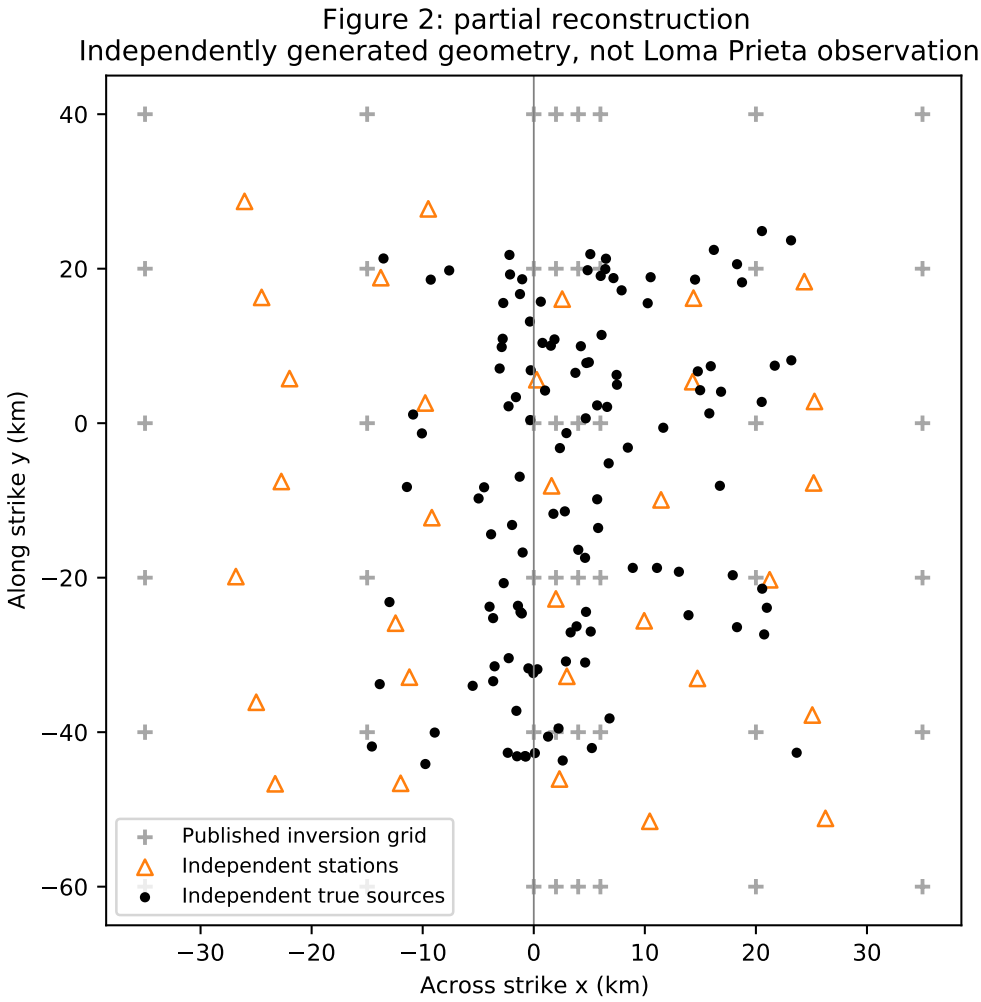



Figure 2: Partial reconstruction of manuscript Figure 2. Crosses are the written inversion grid, triangles are independent stations, and dots are independent true sources. The coordinates are experimental design choices, not estimates extracted from the published map.

Before the main inversions, the median three-dimensional source error was 1.0285 km. The homogeneous 5.50 km/s initial model had mean absolute velocity errors of 0.456 km/s over the full domain and 0.544 km/s in the interior, evaluated at the same saved slice samples as the recovered models.

The standard run ended with median three-dimensional source error 1.7069 km, 0 sources on a model boundary, 1 zero-update iterations, and nodal velocities ranging from 2.41 to 7.05 km/s. LSQR converged in 10 of 10 solves, with worst relative normal residual 8.35e-09. The dd run ended with median three-dimensional source error 0.8222 km, 0 sources on a model boundary, 1 zero-update iterations, and nodal velocities ranging from 3.74 to 6.41 km/s. LSQR converged in 18 of 18 solves, with worst relative normal residual 8.5e-09. The loc1d run ended with median three-dimensional source error 2.9856 km, 11 sources on a model boundary, 1 zero-update iterations, and nodal velocities ranging from 5.50 to 5.50 km/s. LSQR converged in 15 of 15 solves, with worst relative normal residual 9.64e-09. The loc3d run ended with median three-dimensional source error 0.1974 km, 14 sources on a model boundary, 0 zero-update iterations, and nodal velocities ranging from 4.00 to 6.00 km/s. LSQR converged in 18 of 18 solves, with worst relative normal residual 9.32e-09.

Relative to standard tomography, DD changed the full-domain mean absolute velocity error by -0.269 km/s and the median depth error by -0.996 km. Negative differences indicate improvement. These comparisons refer only to the independent experiment. The analytic homogeneous joint-inversion recovery gates **PASSED**. Median three-dimensional location error changed from 0.1821 to 0.0210 km, while median velocity error at the 203 illuminated nodes changed from 0.3000 to 0.0080 km/s. The absolute residual RMS changed from 0.35209 to 0.00635 s. The precise residual RMS changed from 0.09192 to 0.00031 s. There were 12 accepted velocity updates, 12 accepted location updates, and 0 sources on a boundary. LSQR met its normal-residual tolerance in 12 of 12 solves; the worst relative normal residual was 9.15e-09.

The fixed recovery criteria require positive finite velocities, no boundary sources, nonincreasing accepted frozen-weight objectives, at least tenfold reductions of both absolute and precise residual RMS, lower median location and illuminated-node velocity errors, and nonzero updates to both locations and velocity. Illumination means accumulated interpolated path length greater than $10^{-9}$ km on final rays; it does not establish resolution or recovery at every node. The arrivals are analytic and unrounded. This test examines a coupled homogeneous limiting case; it does not establish heterogeneous model recovery. Ray diagnostics compare production traces with half-spacing, 1000-iteration traces at the saved true, final standard, and final DD models and event locations. Source derivatives are compared with central differences of refined travel times using 0.01 km endpoint perturbations. A diagnostic exceedance uses the existing single-ray tolerance of 0.003 s/km; it is not a demonstrated inversion-error bound.

For the truth model, 32 rays had median and maximum absolute refinement changes of 0.054 and 0.892 ms. Production/refined convergence counts were 15/20 out of 32; maximum projected gradients were 0.00868/0.000668 s/km and maximum raw gradients were 0.00868/0.000668 s/km. Thus 17/12 production/refined traces failed the optimizer stopping criterion. Across 24 eligible source-coordinate derivatives, maximum production/refined errors were 0.000584/0.000599 s/km, with 0/0 components exceeding the diagnostic tolerance; 0 components were skipped at boundaries.

For the standard model, 32 rays had median and maximum absolute refinement changes of 0.155 and 0.684 ms. Production/refined convergence counts were 1/3 out of 32; maximum projected gradients were 0.0118/0.00645 s/km and maximum raw gradients were 0.0118/0.00645 s/km. Thus 31/29 production/refined traces failed the optimizer stopping criterion. Across 24 eligible source-coordinate derivatives, maximum production/refined errors were 0.000746/0.000328 s/km, with 0/0 components exceeding the diagnostic tolerance; 0 components were skipped at boundaries.

For the dd model, 32 rays had median and maximum absolute refinement changes of 0.100 and 0.861 ms. Production/refined convergence counts were 1/8 out of 32; maximum projected gradients were 0.0115/0.014 s/km and maximum raw gradients were 0.0115/0.014 s/km. Thus 31/24 production/refined traces failed the optimizer stopping criterion. Across 24 eligible source-coordinate derivatives,

maximum production/refined errors were 0.000589/0.000179 s/km, with 0/0 components exceeding the diagnostic tolerance; 0 components were skipped at boundaries.

Raw and projected gradients differ at active domain boundaries. Small changes in travel time do not establish derivative convergence, and unconverged refined traces are not exact references. These deterministic samples cannot establish global first-arrival optimality or the accuracy of every production ray.

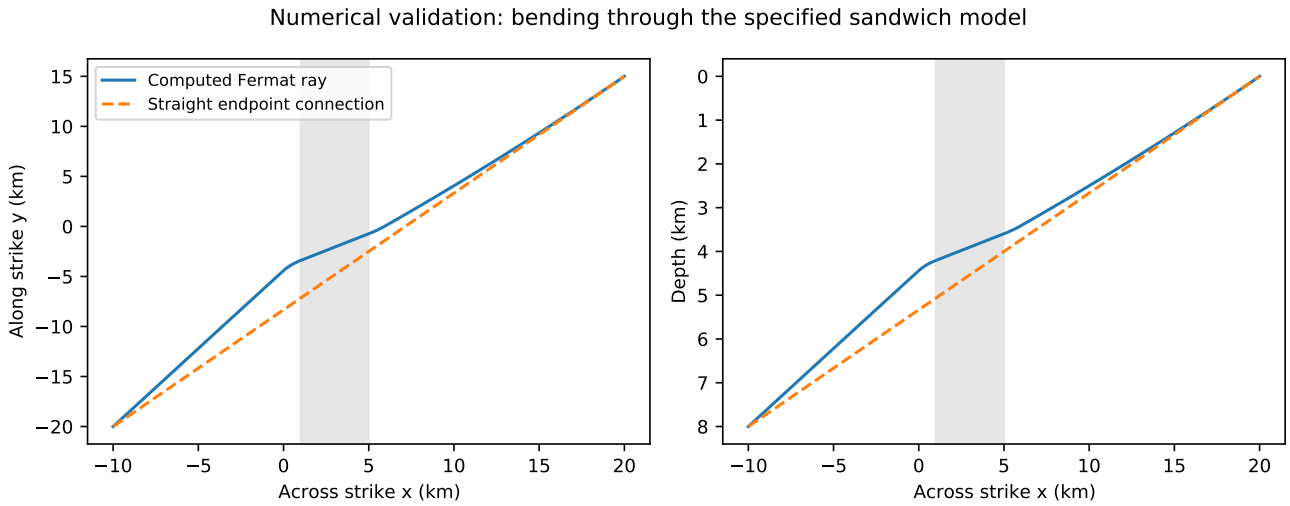



Figure 3: Saved numerical ray through the specified sandwich medium, compared with the straight connection between identical endpoints. The shaded region marks the low-velocity trough. Bending reduces calculated travel time. This is a validation figure, not a manuscript-panel reproduction.

The exact homogeneous travel-time solution, derivative checks, reciprocal ray times, station-error cancellation, and known least-squares solution test different parts of the implementation (Appendix A). They do not establish uniqueness of the nonlinear inverse problem. A larger, deterministic ray sample tests the production settings against half-spacing, 1000-iteration traces. The source finite-difference checks are reported for both the true and inferred media. Small travel-time differences alone do not establish stationarity or accurate source derivatives. Raw gradients can also be nonzero at active constraints, so projected gradients and iteration limits are recorded separately. The homogeneous joint test generates arrivals analytically as distance divided by 6 km/s, independently of the ray tracer, and enables both source and velocity unknowns.

The cross-strike-only medium also admits an independent Snell-law reference for the validation ray, which crosses the layers monotonically. If $R$ is the endpoint separation in the transverse plane, its conserved transverse slowness $p$ satisfies

$$R = \int_{x_a}^{x_b} \frac{p\, v(x)}{\sqrt{1 - p^2 v(x)^2}}\, dx, \qquad T = \int_{x_a}^{x_b} \frac{1}{v(x)\sqrt{1 - p^2 v(x)^2}}\, dx. \tag{4}$$

This scalar reference tests both travel time and the source slowness vector without optimizing path vertices. At the 600-iteration setting, a source-perturbation sweep found finite-difference errors below $1.53 \times 10^{-4}$ s/km at 0.1, 0.03, and 0.01 km, while the 0.001 km difference amplified ray-discretization and optimization errors to 0.00350 s/km. More iterations did not ensure finite-difference convergence. The consistency check therefore uses 0.01 km, matching the larger ray diagnostic sample; the derivative tolerance remains 0.003 s/km. The saved sweep in `data/development/` records this limitation. It is not evidence of derivative convergence in every inferred three-dimensional medium.

DD reduces the full-domain mean velocity error from the initial 0.456 to 0.305 km/s; standard tomography ends at 0.574 km/s. The DD value remains larger than the published reference of 0.178

| Method | Median $\lvert\Delta x\rvert$ | Median $\lvert\Delta y\rvert$ | Median $\lvert\Delta z\rvert$ | $R_{\mathrm{abs}}$ | $R_{\mathrm{cat}}$ | $R_{\mathrm{prec}}$ |
|---|---|---|---|---|---|---|
| standard | 0.5232 | 0.4525 | 1.5176 | 0.3565 | 0.4621 | 0.4619 |
| dd | 0.3690 | 0.3266 | 0.5217 | 0.3526 | 0.4399 | 0.4390 |
| loc1d | 2.0050 | 0.5019 | 0.8753 | 0.6342 | 0.5095 | 0.5081 |
| loc3d | 0.0103 | 0.0221 | 0.1955 | 0.1771 | 0.0578 | 0.0154 |

Table 2: Independent synthetic results. Coordinate errors are in km and arrival residual RMS values are in s. Residual RMS values are unweighted, so they remain comparable when inversion weights change. Fixed-model relocations are denoted loc1d (homogeneous) and loc3d (true synthetic grid). These fixed-model comparisons retain absolute arrivals and do not copy all gauge conventions of hypoDD. The axes are Cartesian, not the latitude/longitude axes of published Table 1.

| Method | Domain | Median | Mean | Standard deviation |
|---|---|---|---|---|
| Initial | Full domain | 0.500 | 0.456 | 0.318 |
| Initial | Interior | 0.500 | 0.544 | 0.393 |
| standard | Full domain | 0.443 | 0.574 | 0.506 |
| standard | Interior | 0.258 | 0.313 | 0.245 |
| standard | Published reference | 0.164 | 0.245 | 0.249 |
| dd | Full domain | 0.252 | 0.305 | 0.262 |
| dd | Interior | 0.153 | 0.214 | 0.181 |
| dd | Published reference | 0.136 | 0.178 | 0.164 |

Table 3: Absolute velocity errors in km/s. Full-domain statistics use saved regular samples on all five depth slices. The interior restricts $x$ to $[-15, 20]$, $y$ to $[-40, 20]$, and depths to 3–11 km. Published values are reference scales only: the article does not specify the averaging mask, and its geometry and starting model differ.

| Model | Depth (km) | Trough | West block | West minus trough |
|---|---|---|---|---|
| standard | 3 | 4.440 | 5.561 | 1.121 |
| standard | 7 | 4.592 | 5.948 | 1.356 |
| dd | 3 | 4.202 | 5.779 | 1.577 |
| dd | 7 | 4.238 | 5.901 | 1.663 |
| Specified true model | 3, 7 | 4.000 | 6.000 | 2.000 |

Table 4: Spatial recovery of the specified trough and western block. All velocities and contrasts are in km/s. Arithmetic means use saved samples with $y \in [-20, 20]$ km, $x \in [1, 5]$ km for the trough, and $x \in [-15, 0]$ km for the western block, including window endpoints. These independently defined windows quantify recovery of the known 2 km/s contrast; they are not published recovered-model averages.

km/s, under the differing sampling and starting conditions described above. Median three-dimensional location error improves from 1.029 to 0.822 km with DD, although its median depth error, 0.522 km, is slightly worse than the initial 0.477 km. Thus improvement depends on both the quantity and the averaging domain.

The recovered DD western-block–trough contrasts are 1.577 and 1.663 km/s at 3 and 7 km depth, respectively, compared with the specified 2 km/s. DD preserves more contrast than standard tomography but still smooths the trough. Its unweighted precise-difference RMS remains 0.439 s, compared with 0.462 s for standard tomography and 0.0154 s with the true velocity fixed. The DD residual is far above the approximately 0.01 s upper bound on differential rounding error: the nearly exact synthetic differences are not fitted to their precision. Grid representation, regularization, source–velocity trade-offs, and robust downweighting are possible contributors; these experiments do not isolate their effects. Improved model-error averages therefore do not imply a close fit to every event pair. Among the three tested starting profiles, the homogeneous start gives the smallest final location and velocity errors; this is a sensitivity result for the independent experiment, not identification of the paper's start. The LOM and LON runs end with median three-dimensional errors of 1.432 and 1.147 km, respectively, both worse than the initial 1.029 km.

All sampled travel-time refinement changes are below 0.9 ms, smaller than the 10 ms rounding interval and 40 ms random-pick scale. None of the sampled source derivatives exceeds the 0.003 s/km diagnostic tolerance. However, most rays in the inferred models do not meet the stricter optimizer stopping criterion, including many refined traces. These checks show close agreement between the numerical settings on sampled paths. They do not certify absolute accuracy, all derivatives, or the first-arrival branch selected by every ray.

Each Figure 3 counterpart shows the five generated grid-depth slices at 0, 3, 7, 11, and 16 km. The published captions do not enumerate plotted depths, so this count is not a verified reconstruction of the published figure layout.

## 5 Discrepancies and reproduction limits

The largest structural difference from the target experiment is its sampling geometry. The synthetic coordinates in the original paper come from actual seismicity and station locations, not from a distribution completely specified in the text. Replacing that geometry changes the null space, ray coverage, and source–velocity trade-off. A homogeneous initial velocity also changes the nonlinear trajectory relative to the referenced layered model. We cannot attribute a mismatch with a published error statistic solely to the inversion algorithm under these changes.

Our forward grid represents the trough edges more finely than the inversion grid. This is a deliberate test of model representation error. The 0.3 s station biases are larger than the random pick errors, and the inverse problem has no explicit station-static parameters. It must accommodate these errors through source shifts, near-station velocity, or residual misfit. Robust weighting may suppress genuine model mismatch as if it were an outlier. A lower weighted objective therefore does not by itself establish a more accurate velocity model; velocity and location errors against the independent truth are reported separately.

During development, rejecting an entire trial update when one source left the domain caused the fixed-model inversion to stall. Projecting individual trial source locations onto the domain corrected that numerical obstruction. Independent review also found that boundary directions within the ray optimizer could block tangential motion, and that collapsed ray segments could corrupt endpoint derivatives. Both cases now have numerical regression checks. A further check found roundoff gradients in coordinate directions where velocity was exactly constant. Ray optimization and column scaling amplified these into spurious source-depth sensitivities. Evaluating each interpolation-gradient component with paired nodal differences preserves this symmetry exactly without changing the mathematical interpolant. Constant-

Figure 3a: Standard tomography
Partial reproduction: independently generated geometry and explicit numerical substitutions

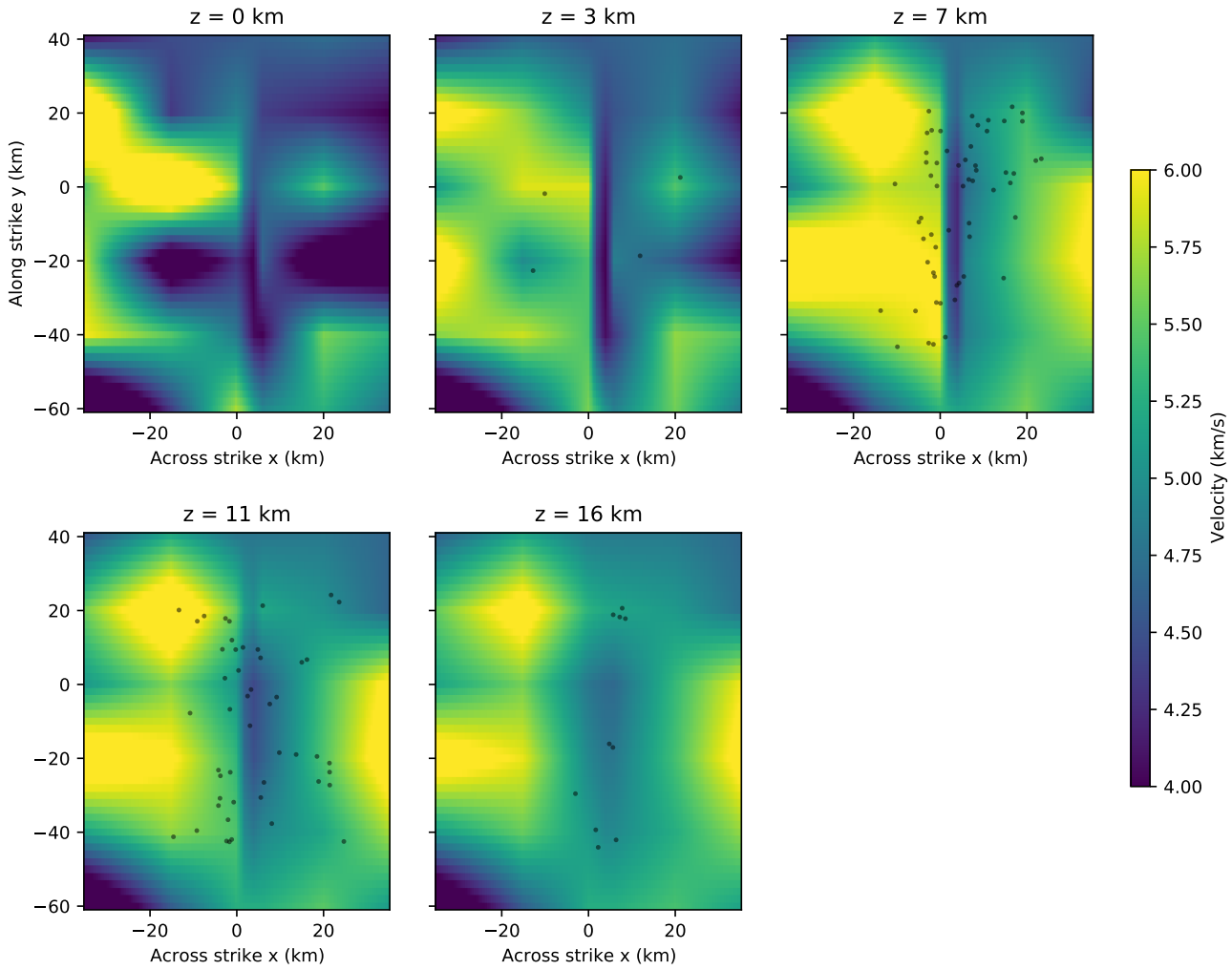



Figure 4: Partial reconstruction of manuscript Figure 3a: standard tomographic velocity from independent observations. Dots are inferred sources selected by adjacent depth midpoints. Colors show 4–6 km/s; values outside this interval saturate and their numerical range is reported in the text. Unconstrained boundaries must not be read as resolved structure.

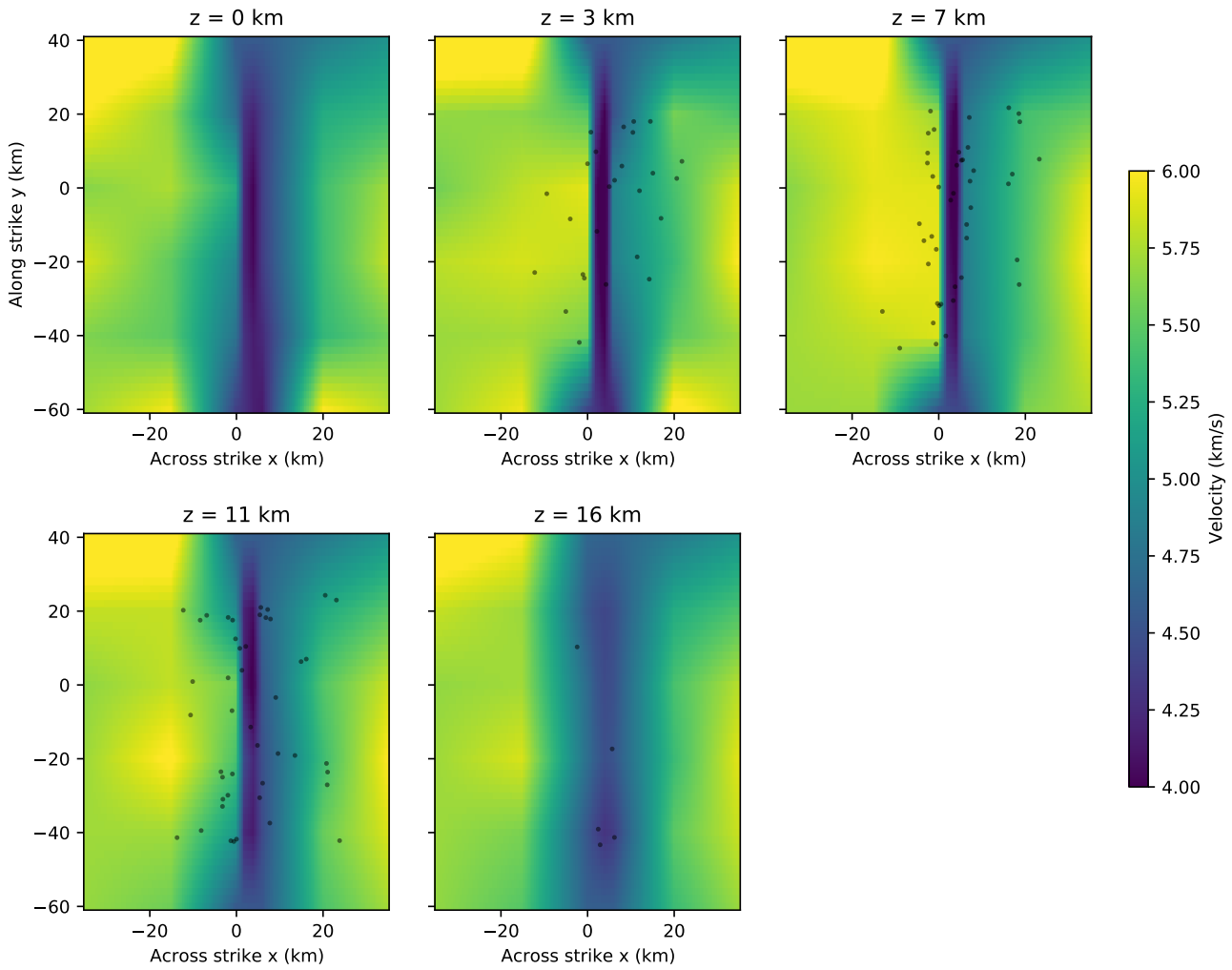



Figure 5: Partial reconstruction of manuscript Figure 3b: DD velocity from independent observations. The initial model, true medium, noise, and inversion grid are identical to those of the standard run. Colors and source-selection conventions follow the preceding figure.

Figure 3c: True minus DD velocity
Partial reproduction: independently generated geometry and explicit numerical substitutions

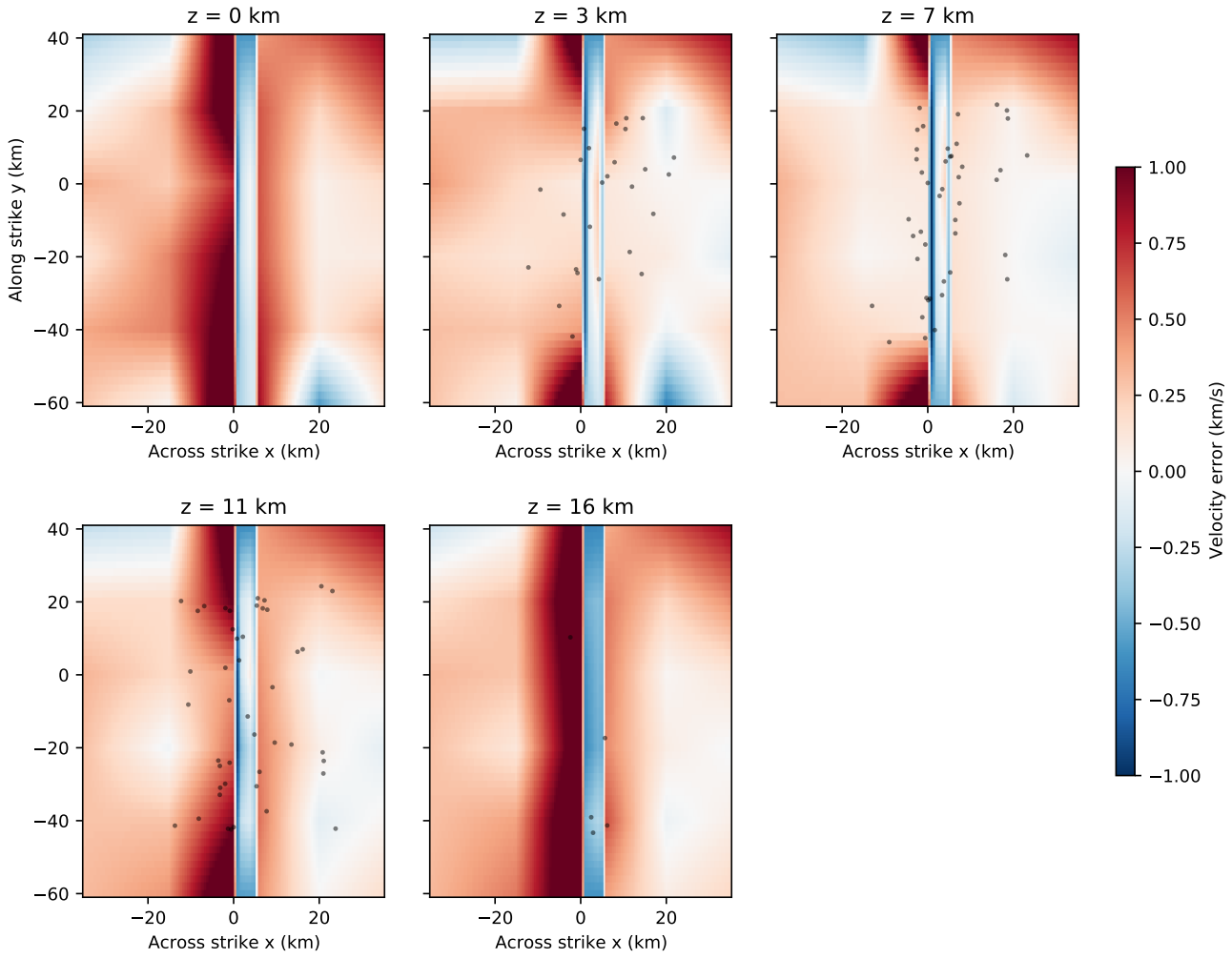



Figure 6: Partial reconstruction of manuscript Figure 3c: true minus DD velocity, following the subtraction order in the manuscript's prose. Blue marks recovered velocities that are too fast, and red marks recovered velocities that are too slow. The fixed color range is ±1 km/s. Saved samples retain unsaturated values.

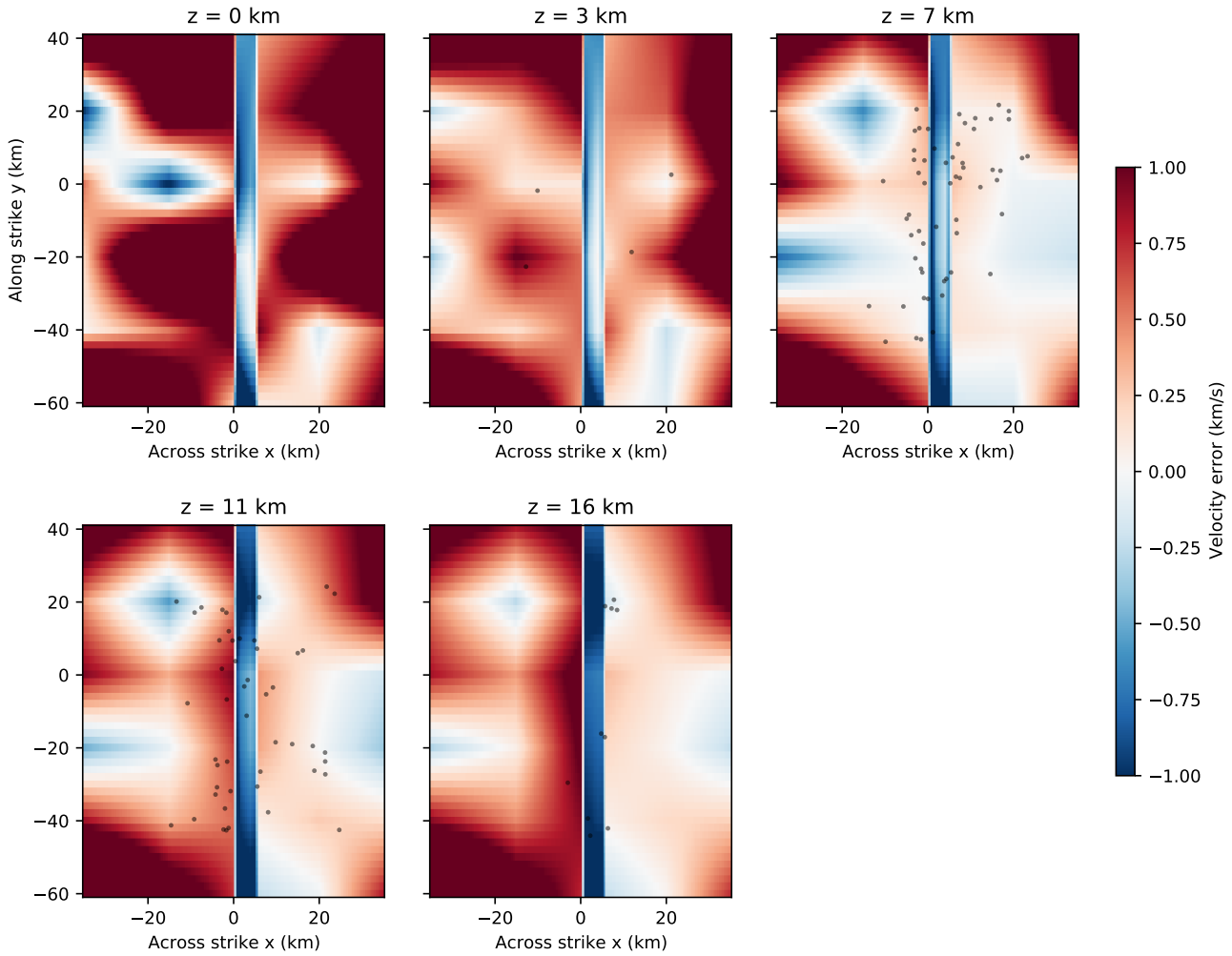



Figure 7: Partial reconstruction of manuscript Figure 3d: true minus standard tomographic velocity, with the sign and color convention of the preceding figure. The spatial patterns depend on the independent geometry and cannot be compared point by point with the original inversion.

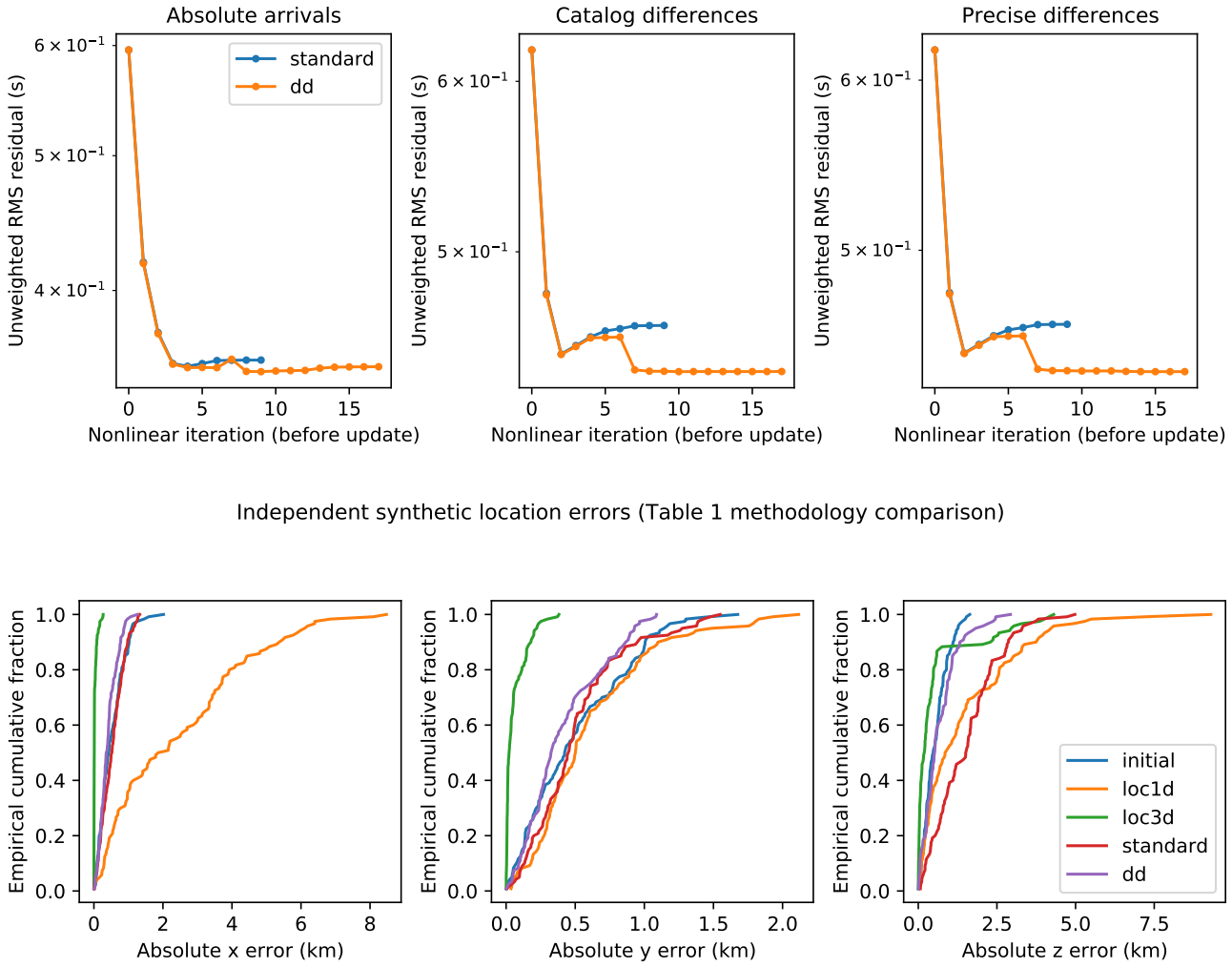



Figure 8: Independent convergence and location-error diagnostics. Top: RMS residuals before each nonlinear update. Stage transitions change the objective, so the unweighted RMS can rise while a weighted objective falls. Bottom: source-coordinate error distributions against saved true coordinates. These are additional method tests, not field-data panels.

axis and surface-ray regression checks cover the correction. The earlier numerical histories are retained in `data/development/`; they are not used for any final numerical panel. Domain projection does not make sources pinned to boundaries physically reliable. In the depth-independent true medium, a source at the surface has zero first-order depth sensitivity to the surface receivers. A linearized fixed-model relocation therefore cannot move such a source back into the interior. The boundary counts record this remaining geometric limitation; the symmetry correction removes numerical drift rather than supplying missing depth sensitivity.

For the initial Hayward Figures 4–11 attempts, the missing dependencies were observational. The field velocity model is the result to be inferred, not an input that can be reconstructed by assuming the reported contrast. We recovered the written grid, projection conventions, sample counts and section definitions, then traced each panel's required data. Those initial attempts stopped at excluded arrivals and event coordinates. In particular, resolution and uncertainty require the actual linearized ray matrix, damping convention and covariance model. A synthetic resolution matrix would not reproduce Figure 11, so none is substituted here.

# 6 Rebuild and artifact provenance

Run the following command from the project root:

```
./rebuild.sh
```

It configures and builds C++, runs validation, generates observations, performs four main inversion comparisons, two starting-profile sensitivities, fixed-model and joint recovery tests, and true/final-model ray checks, computes metrics, generates all figures and audit tables, and compiles this report twice with pdfLaTeX. Its outcome is recorded in `data/rebuild_result.txt` only when all commands have succeeded. Machine-readable numerical checks and iteration histories remain in `data/`; raw console logs are local diagnostics excluded from the shared repository. All caches and build intermediates are confined to the project directory.

For the baseline delivery, the complete clean procedure finished successfully on 9 October 2026 in 801 seconds, with all 32 implementation consistency checks passing and both PDF compilation passes completing. The synthetic baseline numerical results come from that run. `reference/delivery_environment.json` records the compiler, Python packages, TeX version, and parallel settings. Interpretation and plot-label revisions were compiled afterward from those same numerical outputs. Those synthetic inversions have not been regenerated with the subsequent ray-kernel corrections. The later Hayward runs use the revised kernel and have separate provenance; the 801-second record verifies only the synthetic baseline.

Requirements are a C++17 compiler, CMake, OpenMP, Python with NumPy and Matplotlib, and pdfLaTeX with standard packages. No tomography, optimization, or authors' software dependency is required. The saved input CSV files permit rerunning the inverse problem without regenerating random numbers. Standard-library random distributions may differ across platforms, so cross-platform bitwise equality of newly generated observations is not assumed. The experiment configuration, source code, plotting scripts, generated data and figure audit are retained with this LaTeX source.

# 7 Resumed work: authoritative inputs and ray corrections

On 9 October 2026 the user required real earthquake and station observations and metadata from authoritative sources, beginning with USGS and EarthScope. Saved records must retain SEED naming codes and resource identifiers. USGS supplied a Loma Prieta mainshock QuakeML product containing 315 P picks; EarthScope supplied 364 historical NC station records. Provider identifiers, station

epochs, retrieval URLs, raw-file hashes, and identifier attachments are saved in `data/authoritative/discovery/`. These are archive-access checks, not the manuscript's selected event and station set.

The user subsequently authorized a documented authoritative subset with its differences reported, resolving the exact-cohort selection question. Independent NCEDC queries select 1,768 Hayward earthquakes from 24,301 broad-query records. Their preferred origins have 42,236 NC P picks passing the initial quality and identity criteria, across 464 channels at 387 stations. The manuscript's 1,489 events and 52 stations are not forced by sampling. Current catalog revisions, including the geoid depth convention, remain explicit comparison differences.

At the user's request, metadata investigation preceded exclusion. Targeted EarthScope and NCEDC StationXML queries start NC.CMW1 in 2000, but the authoritative NCSN master history explicitly identifies its earlier channels and operation dates. All 142 previously unmatched picks now have unique historical matches; the full join contains 42,236 unique matches. Operation epochs do not establish calibrated response epochs. The examined written studies give no station list establishing inclusion or exclusion of any of the four investigated boreholes. Their manuscript membership remains unknown, and no study-based exclusion is applied. Identified records are preserved for a later additional-data comparison.

NC.CCH1, NC.CGP1 and NC.CSU1 retain provider-declared sensor elevations of 107, 219 and 375 m. StationXML defines channel elevation at the sensor; burial depth is not subtracted again merely because station and channel fields agree. Historical BSL wellhead tables conflict by one burial depth, and their ellipsoid labels differ from NCEDC's stated shared geoid convention. These inconsistencies are recorded without inferring a numerical datum conversion. NC.CMW1 uses its historical ELEV value of 214 m as receiver elevation under an explicit sensor interpretation. Agreement with later StationXML is consistency evidence, not an independent survey. Historical burial depth remains unknown and is not backfilled. The evidence is in `data/authoritative/stations/investigation/`.

The declared quality and model-domain filters retain 1,092 events and 37 physical stations, with 14,262 absolute P picks. All four investigated stations contribute observations. Figure 9 shows this independently selected geometry. Exact manuscript membership and waveform-correlation observations remain unmatched. The later field results use this independent subset and catalog differences only.

Metadata review also found avoidable loss of differential links. Requiring the same metadata epoch split otherwise identical receiver locations: all 62 used NSLCs have one reported position across their used epochs. The initial strict selection and its identified inputs are preserved as a sensitivity subset. The primary pairing instead retains each pick's native epoch and permits links at the same NSLC and exact position. Such pairs are flagged because shared geometry does not establish unchanged timing calibration. An undirected neighbor cap of ten remains an independent pairing choice; its difference-count bound cannot recover the paper's reported count from the reported absolute-pick pool.

The code now preserves provider identities and channel epochs through numerical inputs and outputs, and computes rays only for observed source–receiver pairs. An absolute-data Jacobian exporter and a block-elimination resolution diagnostic have also been implemented. The resolution algebra agrees with a direct small joint-matrix calculation to floating-point precision. The later field diagnostic applies this calculation to absolute arrivals at the catalog DD solution. Its uncertainty calculation assumes that reported symmetric pick uncertainties represent independent one-standard- deviation errors; the archive does not supply confidence levels establishing this calibration. Correlated model errors and regularization bias are excluded.

Independent component work isolated two numerical deficiencies. First, two-point integration of a segment crossing velocity-grid boundaries left a $1.19 \times 10^{-4}$ s/km discrepancy between its variational endpoint gradient and finite differences of computed time. Eight-point integration reduces that example's discrepancy to $5.6 \times 10^{-12}$ s/km and its time error from 33.4 microseconds to numerical roundoff. A separate three-dimensional segment check gives $8.9 \times 10^{-11}$ s/km. This improves integration accuracy; it is not proof of exact gradients for every finite quadrature problem.

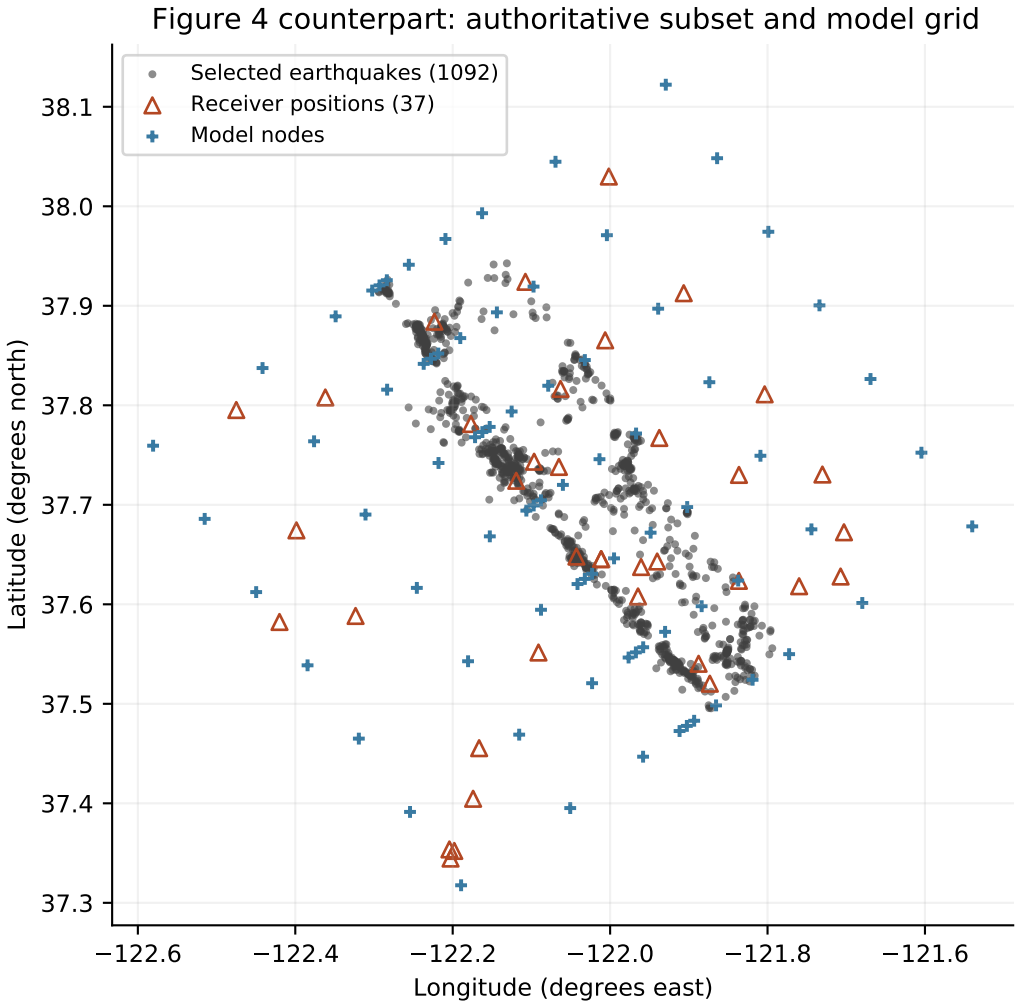



Figure 9: Independent Figure 4 counterpart from identified authoritative inputs: 1,092 selected earthquakes, 37 receiver positions, and 77 horizontal model nodes. The source and receiver selection differs from the published 1,489-event, 52-station cohort. This is a partial geometry reproduction, not an inferred velocity or relocation result. All plotted identities and input hashes are saved.

Second, the homogeneous transverse ray Hessian is a fixed-endpoint path Laplacian, whose condition number grows quadratically with path-node count. Replacing the scalar initial L-BFGS metric with an inverse weighted path Laplacian reduces this stiffness without changing the Fermat objective or admissible paths. Positive segment weights are midpoint slowness divided by segment length. An analytic circular-ray test in $v(z) = 5 + 0.2z$ km/s over 40 km failed to converge within 1,000 baseline updates at spacings 1, 0.5, and 0.25 km. The revised metric converges in 32, 23, and 25 updates with unchanged stopping criteria. Halving spacing reduces the travel-time error by approximately four. A metric-only control preserves this improvement, separating it from the quadrature correction.

All 48 current component checks pass. Forward retracing of frozen baseline models gives the following convergence counts out of 32 sampled rays:

| Frozen model | Production: baseline / revised | Refined: baseline / revised |
|---|---|---|
| Standard | 1 / 5 | 3 / 22 |
| Double difference | 1 / 9 | 8 / 23 |

Nonconvergence remains substantial, and some source finite-difference errors increase despite improved stationarity. These checks do not demonstrate that the 0.439-s baseline DD differential residual is corrected. The synthetic inverse solutions have not been regenerated with the revised kernel. Analytic comparisons, source hashes, all 48 checks, and frozen-model ray tables are in `data/development/ray_optimizer_audit/`. Run the standalone audit with

```
python3 diagnostics/run_ray_optimizer_audit.py
```

### 7.1 A stationary ray can arrive later

An isolated analytic test identifies a remaining branch-selection failure. The invented medium has $v = 4$ km/s for $-1 \le z \le 0$ km and $v = 4 + 0.2z$ km/s below zero. Both endpoints lie at $z = -0.5$ km. At horizontal separations of 20 and 40 km, the production tracer returns the straight path in the constant-velocity layer and declares convergence without an optimizer update. Its projected gradients are below $3 \times 10^{-16}$ s/km. Explicit diving rays obtained from Snell's law arrive 0.0810 and 1.0106 s earlier, respectively. Fine polygonal evaluations of those diving paths agree with their analytic travel times within 0.9 microseconds. Thus the discrepancy concerns the selected ray branch, not the stopping tolerance or a small quadrature error.

Local stationarity therefore does not establish a first arrival, and a derivative consistent with the selected direct branch need not describe the first-arrival travel time. The field starting profile also has a constant above-geoid layer. The current Hayward inversions retain this tracer; the analytic counterexample does not quantify the error of any particular field ray or attribute the inferred source displacements to this mechanism. The test definition, numerical results, and source hashes are saved in `data/development/first_arrival_audit/`. No observational data enter this diagnostic, and it does not modify the field calculations.

Figure 10 tests an experimental first-order fast-marching method (FMM) initializer followed by the unchanged continuous ray bending; it preserves the historical first-order comparison. The original auxiliary-grid audit fails 10 of 61 acceptance checks across 72 cases. Independent refinement of the auxiliary grid and continuous path spacing to 0.0625 km reduces the travel-time errors at 20 and 40 km to less than 1.8 microseconds. However, 4 of 37 checks still fail in the 44-case refinement experiment. At separations 14.68 and 15 km, immediately beyond the 14.6604-km branch crossing, it retains the late direct ray. The time errors are 0.1765 and 3.1896 ms, and the corresponding source-gradient errors are 0.0673 and 0.0699 s/km. Every selected ray in these analytic experiments meets the local stopping rule. Refining local optimization therefore does not, by itself, resolve branch selection. The prototype remains outside the production solver; all failures and the unchanged acceptance thresholds are retained in `data/development/fmm_initialization_refinement/`.

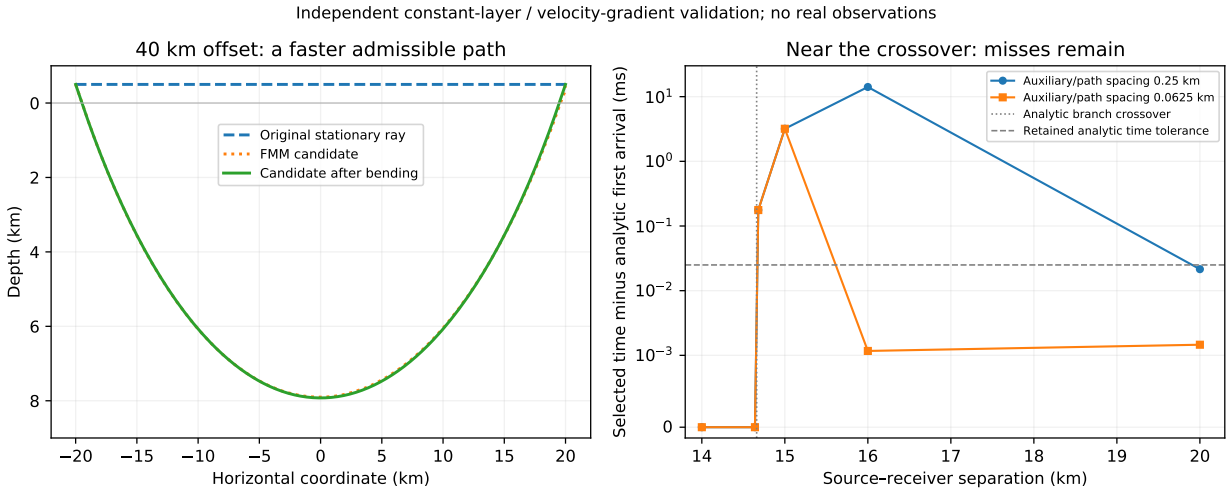



Figure 10: Original first-order FMM diagnostic, not a manuscript counterpart. The medium has $v = 4$ km/s for $-1 \le z \le 0$ km and $v = 4 + 0.2z$ below; both endpoints have $z = -0.5$ km. Left: the stationary straight ray and a faster FMM-seeded, continuously bent path at 40-km separation. Right: travel-time error relative to the analytic earliest branch, with the indicated auxiliary-grid/path spacings. The dotted line marks the branch crossing; the dashed line is the unchanged 0.025-ms tolerance. The symmetric logarithmic scale is linear within $\pm 0.001$ ms. Both spacings miss the earlier branch near the crossing.

A controlled comparison separates the stencil order from source seeding. The optional second-order one-sided derivative is $(3T - 4T_1 + T_2)/(2h)$ when both upstream values are accepted and $T_2 \le T_1$; otherwise that direction retains first order. Here $h$ is the auxiliary spacing, and $T_1, T_2$ are the nearest and next upstream times. The two opposite directions are compared on each axis. A separate seed rule uses the union of corners of every grid cell whose closure contains the receiver. It agrees with the original single-cell rule off grid, but restores seed symmetry when the receiver lies exactly on a grid plane, edge, or node. In 224 component comparisons the original rule retains 28 reflection failures; the closed-cell rule has none, with off-grid fields and paths unchanged.

| Stencil | Receiver seeds | Failed at 0.25 km | Failed at 0.0625 km | Missed offsets (km) |
|---|---|---|---|---|
| First order | Original cell | 10/61 | 4/37 | 14.68, 15 |
| First order | Closed-cell union | 10/61 | 4/37 | 14.68, 15 |
| Second order | Original cell | 9/61 | 2/37 | 14.68 |
| Second order | Closed-cell union | 9/61 | 0/37 | None missed |

Both auxiliary-grid and path spacings take the indicated finest value in each column; missed offsets refer to that finest refined calculation, among the tested offsets. The coarse suite contains 72 comparisons and 61 gates; the refined suite contains 44 comparisons and 37 gates, omitting the already tested homogeneous fixtures. The retained time and source-gradient tolerances remain $2.5\times10^{-5}$ s and $3\times10^{-6}$ s/km. With second order and the closed-cell rule, the finest refined time and source-gradient errors are at most $1.80 \times 10^{-6}$ s and $4.43 \times 10^{-7}$ s/km. All 44 refined paths extract successfully, meet the local stopping rule, retain exact endpoints, and stay in the domain. The coarse suites separately retain three first-order and four second-order strict endpoint discrepancies, at most $8.90 \times 10^{-16}$ km; these are recorded outside the original accuracy-gate counts.

The repeated first-order control reproduces the historical numerical rows and all 61 gates exactly. Frozen source snapshots, output hashes, and the complete controlled comparison are in `data/`

`development/fmm_order_comparison.json`. Passing this finite analytic suite is not a general first-arrival certificate or a demonstration of global second-order accuracy.

A subsequent three-dimensional check uses the positive affine velocity $v(\mathbf{r}) = 6 + \mathbf{g} \cdot \mathbf{r}$ km/s, with $\|\mathbf{g}\| = 0.12$ s$^{-1}$ in a $[-15, 15]^3$ km box. One oblique off-grid geometry, its reflection, and its coordinate permutation are each traced in both directions. The exact circular rays remain at least 3.73 km inside the box. At 0.5-km auxiliary spacing and 0.25-km path spacing, all 72 checks pass across the six directional comparisons. The FMM-seeded, continuously bent paths have maximum time and source-gradient errors of $4.31 \times 10^{-6}$ s and $4.58 \times 10^{-7}$ s/km. Both this candidate and the selected minimum over seeds are tested separately, so a correct direct seed cannot conceal an inaccurate FMM candidate. The evidence is in `data/development/fmm_rotated_affine_baseline/`. These smooth affine cases do not establish branch completeness in arbitrary three-dimensional media. Field validation remains necessary, and the production inversions have not changed.

### 7.2 Independent reference for the initial one-dimensional model

An independent Snell-law calculation enumerates rays in a positive, nondecreasing, continuous piecewise-linear velocity profile. It uses the saved initial Hayward grid within its finite depth interval, $-1 \le z \le 20$ km; the profile is not extrapolated below the model. The candidates include monotone direct rays, all admissible downward turns, tangent paths along accessible constant-velocity intervals, and tangent paths along the finite bottom boundary. Rational coefficients from the binary64 inputs and outward-rounded interval bounds cover the turning parameters. Unresolved roots, tangencies, or boundary membership withhold the first-arrival result. Travel-time ordering still uses ordinary floating-point arithmetic; near ties are retained, and no interval-certified minimum-time gap is claimed.

The reference passes 250 original component checks and 106 added regressions. The added cases exposed and corrected cancellation in nearly flat-layer travel times and circular-path samples, and an incorrectly admitted root immediately beyond the finite bottom. The original 39 failed regression checks remain archived. Stable logarithmic and half-angle formulas remove the cancellation; ambiguous boundary membership now remains unresolved. These corrections change the reference times for the saved 32-ray Hayward sample by at most $8.9 \times 10^{-15}$ s, with branches, source derivatives, and native identity mappings unchanged.

All 32 initial sampled rays have complete enumeration and distance closure within $7.3 \times 10^{-12}$ km. Each has one candidate: 20 direct and 12 turning rays, with no near ties at the recorded $10^{-9}$-s scale. The comparison preserves all 210 observation-to-ray identity associations.

| Numerical time minus reference ($\mu$s) | Minimum | Median | Maximum |
|---|---|---|---|
| Production spacing | 3.82 | 38.87 | 143.95 |
| Refined spacing | 1.32 | 11.79 | 37.48 |

For the 24 saved source-derivative components, the maximum absolute differences are $3.44 \times 10^{-5}$ and $1.26 \times 10^{-5}$ s/km at the production and refined spacings. These differences can contain path discretization and local optimization errors; they do not identify one mechanism. This initial sample does not show millisecond branch errors, but it does not validate every field ray or any updated three-dimensional model. The reference, checks, preserved failures, and identified comparisons are in `data/development/first_arrival_1d_enumeration/`.

### 7.3 Completed DD outputs and a targeted path comparison

The following path comparison uses the saved DD model from the field calculation below.

The preserved first-order fixed-model diagnostic examines every saved DD absolute ray whose source and receiver depths both lie within $[-1, 0]$ km. This selection contains 186 rays and is not representative

of all field paths. With auxiliary and continuous path spacings of 0.5 km, the prototype finds two admissible candidates more than 1 ms faster than the original paths: the reductions are 3.6916 and 2.2385 ms. A third is 0.0725 ms faster. A lower feasible action establishes a late original path in this fixed model; agreement does not certify a first arrival, and this experiment does not quantify source or velocity bias.

This original comparison retains its failures. One FMM extraction has no descending step (solver event index 1043, receiver index 284); its original direct-seed candidate is retained with an explicit failure flag. One selected ray misses the existing local stopping rule. All selected paths stay within the model domain, but 21 original direct paths fail the strict exact-endpoint check at arithmetic round-off, with maximum discrepancy $3.56 \times 10^{-15}$ km. Without relaxing a threshold, the diagnostic saves its outputs and exits with failure status. Observation identities and native epochs are preserved in `data/development/fmm_hayward_audit/`. The unchanged production inversions remain separate; this comparison does not establish the cause of source displacement.

A controlled follow-up repeats the same 186 identified rays with the second-order stencil, closed-cell seeding option, and separately tested extraction correction, still at 0.5-km auxiliary and path spacings. All extractions succeed, but one selected ray still misses local stopping and 22 selected direct paths retain strict endpoint failures below $3.56 \times 10^{-15}$ km. The failed checks remain recorded. Selected times change by at most 0.675 microseconds relative to the historical comparison; two admissible improvements over the straight seed remain above 1 ms, at 3.6916 and 2.2390 ms. All 30 physical receiver groups retain eight seed vertices, so the closed-cell option does not enlarge these seed sets. This comparison combines stencil and extraction changes; it does not isolate one correction. The complete evidence is in `data/development/fmm_hayward_order2_closed/`.

The remaining local-stopping failure has a measurable boundary bottleneck. An isolated audit retains the saved DD ray joining event index 775 to receiver epoch 184, channel `NC.CMO..EHZ`; its source lies on the artificial model top at $z = -1$ km. Instrumented and original functions give identical results in all 12 budget/spacing comparisons. Extending the update budget reaches the unchanged combined stopping rule after 1,249/1,155 updates at 0.5-km spacing and 3,009/2,703 updates at 0.25-km spacing (straight/FMM initialization). This rule requires an accepted action decrease below $10^{-11}$ s and projected gradient below $10^{-5}$ s/km; these paths do not meet the separate $10^{-7}$-s/km gradient threshold.

More than 97% of accepted updates, including all of the final 200 updates in each extended run, follow the same fallback: projecting a full-space quasi-Newton direction at active faces makes it non-descending, so the code clears its curvature memory and uses the unpreconditioned projected negative gradient. No segments merge or line searches fail. Thus the boundary handling repeatedly discards the conditioning improvement measured in the interior tests. Extending the coarse budget changes travel times by only 2–4 ns, while source derivatives change by $2.1\text{–}3.4 \times 10^{-6}$ s/km. Path refinement changes derivatives by up to $1.1 \times 10^{-5}$ s/km. Meeting local stopping alone therefore does not establish derivative accuracy or a first arrival. The original thresholds, failed runs, identities, and direction histories remain in `data/development/ray_stopping_i775_budget_path/`; the production inversions are unchanged.

An isolated direction control solves the principal weighted-Laplacian system on free coordinates, retaining the stiffness of edges adjoining fixed coordinates. Replacing only the original non-descent fallback still fails three of four runs: subsequent retained quasi-Newton directions exhaust the unchanged line-search limit. Using the reduced direction at every update reaches the original stopping rule in all four cases, after 12/12 updates at 0.5 km and 20/21 at 0.25 km (straight/FMM). All 65 directions have negative slope, and all 65 accepted updates strictly decrease the action with one line-search evaluation. Three runs meet the strict projected-gradient criterion; the coarse FMM run meets the combined criterion.

The reduced method does not release coordinates within a direction solve. Wrong-sign quadratic multipliers remain in 10 of 12 coarse and 20 of 21 fine FMM directions, so this method is not the

exact tangent-cone quadratic minimizer. Independent matrix, feasibility, and affine-ray controls pass 560 fallback-variant and 564 pure-metric checks. Artifacts are saved under `data/development/`: `ray_stopping_i775_reduced_fallback/`, `ray_stopping_i775_reduced_all/`, and `reduced_ray_metric_checks/`. These results isolate a direction-choice effect for one identified ray. They establish neither general optimizer accuracy nor a globally first arrival, and they have not changed the production inversions.

### 7.4 Integrated ray correction and field preflights

The separate `first-arrival-integration` branch now combines the receiver-seeded first-arrival initializer with feasible constrained ray directions. When coordinate projection would change an L-BFGS direction, the method instead solves the positive-definite path-metric quadratic problem over directions allowed by the active model faces. It checks feasibility, multiplier signs, and descent before line search. This changes the numerical search direction; the travel-time action, domain, endpoints, quadrature, Armijo condition, and stopping rule remain fixed.

For the previously slow identified Hayward boundary ray, the integrated initializer and constrained direction meet the unchanged combined stopping rule in four updates at both tested discretizations. The 186 retained field entry checks have no initialization or local-stopping failure. Two selected actions decrease relative to the preceding implementation, by at most $5.63 \times 10^{-9}$ s. Analytic layered and rotated-affine controls also pass at their assessed spacings. These checks support the implementation and isolate the boundary-direction defect; they do not certify all first arrivals or nonlinear field recovery. Full-cohort forward and inverse runs are separate requirements, and no completed corrected inverse result is available at this checkpoint.

The independently retrieved Loma Prieta geometry is also prepared for a replacement synthetic experiment: 724 catalog sources, 52 receiver positions, 37,648 source–receiver rays, and all 152,834 event pairs within the stated 20 km separation. The generator retains complete identified ray status and rejects failed rays before writing synthetic observations. The prepared full experiment has 15,932,384 observations, including both catalog and precise differences. LOM and LON starting profiles, source perturbations, and time-rounding differences are explicit assumptions in the preparation record. Geometry preparation and component checks are complete; the identified geometry diagnostic has finished with the failures described below. None of these prepared inputs has yet replaced the synthetic inversion figures earlier in this report.

### 7.5 Identified failures in the full-geometry checks

The larger geometry checks expose limitations that the preceding component suite did not resolve. These failures concern numerical paths in specified velocity models; they are not evidence for excluding any station or event. All identified inputs and failed results remain in the experiment.

The completed third Hayward preflight initializes all 14,262 absolute rays, but 139 fail the local stopping rule at 0.5 km field and path spacing. These failures span 128 events and 80 channel epochs. Of the failed rays, 121 exhaust 1,000 updates; the remaining 18 stop earlier. All 32 rays in the separate refinement sample and all 24 endpoint perturbations meet their local stopping rules. Passing this sample therefore does not establish convergence on the full observation support. The terminal driver records a failed preflight and does not start an inversion. Its complete identified tables, configuration, source snapshot, and failure review are preserved in `data/development/hayward_first_arrival_003_review/`.

Continuous layered-ray enumeration has also completed for all 14,262 initial rays. Every reference is complete and valid, with no near ties; 9,069 first arrivals are direct and 5,193 turn below the source. Production times exceed the reference by a median 37.803 microseconds and a maximum 2.232 ms. Only 603 rays satisfy both existing analytic-control limits. The largest time and source-vector discrepancies

belong to a locally converged ray: event NC/30204761 to NC.JJR..EHZ, absolute row 14099. Its source-vector error norm is 0.033571 s/km. This discrepancy is independent of the 139 failed stopping flags. All reference candidates, original flags, identities, and frozen inputs are in data/development/hayward_first_arrival_003_1d_support/.

Event quakeml:NC.anss.org/Event/NC/390228 and channel NC.CCY..EHZ define Hayward absolute observation row 9291. At 0.5 km field and initial path spacing, the selected FMM-started path rejects its line search after 208 accepted updates. Its time is 3.645674501 s and projected gradient is $2.7990 \times 10^{-3}$ s/km. Increasing the iteration budget from 1,000 to 4,000 leaves this result unchanged. Observation hooks reproduce all eight uninstrumented candidate calculations exactly, including the paths, derivatives, iteration counts, and original stopping flags.

Two adjacent knots approach the 4 km depth interface within $2.18 \times 10^{-13}$ km, leaving an approximately 1.01 km, nearly horizontal segment. Inserting a free midpoint preserves the exact curve and action. The new midpoint's depth derivative is $-0.001645449$ s/km; moving it 0.001 km deeper lowers time by $1.28394 \times 10^{-6}$ s. The velocity increase below the interface reduces slowness, whereas a small depth tent has zero first-order length cost in the horizontal limit. Thus the retained coarse edge omits a feasible descent variation. The calculation uses the saved coordinates without moving them onto the interface.

Across all 38 segment insertions and 190 tested perturbations, the exact depth-only action changes by at most $3.04 \times 10^{-18}$ s on insertion, and the production quadrature agrees with the exact action within $1.06 \times 10^{-15}$ s. This excludes an action-integration discrepancy at the tested scale. At 0.25 km field and initial path spacing, the FMM-started path meets the unchanged combined stopping rule in 41 updates, with time 3.645534035 s and gradient $5.46 \times 10^{-6}$ s/km. This resolves the local stopping failure for this refined calculation. Independent continuous layered-ray enumeration finds a unique first arrival at 3.645479994 s, turning at depth 4.000928869 km. The coarse selected path is therefore late by 194.506 microseconds; the refined FMM path remains late by 54.041 microseconds, with maximum source-derivative error $1.51 \times 10^{-5}$ s/km. Its local flag does not establish continuous-model accuracy. The records are under data/development/initial_ray_first_arrival_003_9291_audit001/ and initial_ray_first_arrival_003_9291_midpoint001/ in the same parent; the independent reference is in initial_ray_first_arrival_003_9291_reference001/.

A further test retains the 0.5 km initialization field and refines only the path. At 0.125 km spacing, the selected FMM-started path stops after 20 updates, with time error 13.075 microseconds and source-vector error norm $2.66 \times 10^{-6}$ s/km. At 0.0625 km, these errors fall to 3.267 microseconds and $7.88 \times 10^{-7}$ s/km. Both selected results meet the existing analytic-control limits of 25 microseconds and $3 \times 10^{-6}$ s/km. The 0.125 km straight-started candidate exceeds the source-derivative limit despite local convergence. All four candidates are retained in initial_ray_first_arrival_003_9291_fine_path001/ under the same development-data directory. This single-ray result motivates further checks; it does not validate a full-cohort discretization.

Path refinement resolves both stopping and accuracy failures on the 139-ray subset (Table 5). Continuous layered-ray enumeration finds a unique first arrival for every ray. At 0.125 km spacing, all time errors satisfy the 25 microsecond control limit, but 16 source-vector errors exceed $3 \times 10^{-6}$ s/km. Three source errors remain above that limit at 0.0625 km. All 139 rays meet both limits at 0.03125 km, with maximum time error 0.818 microseconds and source-vector error $1.83 \times 10^{-6}$ s/km. These calculations retain the frozen executable and original 1,000-update budget. They validate this subset at the initial model, not the full observation support or a later three-dimensional model. The four identified variants and their references are saved under data/development/hayward_initial_failure_refinement_001/ through hayward_initial_failure_refinement_004_reference/.

The locally converged outlier follows a later stationary branch (Figure 11). For this identified geometry, the continuous initial profile admits three turning rays. The first arrival turns at depth 10.896102 km with time 10.775535399 s. A deeper branch turns at 14.139000 km and arrives 2.185310

| Path (km) | Field (km) | Local passes | Both limits | Max time ($\mu$s) | Max source (s/km) |
|---|---|---|---|---|---|
| 0.25 | 0.25 | 134 | 114 | 54.041 | $9.80 \times 10^{-5}$ |
| 0.125 | 0.5 | 139 | 123 | 13.075 | $2.87 \times 10^{-5}$ |
| 0.0625 | 0.5 | 139 | 136 | 3.267 | $4.92 \times 10^{-6}$ |
| 0.03125 | 0.5 | 139 | 139 | 0.818 | $1.83 \times 10^{-6}$ |

Table 5: Refinement of all 139 rays that failed the third Hayward preflight. Counts are out of 139. Accuracy uses absolute travel-time error and the Euclidean norm of the three-component source-derivative error relative to the continuous initial-model reference. The limits remain 25 microseconds and $3 \times 10^{-6}$ s/km. The first row refines both discretizations; the remaining rows retain the same 0.5 km initialization field.

ms later. Refining the path to 0.03125 km while retaining the 0.5 km initialization field brings the numerical action within 0.353 microseconds of the deeper branch, but does not select the first arrival. Both straight and field starts remain on this branch, and increasing the update budget to 4,000 does not change their already-converged results. Exact integration of the saved polygons agrees with production action within $1.44 \times 10^{-14}$ s; finite differences of those fixed polygons agree with the returned source partials within $3.06 \times 10^{-9}$ s/km. These controls isolate branch selection from action integration and fixed-path differentiation for this case.

At the same 0.03125 km path spacing, refining the field to 0.25 km still selects the deeper branch. A 0.125 km field seeds the earlier branch: the bent ray stops after six updates with time error 0.333 microseconds and source-vector error $5.98 \times 10^{-9}$ s/km. Its strict projected gradient is $9.49 \times 10^{-8}$ s/km. The straight-started control remains on the deeper branch. These results motivate a full-cohort check at the finer settings; they do not establish accuracy for other initial rays or updated three-dimensional models. Full resource identities, epochs, paths, source snapshots, and independent comparisons are retained under `data/development/initial_ray_first_arrival_003_14099_audit001/` and the sibling field variants with suffixes `field025001` and `field0125001`.

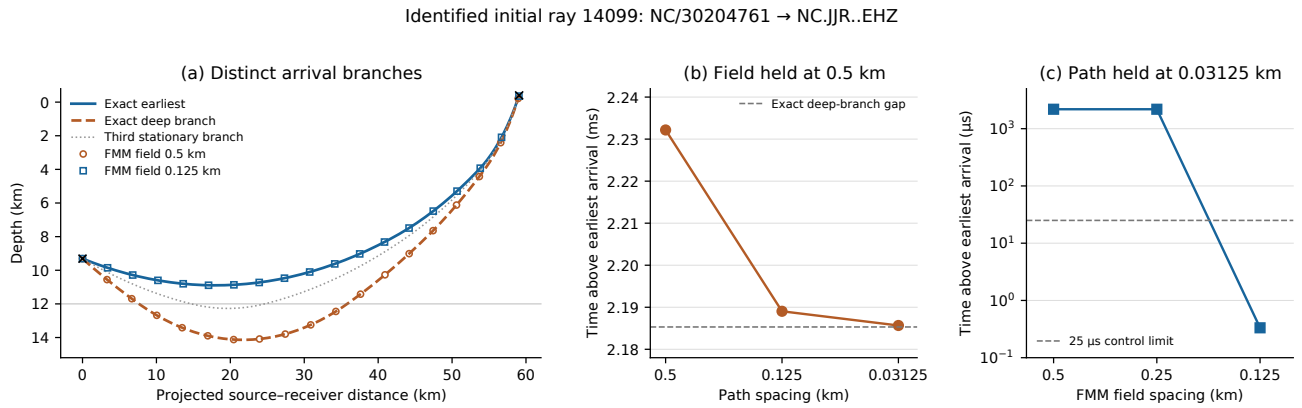



Figure 11: Arrival-branch failure for event `NC/30204761` and channel `NC.JJR..EHZ`, initial absolute observation row 14099. (a) Continuous layered reference curves and saved numerical paths projected onto the endpoint vertical plane; crosses mark the fixed endpoints. Open symbols use 0.03125 km path spacing and the indicated fast-marching field spacing. The horizontal line marks the 12 km velocity-profile interface. (b) Path refinement at fixed 0.5 km field spacing approaches the deeper stationary branch; the dashed line gives its 2.185310 ms arrival delay. (c) Field refinement at fixed 0.03125 km path spacing recovers the earlier arrival. Values in (b,c) select the smaller action from straight and field starts with the unchanged 1,000-update budget. Every selected candidate reports local convergence. This is an independently generated numerical diagnostic, not a reproduced manuscript panel.

Tightening only the stopping rule does not resolve every source sensitivity. For the near-interface ray from event NC/321644 to NC.CBR..EHZ, row 8147, a separate control removes the combined small-action-change stopping condition while retaining the strict $10^{-7}$ s/km gradient criterion. The eight stricter calculations use both starts, path spacings 0.125 and 0.0625 km, and budgets of 1,000 and 4,000 updates. Six exhaust their budget; two reject the line search after 597 updates. None satisfies strict stationarity or the source-vector accuracy limit, although every time error remains within 25 microseconds. Further optimization therefore does not establish sensitivity accuracy at these path spacings. The unresolved stationarity also prevents attributing these errors solely to the finite polygon representation. The full comparison is in data/development/initial_ray_first_arrival_003_8147_strict_comparison001/.

The Loma Prieta check also separates local convergence from arrival-branch selection. The terminal record retains all 104 requested samples. For 38 rays, an independently integrated path that grazes the maximum-speed plane $x = 0$ is admissible. Twenty-three production and refined actions exceed that candidate. Three production paths report local convergence while remaining 25.693, 20.765, and 37.159 ms slower: event NC/20091683 to NC.HLT..EHZ and NC.HQR..EHZ, and event NC/256583 to NC.JSC..EHZ, respectively. At refined spacing their excesses fall to 0.166, 0.187, and 0.133 ms, but none meets the local stopping rule. The explicit candidate is an upper bound on the first arrival, not a complete enumeration of branches. Its absence for another ray does not justify excluding that ray. Full native identities, epochs, selector flags, and captured rows are retained in data/development/loma_geometry_preflight/all_guides_002/.

A subsequent continuous reference enumerates direct, turning, and maximum-speed guide branches for all 104 sampled Loma geometries. Every reference passes its completeness and distance-closure checks, with no near ties: 73 first arrivals are direct, 23 follow a critical guide, six turn, and two are direct paths in a constant-speed region. The 23 critical guides confirm the earlier feasible candidates as first arrivals for those rays. Production and refined root-mean-square time errors are 4.879 ms and 67.366 microseconds; their maxima are 37.159 ms and 186.569 microseconds. Even among locally converged production rays, the maximum error is 37.159 ms. The reference applies to the specified laterally varying synthetic truth model within its finite box and does not validate the production path solver. All identities and comparisons are saved in data/development/loma_geometry_preflight/layered_reference_001/.

The completed Loma Prieta diagnostic retains all 104 requested rays. Initializers succeed for every ray at both spacings, but 31 rays fail the local stopping rule at each spacing. The largest time change on refinement is 37.026 ms. Endpoint finite-difference comparisons reach approximately $1.26 \times 10^{-3}$ s/km, but only 13 of 24 positive perturbations and 12 of 24 negative perturbations converge locally. These discrepancies cannot be interpreted as validated derivative-error estimates. No full synthetic observation set or Loma Prieta inversion follows this check. The identified terminal tables are in data/loma_lom_001/; the review and execution provenance are in data/development/loma_geometry_preflight/001/.

A separate exact control identifies a defect in using an ordinary gradient at an interface. For $v(x) = 6$ km/s on $x \le 0$ and $v(x) = 6 - 2x$ km/s on $0 \le x \le 1$ km, a straight 2 km path on $x = 0$ attains the lower bound $T = 1/3$ s. The current trace returns this action but reports nonconvergence, using a midpoint normal gradient of $1/18$ s/km from the right-hand cell. The true action has unequal one-sided derivatives there. The returned source-normal partial is $1/36$ s/km, whereas the optimized continuous arrival has zero source-normal derivative at this control. Relabeling its stopping flag would therefore leave the sensitivity problem unresolved. The reproducible control is in data/development/interface_stationarity/. This component defect is distinct from the off-interface, slowly optimized Loma paths already examined. No convergence-only correction has been applied, and these checks have not produced a completed corrected inversion.

### 7.6 Full initial Hayward accuracy after separate refinements

The fourth Hayward calculation applies the 0.03125 km path spacing and 0.125 km initialization-field spacing to all 14,262 initial observed rays. Every initializer succeeds and every local stopping flag is true. All rays also satisfy the unchanged continuous-reference limits of 25 microseconds in time and $3 \times 10^{-6}$ s/km in source-vector norm. The largest errors are 0.959 microseconds and $1.782 \times 10^{-6}$ s/km; their medians are 0.333 microseconds and $2.422 \times 10^{-8}$ s/km. In the preceding calculation, only 603 rays satisfied both accuracy limits. Figure 12 compares the complete distributions without removing failed rays.

The reference cache is reused only after exact equality of the complete initial geometry, model, observation tables, and numerical reference source has been verified. The original native identities, branch certificates, and local flags are retained in `data/development/hayward_first_arrival_004_1d_support/`. The separate sampled refinement also passes its independent accuracy review. All 32 production and refined travel times satisfy the same reference limit; their maximum errors are 0.717 and 0.181 microseconds. All eight sampled source vectors pass, with maximum errors of $8.55 \times 10^{-8}$ and $1.83 \times 10^{-8}$ s/km. Reference evaluations at the 48 displaced sources separate finite-difference truncation from tracing error; the largest truncation is $3.13 \times 10^{-6}$ s/km, whereas the largest saved-minus-reference finite-difference discrepancy is $3.27 \times 10^{-8}$ s/km. These checks are retained in `data/development/hayward_sampled_accuracy_004_002/`.

The reviewed configuration started a separate guarded catalog-DD attempt on 10 October 2026 at 07:37 UTC. It retains all 58,980 catalog observations and uses no waveform differences. The launch record is in `data/development/hayward_guarded_dd_launch_plan_001/`; numerical output belongs to `data/hayward_first_arrival_guarded_001/`. The calculation stopped after 1 h 53 min at iteration zero's first line-search trial. All 14,262 initializers succeeded, but 181 rays failed local stopping; 106 exhausted the 1,000-update budget and 75 returned earlier. All 181 selected the FMM candidate. The history contains only its header: no inverse update was accepted. The proposed full step moves sources by up to 4.448 km and changes nodal velocities by $-2.635$ to $+1.630$ km/s. The last accepted state is numerically identical to the initial state. Smaller steps were not evaluated; the failure does not establish that every step would fail. The rejected trial and last accepted states, native failure roster and paths are preserved in `data/development/hayward_guarded_failure_001/`. Passing the initial depth-dependent reference therefore did not establish local convergence at the first three-dimensional trial model. The guard prevented those failed rays from entering its trial objective. This attempt supplies no corrected field result or final-model resolution.

A matched replay evaluates two rejected rays to `NC.CMJ..EHZ`, events `NC518943` and `NC519264`, using the recorded trial model and coordinates. One original receiver field supplies both FMM starting paths. Before applying the adopted optimizer, the original implementation reproduces every recorded 17-digit status field and selected-path coordinate exactly. The adopted version then uses those same starting paths and retains the original lower-action selection. Both selected rays still fail local stopping after 1,000 updates; their projected-gradient norms are $4.96586 \times 10^{-4}$ and $5.54354 \times 10^{-4}$ s/km. All four starts in the adopted version exhaust that budget, with no merges. Reported selected actions change by $+5.53 \times 10^{-11}$ and $-2.19 \times 10^{-10}$ s; different outer summation arithmetic prevents interpreting these differences as certified physical action orderings. No endpoint or model derivatives were requested. The evidence in `data/development/hayward_trial_cmj_old_D_replay_001/` therefore excludes this optimizer change alone as a remedy for these two local failures, without diagnosing the other 179 rays.

The two original paths contain 31 and 32 internal nodes within one millimetre of the internal plane $x = 0$. At each node's projection onto that plane, exact rational interpolation of the saved binary64 grid gives positive left and negative right normal velocity slopes. Thus reciprocal velocity has a normal cusp minimum at the face (Figure 13). The smallest saved normal offsets are $3.34 \times 10^{-11}$ and $1.54 \times 10^{-10}$ km. These describe numerical coordinates, not observational position accuracies. The one-millimetre inventory is descriptive: no point is moved and no stopping tolerance changes. The calculation

in `data/development/hayward_cmj_interface_geometry_001/` identifies a nonsmooth model feature beside the failed paths, but does not establish causal attribution, polygon stationarity or first-arrival accuracy.

Reevaluating each saved polygon's free-knot gradient reproduces every recorded raw and projected norm at 17 digits, without tracing or changing a path. The $x$ components of those 31 or 32 near-face nodes contribute 99.513% and 99.295% of the original squared norms, and 99.446% and 99.638% after the optimizer change. Exact rational sums of the exported binary64 components give these fractions; raw and box-projected results coincide. Thus the recorded ordinary gradient is concentrated in normal motion beside the face. This calculation, saved in `data/development/hayward_cmj_saved_gradient_001/`, localizes the norm but does not supply a nonsmooth stationarity test or justify changing a failed flag.

Invented exactly face-aligned paths clarify the distinction. With $v(x) = 6 - |x|$ and endpoints $(0, \pm 1/2, 0)$, the straight face path attains the global length/speed lower bound $T = 1/6$ s. Its selected right-cell midpoint partial is nevertheless 1/72 s/km. Conversely, $v(x) = 5 + |x|$ admits normal descent in both directions even though zero lies in the convex hull of the two ordinary gradient limits. A separate coupled endpoint-direction example passes all signed coordinate probes but retains a decreasing combined direction; endpoint supports cannot be chosen independently. The 37 exact rational and 62 numerical checks in `data/development/internal_ridge_stationarity_controls_001/` use fixed invented geometry, with no trace or optimizer call. The variable trilinear face formula applies piecewise between other grid-plane crossings. It is not a continuous certificate for general paths, and these exact-face controls cannot relabel the merely nearby native nodes.

Within each grid cell, velocity along a straight edge is a polynomial of degree at most three. For positive nodal velocities, exact grid-plane splitting and bounded polynomial sums enclose reciprocal-velocity action and its first variation; an aligned-face variation uses the incident cell selected by its normal motion. All 80 invented checks pass: 65 mathematical checks and 15 source/file hash checks. Independent saved readback verifies 18 exact comparisons, including both adjacent edges of a tangent variation. The controls retain coupled descent missed by coordinate probes and deliberately unresolved bounds. No native geometry, trace or optimizer was evaluated by this suite, and no production flag changes. Results are in `data/development/trilinear_action_controls_001/RESULTS.md`.

The completed native calculation forms two separately identified candidates by setting only the previously selected 31 or 32 node $x$ coordinates to exact zero. All other coordinates, node order and endpoints remain fixed. Bounded continuous integration resolves a decrease in action for each candidate (Table 6). The 30 and 31 aligned edges split into 32 and 33 face pieces; independent quadratic extrema give positive normal slowness jumps over every complete piece. However, both candidates retain ordinary tangent descent. The complete available tangent components, including both edges incident on each free node, have norm lower bounds above $10^{-5}$ s/km. No free coordinate lies exactly on the physical box boundary. These subset lower bounds apply to any full coupled support residual sharing the tangent components; their upper bounds do not bound that unknown full residual. Thus face alignment alone does not meet either existing gradient scale. The first prescribed coordinate witness is at node one, far from the modified nodes, so that witness does not localize the cause of the original failure. No displacement step or optimizer was run, and no endpoint/model derivative, stationary path or first arrival is certified. Original flags remain false; candidate convergence is unevaluated. Full native identities, rational bounds and independent saved readback are in `data/development/hayward_cmj_face_action_001/RESULTS.md` and `data/development/hayward_cmj_face_action_review_001/`.

One subsequent prescribed tangent step from each face candidate gives a further resolved action decrease (Table 9). The direction uses the existing positive edge metric and a principal system with unavailable face-normal displacements set to zero. This defines a search subspace, not a physical constraint or a reduced stopping test. All available tangent components enter the nominal slope bound and the separate bound for the actual rounded displacement. Both steps have dimensionless length one and

Table 6: Prescribed face candidates for NC.CMJ..EHZ. Values are rounded displays of rational enclosures for the recorded binary64 model. The tangent norm omits 31 or 32 unavailable face-normal components; it is not a full stopping test.

| Event | $\Delta T$ (s) | Tangent norm (s/km) |
|---|---|---|
| NC518943 | $-4.3891 \times 10^{-11}$ | $3.4637 \times 10^{-5}$ |
| NC519264 | $-8.1650 \times 10^{-11}$ | $4.1306 \times 10^{-5}$ |

satisfy the fixed conservative Armijo check. The full trilinear action uses the same degree-eight, one-subdivision, 128-bit enclosure as the preceding calculation. The run evaluates 2,058 new edge actions, two metric solves and two rounded trials. Independent saved readback verifies the exact accounting and complete native identities. A separate evaluation of all 2,058 tangent edges reduces the available gradient norms by about 91% and 69%, with unchanged availability masks. Both lower bounds remain above $10^{-7}$ s/km; the second also remains above $10^{-5}$ s/km. No choice of unavailable normal supports can remove these tangent components. Neither preceding accepted action change is below the combined stop's $10^{-11}$ s scale. Independent saved readback verifies the complete incident-edge sums, projections and exact squared-norm bounds. The original false flags and null candidate convergence are preserved; stationarity, source accuracy and first arrival remain unresolved. Results and the preserved signed-zero serialization control failure are in data/development/hayward_cmj_tangent_step_001/RESULTS.md. The subsequent gradient calculation and its independent review are documented in data/development/hayward_cmj_post_step_tangent_001/RESULTS.md.

The next prescribed full-size trials are rejected. Their complete nominal slopes are $-3.7906 \times 10^{-10}$ and $-5.5922 \times 10^{-10}$ s per dimensionless step, but their finite action changes are $+3.0212 \times 10^{-10}$ and $+2.7020 \times 10^{-10}$ s. The action enclosures have widths below $6.0 \times 10^{-32}$ s, so the increases are resolved. Negative first variation does not guarantee decrease at unit step length; these results motivate a smaller fraction along each saved direction. The first accepted paths remain the starting state for that comparison. Independent saved readback confirms the exact cache, slope, rounding, action and rejection accounting in data/development/hayward_cmj_tangent_step_002/RESULTS.md.

Reusing each direction at half length gives action changes of $-1.9272 \times 10^{-11}$ and $-7.5767 \times 10^{-11}$ s, with enclosure widths below $6.0 \times 10^{-32}$ s. Both trials satisfy the same conservative Armijo check. Maximum node movements are 0.04553 and 0.02491 m, and these paths supply the subsequent refinement's starting state. The calculation uses no new gradient or metric solve: it evaluates two rounded trials and 2,058 new edge actions. Independent saved readback confirms the geometry, action, acceptance and native identity joins. Both decreases still exceed the combined stopping scale of $10^{-11}$ s in magnitude. Results are in data/development/hayward_cmj_tangent_half_step_001/RESULTS.md.

A separate calculation on the earlier first-step paths proves $v_{-,n} \geq 0 \geq v_{+,n}$ over all 65 complete aligned face pieces. Because velocity is positive and continuous there, the same pointwise convex mixture of the two normal slowness gradients can be zero for both endpoint moments. All ordinary adjacent-edge contributions are retained, including the entry and exit terms. The resulting feasible full-vector norms are $4.4155 \times 10^{-5}$ and $2.4963 \times 10^{-5}$ s/km. A large norm for one feasible choice gives no lower bound on the minimum support residual, so this construction does not certify or exclude stationarity. Its geometry is the first accepted step, not the subsequent half step. The first saved-data checker failed on equivalent hexadecimal spellings; canonicalizing those spellings makes the independent readback pass without changing or repeating the numerical calculation. The retained failure and corrected review accompany data/development/hayward_cmj_post_step_zero_support_001/RESULTS.md.

On the accepted half-step geometry, a new bounded evaluation of all 2,058 edges gives available gradient norms of $2.5528 \times 10^{-6}$ and $6.7473 \times 10^{-6}$ s/km. These are 0.8212 and 0.5306 times their preceding values, with unchanged availability masks. Both are below $10^{-5}$ s/km, but their lower bounds still exceed $10^{-7}$ s/km. The preceding accepted changes exceed $10^{-11}$ s, so these reductions do not satisfy

the combined stopping condition either. All 65 complete face pieces again admit zero aligned-normal support. Retaining the ordinary entry and exit terms gives feasible full-vector norms of $4.4053 \times 10^{-5}$ and $2.2711 \times 10^{-5}$ s/km; these remain inconclusive about the minimum support norm. Independent saved readback checks the current geometry, incident-edge sums, exact polynomial signs and full native identities. Results are in `data/development/hayward_cmj_post_half_step_tangent_001/RESULTS.md`.

The existing C++ evaluator is compared with those saved action enclosures on eight fixed polygons: the face candidate, first accepted step, rejected full step and accepted half step for each event. Its eight-point segment integration and sampling are unchanged from the original retry kernel. Paired sums of the emitted segment changes reproduce all four full-step rejection and half-step acceptance decisions, with error below $4.0 \times 10^{-17}$ s. On the face and first-step polygons, the available gradient-vector errors are below $1.057 \times 10^{-15}$ s/km. These errors are far smaller than the retained tangent residuals and resolved action changes on this roster. Each path evaluation takes 1.84–3.42 ms in this single batch. This comparison supports using the existing evaluator for further bounded tangent relaxation; it does not establish its accuracy on every later polygon or certify endpoint derivatives, stationarity or first arrival. Raw hexadecimal values, reference joins and numerical comparisons are retained in `data/development/hayward_cmj_fixed_polygon_kernel_001/`. After independent review of the half-step bounds, a separate saved-data comparison joins them to the already emitted C++ half-step gradients. The two available-vector errors are below $1.020 \times 10^{-15}$ s/km. This adds no evaluator or reference-integration call and leaves the frozen benchmark's original oracle inventory unchanged. Evidence is in `data/development/hayward_cmj_half_gradient_comparison_001/RESULTS.md`.

A bounded iteration then relaxes the same available coordinates from the half-step paths. The existing edge metric supplies each direction, with the face-normal search mask unchanged. A paired sum of segment-action changes controls the finite-step Armijo decision; the actual rounded displacement must also have negative computed slope. The two paths accept 16 and 38 updates in 32 and 152 trials, respectively. Every accepted fraction is 1/2 for the first event and 1/8 for the second. Their computed available norms reach $8.1821 \times 10^{-8}$ and $9.6184 \times 10^{-8}$ s/km. Independent saved readback reconstructs all 184 rounded trial geometries and decisions. Intermediate contractions and paired sums remain recorded evaluator quantities; this replay does not independently integrate each accepted step. The final continuous action and full coupled support require separate bounds. Original false flags and null candidate convergence remain unchanged. The complete iteration and its limitations are in `data/development/hayward_cmj_tangent_iteration_001/RESULTS.md`.

The final audit bounds a complete supporting vector for the first variation on those exact final polygons. For each aligned face piece, write its edge parameter as $u = \ell + \delta t$, $0 \le t \le 1$, and let $h_0 = 1 - u$ and $h_1 = u$. The endpoint normal moments from either neighboring cell are

$$M_j^{\pm} = -L\delta \int_0^1 \frac{h_j(t) v_{n,\pm}(t)}{v(t)^2}\, dt. \tag{5}$$

Here $L$ is the complete edge length and $v$ is its continuous face velocity. Exact polynomial extrema establish a convex normal slowness jump over every one of the 65 face pieces. A single coefficient $0 \le \theta \le 1$ per piece mixes $(1-\theta)M_j^- + \theta M_j^+$ for both endpoints. This shared coefficient preserves their coupling. All ordinary contributions from adjacent edges, including the entry and exit normal terms, remain in the assembled vector.

A bounded midpoint fit proposes the coefficients; exact promotion of those binary64 values and outward interval assembly determine the final bound. The first fit reaches its 10,000-sweep cap; the second stops after 1,894 sweeps with unchanged coefficients. Neither outcome certifies a minimum. Nevertheless, both assembled feasible vectors have norm upper bounds below $10^{-7}$ s/km (Table 7). For every fixed-endpoint displacement $d$, the supporting vector $G$ satisfies $DT[d] \ge G \cdot d$; a small $\|G\|$ therefore bounds the possible first-order decrease on this fixed polygon topology. No free coordinate lies

on the physical box boundary. The same audit resolves a net action decrease from each accepted half-step path, with enclosure widths below $6.0 \times 10^{-32}$ s. It does not certify every intermediate approximate Armijo decision.

Table 7: Final prescribed CMJ polygons. Available norms are rounded displays of interval bounds; the full-vector entries are conservative upper bounds. Action changes compare final and half-step geometry. These are fixed-polygon first-order results, not first-arrival or source-accuracy certificates.

| Event | Available norm (s/km) | Full norm upper bound (s/km) | $\Delta T$ (s) |
|---|---|---|---|
| `NC518943` | $8.1821 \times 10^{-8}$ | $8.1822 \times 10^{-8}$ | $-3.3921 \times 10^{-11}$ |
| `NC519264` | $9.6184 \times 10^{-8}$ | $9.6185 \times 10^{-8}$ | $-5.8997 \times 10^{-11}$ |

Twenty invented controls test the moment formula, shared endpoint coupling, variable face velocity, convexity, ordinary terms and interval projection. Independent saved readback then passes 90,213 checks of the native geometry, identities, polynomial signs, moment scaling and full-vector assembly. Its primitive action and component intervals remain tied to their reviewed source and controls; the checker does not reintegrate them or repeat the fit. These results establish small feasible residuals on the prescribed face paths. They do not supply automatic contact discovery, source/model derivative accuracy or first-arrival selection, and the original false flags remain. The calculation and its complete limitations are in `data/development/hayward_cmj_final_tangent_support_001/RESULTS.md`.

A separate exact inventory of the two original off-face polygons finds 30 open-segment grid-plane crossings per path. Scalar replay of the frozen integration predicates inserts all 30 in each case, with no omitted or spurious cut, short-edge skip or duplicate cut. Thus these cutoff omissions do not explain the two saved original failures. Retained cut parameters are still rounded; this inventory does not assess quadrature, sampled-cell selection or arbitrary future off-face gradients. The identified records are in `data/development/hayward_cmj_original_cut_inventory_001/RESULTS.md`.

An invented contact experiment separates event construction from stopping. For $v = 6 - |x|$ km/s, two successive exact event proposals place adjacent internal vertices on $x = 0$ and reduce the action by $4.2569 \times 10^{-4}$ and $1.9651 \times 10^{-4}$ s. The first contact normal is retained for one search proposal only; its ordinary derivative still admits descent. For $v = 6 + |x|$ km/s, three unit edges initially on the face have zero tangent derivatives, yet simultaneous positive or negative unit-normal release of the two internal vertices gives $DT = -1/18$ s/km. The declared positive release lowers the action by $8.2554 \times 10^{-4}$ s. All 30 checks pass, including independently bounded action and true directional changes. These controls establish finite contact and release examples, not a full supporting vector or automatic convergence on native paths.

The same experiment exposes a separate integration failure (Table 8). Omitting a tiny central crossing leaves an endpoint-gradient error of $1.1322 \times 10^{-4}$ s/km, although the action error is only $5.2427 \times 10^{-17}$ s. The ordinary derivative weights the velocity-gradient jump on both sides of the face; accurate integration of the nearly constant slowness does not resolve that jump. Thus action agreement alone cannot validate the variational derivatives. The unchanged kernel and complete saved results are in `data/development/internal_face_contact_controls_001/RESULTS.md`. This synthetic omission does not explain the two original CMJ paths with complete crossing inventories.

A separate diagnostic preserves every exact open crossing and fixes each integration piece's cell from its exact interior geometry. It retains the eight Gauss nodes and weights, but prepares local interpolation and endpoint shape weights before rounding them to binary64. This combined arithmetic change reduces the tiny-crossing gradient error to $7.4166 \times 10^{-17}$ s/km. A ridge translated to $x = 1$ also improves substantially. In that case the old evaluator omits the crossing as well, so the comparison does not isolate the effect of cell selection from partitioning. Constant-speed, reversed, aligned-face and genuinely trilinear controls complete the nine-case roster; 59 structural checks pass. The three-

dimensional field is $v = (2 + x)(3 + y)(4 + z)$ with coordinates in km and velocity in km/s, evaluated between $(-1/4, 0, -1/4)$ and $(1/4, 1, 1/4)$. Its reference uses fixed rational bounds with reciprocal-series degree 16 (17 retained powers), four subdivisions and 192 bound bits. All approximation-error intervals are retained in `data/development/strict_cell_segment_controls_001/RESULTS.md`. These values approximate continuous endpoint variations; they are not exact derivatives of the finite floating-point quadrature sum. General native geometry, optimized paths and optimized-ray source sensitivities remain untested by this diagnostic, and the maintained kernel is unchanged.

The unchanged original CMJ polygons provide a separate native test of the ordinary gradients. For `NC518943` and `NC519264`, the independent free-knot gradient norms are $4.9649 \times 10^{-4}$ and $4.9379 \times 10^{-4}$ s/km. The saved C++ vectors differ from the independent bounds by only $1.0564 \times 10^{-15}$ and $1.0436 \times 10^{-15}$ s/km. All 6,168 internal components are ordinary; no exact physical-box projection is active. Thus the large original residuals persist under independent evaluation and cannot be attributed to these component errors. The audit retains the historical activity-band norm separately and passes 59,290 saved-data checks over all 2,058 edges. These results motivate testing exact contact proposals from the original paths, while leaving automatic search, optimized-ray source sensitivities and first-arrival selection unresolved. Both original failure flags remain unchanged. The complete error intervals and identified records are in `data/development/hayward_cmj_original_gradient_bounds_001/RESULTS.md`.

Table 8: Invented segment gradient errors, in s/km. Norms compare all six ordinary endpoint components with independent affine references or, for the last row, rational trilinear bounds. The original and separate explicit-cell evaluators use the same eight-point quadrature. Displayed values round their saved error intervals; passing the structural checks imposes no numerical accuracy threshold.

| Case | Original cut omitted? | Original norm | Explicit-cell norm |
|---|---|---|---|
| Below displacement cutoff | Yes | $1.1322 \times 10^{-4}$ | $7.4166 \times 10^{-17}$ |
| Above displacement cutoff | No | $6.1085 \times 10^{-17}$ | $6.1085 \times 10^{-17}$ |
| Below endpoint cutoff | Yes | $3.2342 \times 10^{-12}$ | $5.7856 \times 10^{-18}$ |
| Above endpoint cutoff | No | $5.0938 \times 10^{-18}$ | $3.8688 \times 10^{-18}$ |
| Translated nextafter edge | Yes | $7.5209 \times 10^{-3}$ | $2.2902 \times 10^{-17}$ |
| Three-dimensional field | No | $1.2275 \times 10^{-17}$ | $1.2542 \times 10^{-17}$ |

Table 9: One tangent step from each exact-face candidate at `NC.CMJ..EHZ`. The action change is relative to that candidate, not to the original failed path. Displayed values round rational bounds. The final column is the subsequent available tangent norm, not a full stationarity test.

| Event | $\Delta T$ (s) | Maximum move (m) | Tangent norm (s/km) |
|---|---|---|---|
| `NC518943` | $-2.4666 \times 10^{-10}$ | 0.05169 | $3.1087 \times 10^{-6}$ |
| `NC519264` | $-4.6728 \times 10^{-10}$ | 0.14095 | $1.2717 \times 10^{-5}$ |

### 7.7 Exact branch seeds and continuous synthetic truth

Supplying the exact earliest branch separates initialization error from path representation and local optimization. Four identified Loma rays, diagnostic rows 5, 49, 57, and 88, have continuous first arrivals along the critical guide at $x = 0$. Circular arcs and the straight guide are sampled at five spacings from 0.5 to 0.03125 km. One calculation uses the unchanged tracer, including its initial uniform resampling. A second copied tracer retains the supplied nodes initially; all interior nodes remain free during optimization. The action, derivatives, line search, and 1,000-update budget remain unchanged.

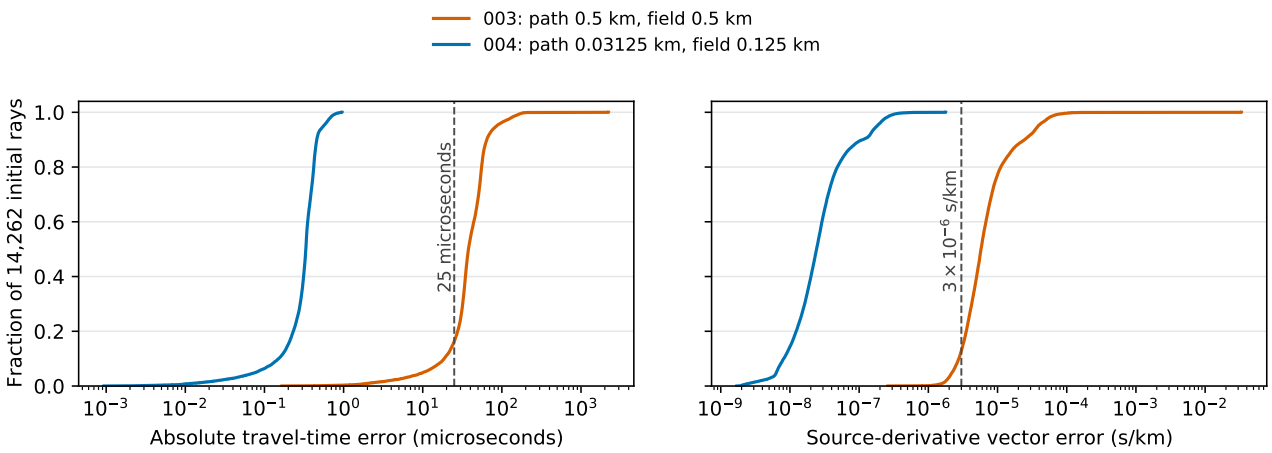



Figure 12: Initial Hayward errors relative to independently enumerated continuous first arrivals. Both curves retain the same 14,262 native observations and references. The left panel shows absolute travel-time error; the right shows the Euclidean norm of all three source-derivative errors. Dashed lines mark the unchanged accuracy limits. Refining path spacing and the initialization field together brings every fourth-run ray within both limits. This diagnostic does not validate a recovered velocity model.

Only one of twenty optimized uniform-seed paths meets the local stopping rule; none of twenty retained-seed paths does. At 0.03125 km, all four uniform-seed results nevertheless have time errors below 4.750 microseconds and source-vector errors below $4.35 \times 10^{-7}$ s/km. The finest retained row-5 path accepts no update, despite time error 4.696 microseconds and source-vector error about $2 \times 10^{-15}$ s/km. Its ordinary gradient norm is 0.0751 s/km, and collective normal displacements of its interface nodes have unequal one-sided action slopes. The correct arrival branch therefore does not resolve the nonsmooth optimization problem. These controls also show that initial uniform resampling can damage a coarse guide seed; its additional row-5 time error falls from 473.730 microseconds at 0.5 km to 0.05368 microseconds at 0.03125 km. Exact polygon integration agrees with production action within $2.25 \times 10^{-13}$ s. All paths, original flags, identities, and isolated source changes are retained in `data/development/loma_geometry_preflight/reference_seeds_001/`.

A separate directional calculation resolves two contributions to the large row-5 gradient. For an interface-aligned segment, the perturbed normal coordinate interpolates both endpoint directions. Its first variation must integrate the appropriate one-sided slowness derivative on each side of any directional zero; choosing one incident cell for the entire segment is insufficient. Homogeneous, fast-plateau, and reversed-jump cusp controls check this construction. The saved finest retained polygon also admits a small tangential descent. Independent 80- and 100-digit exact-segment evaluations agree on action reductions of $4.532 \times 10^{-12}$ s and $2.028 \times 10^{-10}$ s for two saved feasible directions. Thus the interface convention dominates the large ordinary gradient, but the polygon retains actual optimization error. Smaller normal-direction signs near $10^{-17}$ s are inconclusive. Neither finitely many probes nor accurate travel time justifies changing the original failed stopping flag. The identified paths, directions, and reviewed classification are saved in `data/development/interface_directional/001/`.

An interval supporting-vector calculation bounds the continuous first variation for every displacement of the retained interior knots: $F(d) \geq -B\|d\|$, with $B \leq 6.825849 \times 10^{-6}$ s/km; the arithmetic interval-radius norm is below $2.444 \times 10^{-37}$ s/km. Reciprocal-affine segment primitives and rational arithmetic bounds replace fixed quadrature. A feasible convex-guide support removes the incident-cell normal artifact, while the saved tangent still has strictly negative first variation. Proofs, controls, and independent reviews are retained in `data/development/interface_stationarity_bound/` and

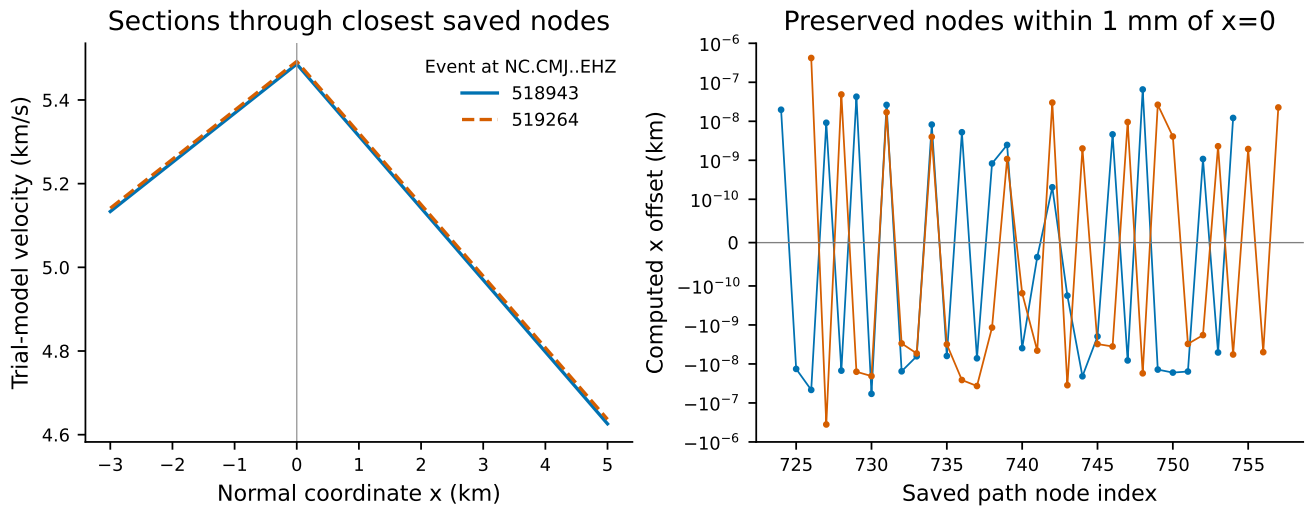



Figure 13: Geometry beside two failed rays in the rejected Hayward trial. Left: velocity sections at fixed $y, z$ through each path's closest saved node to $x = 0$. The face is a strict normal velocity maximum in both sections. Right: all saved nodes within one millimetre of that plane, with a symmetric logarithmic vertical axis. Colors identify the same two events at `NC.CMJ..EHZ` in both panels. Numerical normal offsets are not observational accuracies. This diagnostic preserves both failed flags and supplies no accepted ray or inverse model.

`data/development/native_interface_support/`.

Two isolated steps test the complete first variation, including its convex-cusp term, before interval Armijo acceptance. The negative supporting-vector direction and its tangent projection first pass after 19 halvings. Action decreases by about $4.673 \times 10^{-12}$ s, but new support-norm bounds increase to about $1.842 \times 10^{-5}$ s/km. All 40 trials, identified rational polygons, and original false flag remain in `data/development/native_interface_step/001/`. Exact endpoint-moment analysis shows that the selected support already minimizes the Euclidean norm for the original and full-step polygons; changing normal support cannot remove their smooth residual. The tangent-step sign criterion is inconclusive. The proof and 156 controls are in `data/development/interface_moment_set/`. No production stopping rule, optimizer, or first-arrival certificate follows from these isolated results.

A one-sided second variation along the saved tangent direction evaluates all 2,704 edge contributions, including transverse endpoints on velocity-gradient breaks. Fifty-six checks pass, including three against the saved native evidence. The resulting scalar curvature is 4.5882131 s/km$^2$, whereas the existing isotropic metric assigns 15.3013125 s/km$^2$ to that direction. Replacing midpoint slowness by exact segment-mean slowness changes the metric energy by only $4.45 \times 10^{-6}$ relative; its longitudinal contribution is 10.7241988 s/km$^2$. The saved action secants approach half the one-sided curvature. This comparison concerns one direction and determines neither a Hessian spectrum nor the cause of production convergence failure. The earlier incomplete calculation and the reviewed complete extension remain in `data/development/native_interface_conditioning/`.

Continuous layered enumeration was then extended to all 37,648 pairs in the prepared 724-source, 52-receiver geometry. The first attempt completed every candidate enumeration, but eleven direct candidates failed the unchanged $10^{-8}$ km distance-closure requirement. Their first-arrival values remain blank. All eleven use source 357, event `NC/20091139`, at $x = -0.011677621261526383$ km, just inside the 6 km/s constant-speed region. Solving for $pv_{\max}$ close to one loses the small critical offset in binary64 arithmetic. A separate correction solves for $\delta = 1/p - v_{\max} \geq 0$ and forms the phase speed by exact rational addition before evaluating the segment integrals. Twenty-one isolated cases, including all eleven failed geometries and ten invented near-interface controls, close within $7.11 \times 10^{-15}$ km. Independent

80-digit integrals agree in time within $3.91 \times 10^{-15}$ s and in source-vector norm within $2.87 \times 10^{-17}$ s/km. The failed first attempt is preserved under `data/development/loma_layered_full_001/`.

The corrected second enumeration passes the full geometry and independent native-identity review: all 37,648 references are valid, with no near ties. Their first arrivals comprise 25,724 direct paths, 5,951 turning paths, 5,474 critical guides, and 499 direct paths in constant-speed regions. The maximum closure residual across all 66,302 candidates is $3.13 \times 10^{-9}$ km. The eleven formerly invalid references are the only validity changes. Previously valid branch kinds remain unchanged; their largest time change is 1.55 ns. No source or receiver is removed. Both compressed identified streams, the executed reference source, and the separate completion, physical-consistency, and independent identity reviews are retained in `data/development/loma_layered_full_002/`.

An explicit analytic-truth input mode permits a separately reviewed, complete continuous reference to supply synthetic travel times. The C++ consumer verifies the model, ordered source–receiver support, every native metadata field, finite values, unique valid references, and physical speed bounds. It preserves the existing random-draw, quantization, and differential observation order. Its invented contract suite passes 55 checks, and six default-mode output files remain byte-identical to the frozen preceding executable. Analytic mode writes reference status separately and leaves polygon-ray status fields blank. This changes the declared synthetic truth calculation; it does not repair or accept the inverse forward operator.

The actual LOM-profile simulation now writes all 37,648 absolute observations, 7,947,368 catalog differences, and 7,947,368 precise differences, totaling 15,932,384 rows. A separate streaming audit checks every arrival and observation, including order, both native event identities, receiver epoch, quantization, station bias, and saved-noise arithmetic. All checks pass. The largest saved catalog-subtraction discrepancy is below $1.01 \times 10^{-10}$ s, within the explicit bounds from 12-significant-digit serialization. This rounding bound does not relax a scientific accuracy criterion. The raw run remains preserved, and `data/published/loma_lom_analytic_001/` contains a lossless ten-part archive of 504,089,284 compressed bytes. Every member was read back and verified.

The independently audited LON-profile simulation contains the same number of rows. Full-file comparison establishes byte-identical synthetic observations, truth arrivals, and perturbed event geometry. Its starting grid differs at all 288 velocity nodes; the numerical configuration changes only the declared starting-profile selector. The distinct source snapshot, configurations, execution records, and reviews are retained. Identical simulated observations do not establish equivalence of inversions from the two starting models.

The inverse calculation starts from perturbed sources and a depth-dependent grid, requiring references distinct from the $x$-dependent synthetic truth. All 37,648 LOM initial references pass their independent reviews, with unique first arrivals and defined source derivatives; 59 pairs use the finite-bottom branch. A separate LON enumeration preserves one invalid reference among the same 37,648 pairs. Pair 8156, event `NC/255127` at `NC.JSC..EHZ`, has a turning candidate just below the 11 km knot and distance residual $1.0784 \times 10^{-8}$ km, above the unchanged $10^{-8}$ km requirement. Its selected time and derivatives remain blank. No tolerance is relaxed. Both full identified reference streams and their reviews are retained in `data/development/loma_initial_reference_001/` and `data/development/loma_lon_initial_reference_001/`.

The LON failure arose when reconstructing turning speed from the isolated root. Rounding $u = \sqrt{u_0^2 + r^2}$ to binary64 loses part of the small offset above the layer-knot speed $u_0$. A separately reviewed variant retains a rational square-root enclosure midpoint while evaluating each turning candidate. It leaves the isolated roots, physical formulas, and acceptance limits unchanged. All 37,648 references now pass in the separate second attempt. Every one of the 37,752 candidate signatures is retained, with no change to a previously selected branch. Pair 8156 is the only validity change; its original failed values remain blank in the first attempt. The maximum candidate-time change is 1.701 nanoseconds, and the maximum source-vector change is $1.321 \times 10^{-13}$ s/km. Maximum distance closure residual

decreases to $9.652 \times 10^{-10}$ km. The full identified comparison and independent readback are retained in `data/development/loma_lon_initial_reference_002/`. This corrects the reference arithmetic; it does not validate a production ray or inversion.

Separate frozen LOM and LON initial-forward preflights assess these inputs. The LON preflight uses the corrected complete reference and the same numerical settings as LOM. Seven bounded controls verify unchanged matrix-only outputs between the earlier LOM executable and the guarded executable used for LON, including retention of false local-convergence flags. These controls and the launch eligibility review are saved in `data/development/loma_lon_initial_preflight_adoption_001/`. Each preflight exports full source derivatives and exact endpoint/domain checks from the rays used to count matrix entries. A separate sampled refinement retains all 104 declared samples and 24 valid source finite differences while limiting cached fine receiver fields. It performs no inverse solve. Neither simulation completion nor a matrix storage estimate establishes inverse convergence.

The LOM matrix census has completed. A full comparison with its continuous initial-model references finds all 37,648 time errors below 25 microseconds, with maximum 2.8984 microseconds. However, 48 source-vector errors exceed $3 \times 10^{-6}$ s/km, and one additional ray retains a false local flag. Only 37,599 rays pass all full-ray comparison criteria. All derivative failures involve seven sources at the 16 km bottom face and finite-bottom reference branches; their errors lie in the vertical component and reach $1.5851 \times 10^{-5}$ s/km. The complete identified comparison and its independent review are in `data/development/loma_lom_initial_accuracy_001/`. The LON matrix census has also completed using its earlier frozen optimizer. All 37,648 local flags and time comparisons pass; the largest time error is 4.877 microseconds. However, 50 source-vector errors exceed $3 \times 10^{-6}$ s/km, with maximum $9.8634 \times 10^{-6}$ s/km. Thirteen failures involve four bottom-boundary sources, whereas 37 involve 12 interior sources on turning branches. All 50 have negative vertical error. The largest interior error is $4.8071 \times 10^{-6}$ s/km. Thus the LON failure set cannot be explained solely by the LOM bottom-boundary mechanism. Only 37,598 LON rays meet all full-ray comparison criteria. The complete identified comparison and independent readback are in `data/development/loma_lon_initial_accuracy_001/` and `data/development/loma_lon_initial_accuracy_review_001/`. Both sampled preflights remain active. No inverse calculation is accepted.

Observation-only replays preserve both initializers and their original numerical results for the two distinct failures. Pair 3426, event `NC/10090782` at `NC.JSJ..EHZ`, exhausts all 24 line-search evaluations after 53 accepted updates; it does not reach the 1,000-update limit. Its selected path has projected gradient $1.3762 \times 10^{-5}$ s/km near the 7 km velocity-gradient break. Its accurate time and source vector do not justify changing the false stopping flag. Pair 24104, event `NC/20091537` at `NC.HQR..EHZ`, retains a true local flag after 33 updates but fails source accuracy. The complete paths, histories, native identities and independent saved-record reviews are in `data/development/loma_initial_failure_review_001/` and the two linked observer directories.

An exact calculation on the two segments adjacent to node 484 confirms its vertical partial derivative, $1.03120815 \times 10^{-5}$ s/km, to within $4.59 \times 10^{-21}$ s/km. The earlier $10^{-6}$ km central difference crosses the 7 km velocity-gradient break and has the opposite sign. Within-piece displacements recover the derivative and include certified decreases. This saved-node calculation is retained in `data/development/loma_node_action_001/`; it does not change the path.

A second neutral replay captures all 24 actual rounded trial polygons from the failed search. Its five original numerical tables remain byte-identical. Reciprocal-affine action enclosures then show that every trial satisfies the original Armijo condition, although the binary64 calculation rejects all of them (Figure 14). The first trial decreases the exact action by $6.76195 \times 10^{-15}$ s, while the recorded action increases by $1.24345 \times 10^{-14}$ s. Even the last rounded path has a certified decrease of $3.36321 \times 10^{-22}$ s. Both the recorded and independently enclosed directional derivatives give the same classifications. The maximum difference between recorded and enclosed gradient components is $1.17 \times 10^{-16}$ s/km. An independent 80- and 100-digit calculation of the first and last trials agrees with the enclosures.

The replay, exact intervals, native identities and independent reviews are in `data/development/loma_failed_search_3426_001/` and `data/development/loma_failed_trial_exact_001/`. This identifies false numerical rejection in this captured search. It does not establish convergence of a corrected optimizer or validate an inversion.

Further evaluation of the same saved paths isolates arithmetic contributions. Exact summation of the original binary64 segment values recovers 17 of the 24 decrease decisions, including the first full step. Reevaluating segments and accumulating their paired differences with a 64-bit significand gives 21 approximate passing comparisons; the last three changes remain zero or positive despite their certified decreases. All original double segment values and path totals are reproduced exactly by this diagnostic. Outer accumulation therefore explains the first false rejection, while local segment arithmetic also limits the smallest steps. The saved comparison in `data/development/loma_trial_accumulation_001/` changes neither the optimizer nor any stopping flag.

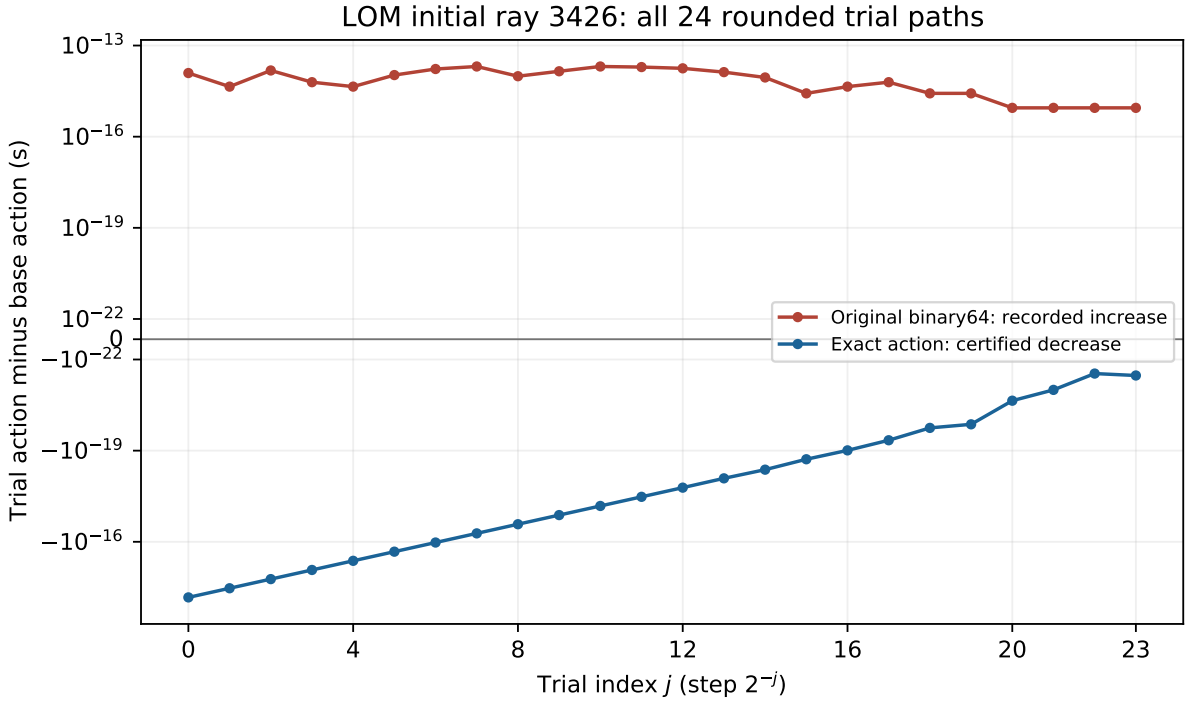



Figure 14: Travel-time changes for the 24 saved LOM trial polygons of pair 3426, using the unchanged depth-only initial velocity model. Positive values are increases from the original binary64 evaluation; negative values are decreases enclosed by rational integration of the same rounded coordinates. The vertical axis is symmetric logarithmic, with a linear interval within $\pm 10^{-22}$ s. Enclosure widths are smaller than the symbols. All exact comparisons pass the original Armijo requirement, and all original numerical evaluations reject it. This diagnostic figure does not promote a manuscript panel or change the original false local-convergence flag.

A subsequent experiment changes only the outer path-action accumulator from `double` to `long double`; the function still returns a binary64 total. Segment evaluation, gradients, initialization, line search, curvature-pair acceptance and both stopping rules remain unchanged. The two frozen builds pass 182 homogeneous and saved-path checks, and both modified native runs agree numerically with their unobserved counterparts. This change removes the early line-search termination but does not establish convergence (Figure 15). The FMM start accepts all 1,000 permitted updates and retains a false local flag, with final projected gradient $4.7200 \times 10^{-5}$ s/km. Its travel-time error is 0.860614 microseconds and its source-vector error is $1.41175 \times 10^{-7}$ s/km; both accuracy limits pass. The straight start retains

a true combined-stopping flag after 54 updates, with errors 1.36618 microseconds and $4.67852 \times 10^{-8}$ s/km. Its later arrival does not replace the selected FMM result.

The modified FMM history contains 436 strict decreases and 564 equal values of the recorded binary64 action. It accepts 40 curvature pairs, with the last at update 435, and retains eight pairs thereafter. The original run accepted 34 pairs before its failed search. These records motivate a separate conditioning diagnosis; they do not establish that rejected curvature pairs cause the remaining failure. Equal recorded totals also do not establish equal continuous actions. The single-change source diff, controls, complete histories, native identifiers and independent terminal review are retained in `data/development/loma_outer_accumulation_001/` and `data/development/loma_outer_accumulation_native_001/`. Neither the update budget nor the false stopping flag is relaxed, and no modified inversion is accepted.

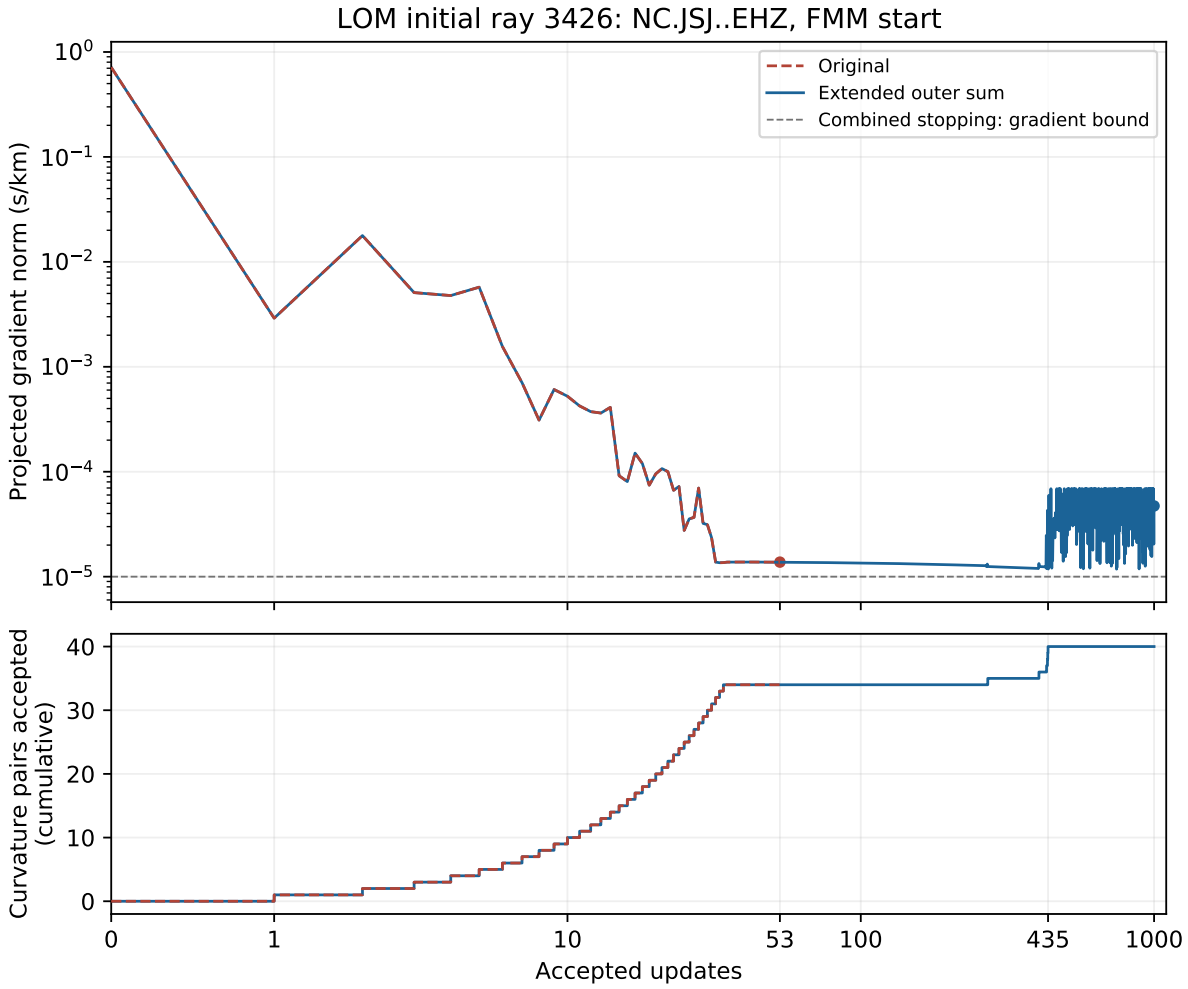



Figure 15: Original and modified-accumulator FMM histories for LOM initial pair 3426, event `NC/10090782` at `NC.JSJ..EHZ`. Upper panel: projected gradient norm after each accepted update. The dashed horizontal line is the gradient bound in the combined stopping rule, which also requires a sufficiently small recorded action change. Lower panel: cumulative accepted curvature pairs. The horizontal axis is logarithmic above one update. The original search fails after 53 updates; extended outer summation reaches the unchanged 1,000-update budget with a false flag. Overlapping curves retain both saved histories. The action sum changes arithmetic accuracy, but this experiment does not resolve local stationarity.

A subsequent logging-only replay reproduces all five existing diagnostic tables byte for byte and exports every curvature pair. Here $s$ is the stored step and $y$ is the gradient change. All 960 discarded pairs have positive $s^T y$, between $2.06 \times 10^{-20}$ and $8.91 \times 10^{-15}$ s, below the unchanged $10^{-14}$ s retention cutoff. Exact rational sums of the saved binary64 components retain the same signs and cutoff classifications. Replacing the stored step by the actual rounded coordinate displacement also leaves every classification unchanged. The last retained pair is iteration 434 (update 435); all 565 subsequent pairs are discarded. The eight retained pairs are 565–967 iterations old at the final update.

Independent readback streams all 1,132,000 component records and recomputes exact sums for 22 representative iterations, including the smallest pair and both sides of the cutoff. These results isolate the implemented retention decision, but positivity alone does not establish useful curvature or a remedy for the failed local flag. The exact sums concern stored steps and gradients, not a continuous Hessian or exact path-action differences. The complete identified capture, saved initializer, terminal receipts and review are retained in `data/development/loma_curvature_capture_native_001/` and `data/development/loma_curvature_capture_review_001/`.

The terminal modified-accumulator polygon approaches a velocity-gradient interface: vertices 484 and 485 lie on opposite sides of 7 km, within $4.67 \times 10^{-12}$ km of the plane. Three prescribed candidates place either vertex, or both, exactly on that interface while retaining every other coordinate and both endpoints. Rational integration gives action changes of $-1.46038 \times 10^{-16}$, $+4.31712 \times 10^{-17}$, and $-2.11958 \times 10^{-16}$ s, respectively (Figure 16). Moving vertex 485 alone therefore increases the action. These values concern explicit candidates, not accepted optimizer steps or a rule for moving vertices onto interfaces.

The two-vertex candidate has exactly one aligned convex-kink edge in the depth-only velocity field. Let $c = L(s'_+ - s'_-) > 0$ for that edge. Its interface first variation admits normal increments $A$ and $B$ satisfying $(A + B)^2 \leq 2cA$ and $(A + B)^2 \leq 2cB$. The pinned moment-set derivation retains their coupled dependence; the two endpoint increments cannot be chosen independently. Outward rational bounds give strictly positive margins for the increments that cancel both internal normal components. All other internal components remain fixed. The complete supporting-vector norm is enclosed near $5.57903 \times 10^{-6}$ s/km, compared with $6.88937 \times 10^{-5}$ s/km for the lower-side support on the same candidate. An independent integration in physical coordinates at 80 and 120 decimal digits agrees with every one of the 3,396 component enclosures.

This establishes the existence of a small supporting vector for the prescribed polygon and its fixed positive-length topology. It does not select an ordinary gradient at the kink, find a minimum-norm vector, certify the original off-interface polygon, or supply a general optimizer correction. Source and velocity-model derivatives are not changed. The original modified-accumulator ray retains its false flag. The complete identified candidates are retained in `data/development/loma_interface_candidates_002/`. Supporting-vector enclosures are saved in `data/development/loma_candidate_support_001/`; the independent review is in `data/development/loma_candidate_support_review_001/`.

An isolated replay tests the role of the absolute curvature cutoff without repeating field initialization. Loading the complete saved initializer first reproduces all five B numerical tables byte for byte. A second replay changes only $s^T y > 10^{-14}$ s to $s^T y > 0$ in the actual and observed optimizers. The action accumulation, gradients, resampling, line search, stopping rules, and update budget remain unchanged. This C replay retains all 145 curvature pairs and satisfies the existing combined stopping rule after 145 updates (Figure 17). Its final projected gradient is $9.734 \times 10^{-6}$ s/km and its last action change is $7.86 \times 10^{-13}$ s; it does not meet the separate $10^{-7}$ s/km gradient condition. Its travel-time and source-vector errors are 0.861 microseconds and $3.18 \times 10^{-8}$ s/km. The straight-initialized result remains unchanged at 54 updates.

This comparison supports a role for the fixed cutoff in this ray's stagnation. It does not justify general adoption of positivity alone. Invented controls retain reciprocal overflow for positive subnormal curvature and acceptance of positive infinity by both predicates. The C history also contains 72 equal

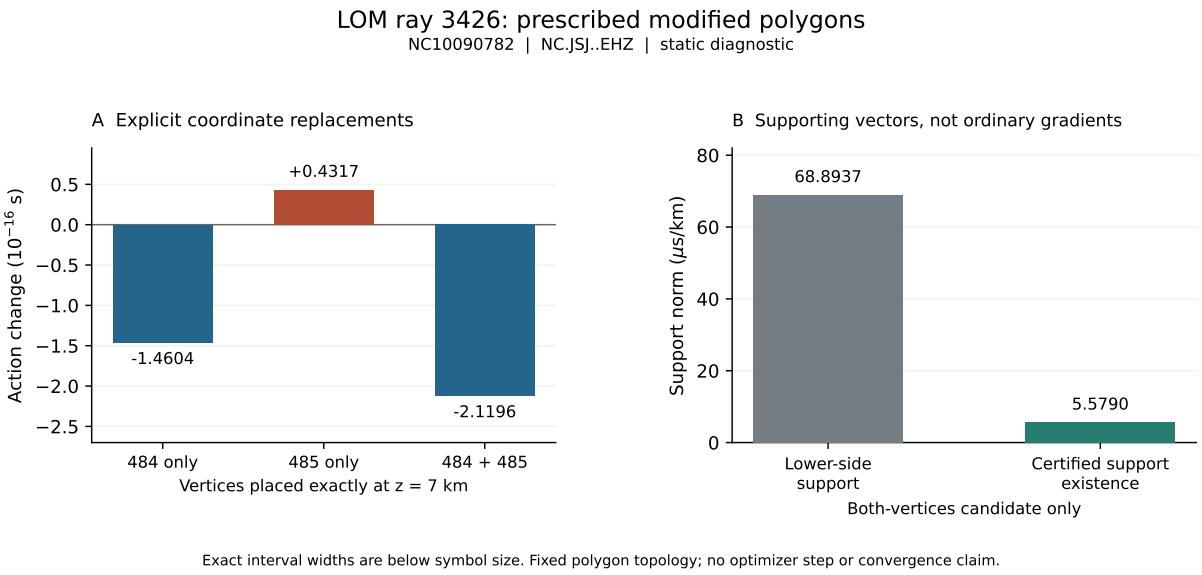



Figure 16: Static interface candidates for LOM initial pair 3426, event `NC/10090782` at `NC.JSJ..EHZ`. Left: exact-action changes when vertex 484, vertex 485, or both are placed at 7 km, retaining all other coordinates. Right: norms of two admissible supporting vectors on the two-vertex candidate. The second norm is an existence bound obtained from the coupled interface moments, with every other internal component retained. The rational enclosure widths are smaller than the plotted symbols. These are prescribed polygons and supports, not accepted optimizer steps or ordinary gradients. No convergence flag or manuscript panel status changes.

recorded action totals, whose exact action changes were not classified. The experiment changes no production source or manuscript panel status. The identified replays and independent reviews are in `data/development/loma_seed_loaded_b_native_001/` and `data/development/loma_positive_curvature_native_001/`.

A further isolated replay admits only finite positive computed curvature with a finite positive reciprocal. It checks the current path metric and the intermediate two-loop quantities before using stored curvature; unusable memory invokes the existing cone-metric fallback and clears memory. An unusable fallback retains its existing exception. These checks introduce no curvature-magnitude floor and change neither the action nor the stopping rules. The guarded and unguarded variants each pass 183 invented component checks. On the native ray, all five C numerical tables remain byte-identical, including both initializers, complete histories, paths, source derivatives, and cone diagnostics. This D result is saved in `data/development/loma_usable_curvature_native_001/`. Computed positive curvature is still not an exact-sign or information certificate; cancellation can reverse its sign. This diagnostic result does not validate a full ray cohort.

The same numerical correction subsequently passed a bounded integration test: 48 core checks, the inversion-guard and external-input components, all 37 declared finest-grid layered checks, 72 oblique checks, and 64 exact production-entry comparisons. At field/path spacing 0.0625/0.0625 km, the largest layered time and source-vector errors are 1.798 microseconds and $4.43 \times 10^{-7}$ s/km. A retained coarser 0.125/0.125 km comparison at 14.68 km offset has source-vector error $3.24035 \times 10^{-6}$ s/km, exceeding the unchanged $3 \times 10^{-6}$ target. The declared accuracy gates apply to the finest grid, so their success does not remove this coarse failure.

The `first-arrival-integration` branch now adopts the exact tested outer accumulator and usable-curvature helper. Segment actions, derivatives, line search, fallback, stopping rules and configurations remain unchanged. The failed Hayward attempt and running LOM/LON preflights retain their earlier frozen binaries. Their results cannot validate the new optimizer. The bounded tests and adoption record are in `data/development/usable_curvature_integration_checks_001/` and `data/development/usable_curvature_adoption_001/`. No new full-cohort validation or corrected inverse result follows from adoption.

A bounded LON replay tests the adopted D correction on the worst interior failure, pair 8670 at `NC.JHL..EHZ`, and a passing control, pair 8674 at `NC.JRG..EHZ`. Both use event `NC/10090456`; both were selected from straight initialization in the original census. The archived public trace first reproduces all 100 recorded physical, status and native fields per pair at their original 12-significant-digit precision. No receiver field is rebuilt. The old and D calculations then produce byte-identical complete final paths and model partials, with unchanged source vectors, local flags and 178/207 updates. Their source-vector errors remain $4.80706 \times 10^{-6}$ and $1.05998 \times 10^{-6}$ s/km: the first still fails and the control still passes. Only the outer action totals change, by less than $9 \times 10^{-16}$ s. The evidence in `data/development/loma_lon_straight_curvature_replay_001/` shows no sensitivity to this adopted correction on these two cases. It does not isolate all optimization or stopping effects, explain the remaining interior failures, or establish full-cohort accuracy.

The saved first edge permits a separate integration check with its other endpoint held fixed. Both edges lie entirely within the same affine layer, $v(z) = v_0 + gz$, between 0 and 3 km depth. Let $\mathbf{a}$ be the source, $\mathbf{b}$ the first saved knot, $L = |\mathbf{b} - \mathbf{a}|$, $v_a = v(\mathbf{a})$ and $v_b = v(\mathbf{b})$. The exact straight and curved connector actions are

$$T_s = L\frac{\log(v_b/v_a)}{v_b - v_a}, \qquad T_c = \frac{2}{|g|}\operatorname{asinh}\left(\frac{|g|L}{2\sqrt{v_a v_b}}\right). \tag{6}$$

Exact geometric checks place both curved arcs inside the affine layer and finite model box. Differentiating $T_s$ with respect to $\mathbf{a}$ recovers the saved source partials within $1.63 \times 10^{-17}$ s/km. Replacing that partial by the derivative of $T_c$ slightly increases the discrepancy from the full layered reference: $(4.80706 \to 4.84264) \times 10^{-6}$ s/km for pair 8670 and $(1.05998 \to 1.06845) \times 10^{-6}$ s/km for pair 8674.

The local action decreases by only 2.52 and 1.39 ns, respectively. Four formula controls and an independent calculation support these values; 80/120-digit agreement checks arithmetic sensitivity, not a rigorous transcendental-roundoff bound. The identified inputs, component vectors and review are saved in `data/development/loma_lon_first_edge_decomposition_001/`. This excludes a material integration error in these saved first edges. It does not reoptimize the first knot or the remaining polygon, replace model sensitivities, or distinguish residual optimization error from approximation elsewhere along the path.

Removing only the combined small-action stopping alternative tests residual optimization at the same initial spacing, 0.03125 km. The strict diagnostic retains the original $10^{-7}$ s/km projected-gradient criterion, 24 Armijo trials, straight initialization and 1,000-update budget. Pair 8670's source error decreases to $2.51167 \times 10^{-6}$ s/km, while the passing control's error increases to $1.61036 \times 10^{-6}$ s/km. Both errors satisfy the unchanged $3 \times 10^{-6}$ s/km limit, but both runs exhaust their budget and return false convergence flags. No nodes merge; the paths retain 516 and 552 points. The first ray's improved accuracy demonstrates sensitivity to early termination. The control's larger derivative error despite a smaller action prevents a general inference that more optimization improves source derivatives. These results, including all native identities and failed flags, are saved in `data/development/loma_lon_strict_straight_replay_001/`.

A second diagnostic changes only that budget to 4,000 updates. Both rays again exhaust it with false strict flags, unchanged node counts and no merges. Their source errors are $1.78051 \times 10^{-6}$ and $1.88236 \times 10^{-6}$ s/km; their projected-gradient norms are $1.82806 \times 10^{-7}$ and $3.46479 \times 10^{-6}$ s/km. Travel-time errors remain 1.030 and 1.033 microseconds. These further decreases in action preserve the mixed source-error trend: the failing ray improves while the control worsens. The outputs in `data/development/loma_lon_strict_straight_4000_replay_001/` still do not establish strict stationarity or isolate the fully optimized polygon error. No additional budget increase, production stopping change, full-cohort acceptance or inversion follows from these tests.

For the captured boundary ray, the first segment is horizontal at 16 km. Let $L$ be its length and $s'_z$ the local depth derivative of slowness. Holding its other endpoint fixed gives $\partial T_{\mathrm{edge}}/\partial z_{\mathrm{source}} = Ls'_z/2$. With $L = 0.03125726$ km, this predicts the saved nonzero derivative $-1.5850656 \times 10^{-5}$ s/km, whereas the continuous boundary reference has zero normal derivative. Subdividing that edge leaves the saved action unchanged and reduces the endpoint partial when inserted knots are fixed. Moving those knots affinely with the source recovers the original derivative within $2.78 \times 10^{-17}$ s/km. These two perturbation families differ; subdivision alone is not a correction to an optimized travel-time derivative. Twenty-five closed-form controls and all five native subdivision levels are retained in `data/development/loma_boundary_endpoint_001/`. No production source derivative or convergence flag has changed.

An isolated extension retains explicit endpoint meshes while leaving the optimizer equations unchanged. Invented affine-bottom-guide, homogeneous oblique-ray and near-transition cases pass 836 action, derivative and initialization-parity checks. However, their 80 optimized records retain two false local flags and 32 source-vector accuracy failures. In the smallest positive-guide perturbation, projected gradients near $3.3 \times 10^{-9}$ s/km produce zero updates despite inaccurate endpoint sensitivities. Near a branch transition, additional descent after source displacement changes the forward error more than the true travel-time increment, invalidating a small-step secant. These preserved controls in `data/development/loma_boundary_endpoint_mesh_controls_001/` therefore do not support refinement with unchanged stopping as a sufficient correction. The proposed native extension is not executed.

Repeating the same invented mesh roster with the reviewed D optimizer leaves all 240 non-time evaluation records and all 5,058 final path rows unchanged. All 836 analytical checks and 18 parity comparisons pass, while the two false local flags and 32 source-accuracy failures remain. Only 85 reported action totals change, by at most $1.12 \times 10^{-16}$ s, from the altered outer sum. Intermediate histories were not captured, so this is a final-output comparison. The repeated evidence in `data/development/loma_boundary_endpoint_mesh_d_controls_001/` does not support another na-

tive mesh ladder as a remedy for this mechanism.

A separate action-level comparison replaces the first straight element $S(a, x_1)$ by the constrained curved connector $G(a, x_1)$ in the same affine bottom layer. Both source and first-vertex derivatives must then come from $G$; changing the source derivative alone would differentiate a different travel-time function. Before the native component calculation, 304 invented checks cover guide, interior and homogeneous branches, both endpoint derivatives, reciprocity and exact branch transitions. Their tiny-step finite differences evaluate an 80-digit continuous action, not a binary64 production ray. Oblique cases retain nonzero normal derivatives and exclude blanket zeroing at the bottom face.

For the saved pair 24104, the first connector is a positive horizontal guide. Its mathematical action equals the original element action exactly. Both inward normal endpoint derivatives of this connector are zero. The complete first-vertex derivative includes the unchanged second element, however, and its normal component changes from $-3.17013 \times 10^{-5}$ to $-1.58507 \times 10^{-5}$ s/km. The connector source vector differs from the saved continuous reference by $7.83 \times 10^{-17}$ s/km. The exact grid-derived affine ratio and full component accounting are retained in `data/development/loma_endpoint_connector_native_001/`. The independent 120-digit readback is saved in `data/development/loma_endpoint_connector_review_001/`. This evaluates a fixed component and does not reoptimize the ray or promote its accuracy flags. A production correction would require one connector definition in the actual trilinear medium, with consistent endpoint and nodal-model derivatives throughout inversion. The affine calculation alone does not supply that definition or Jacobian.

A restricted two-segment family tests part of that requirement in an invented eight-node trilinear cell. An interior source connects through one movable bottom contact to a fixed bottom receiver. All nodal perturbations use the same reciprocal-velocity action and original eight nodal sensitivities. For the unperturbed affine profile, the optimized family has the correct zero inward derivative limit, but its leading excess action is $3/(2\sqrt{2}) = 1.06066$ times the leading excess action of the continuous curved path. A correct boundary derivative limit therefore does not establish the correct local action. Eighty- and 120-digit calculations pass 87 recorded controls; the largest fixed-curve and reoptimized-candidate model-column discrepancies are $1.44 \times 10^{-11}$ s at the tested finite differences. These are agreement checks, not certified error bounds. All 192 perturbed interior-source envelope comparisons and 224 derivative-error certificates remain unresolved, and the fixed receiver's normal partial remains biased. The controls in `data/development/loma_endpoint_scalar_contact_controls_001/` provide no native-ray or production correction.

A shooting calculation then uses curved stationary candidates in the same original eight-node trilinear velocity field for every perturbation. Direct and bottom-guide branches include explicit zero-length caps. The Hamiltonian equations integrate the action and nodal kernels along each candidate; endpoint momenta give its endpoint derivatives. The selected value is the least action among successfully matched admissible roots. One attempted root per topology, sampled domain and reaction checks, and a restricted branch roster do not establish a global minimum.

Across both declared integration tolerances, all 40 affine reference comparisons pass. The largest travel-time and full endpoint-vector errors are $1.97 \times 10^{-11}$ s and $1.03 \times 10^{-13}$ s/km. At the three resolved source offsets, the excess action agrees with the curved affine reference to within $4.4 \times 10^{-9}$ in relative ratio. All 160 perturbation comparisons have matched candidates, but this does not mean all derivative discrepancies pass. The 56 central endpoint comparisons differ by at most $6.14 \times 10^{-8}$ s/km. All 24 inward bottom-normal secants exceed $3 \times 10^{-6}$ s/km at the tested steps. Their discrepancies decrease by factors 3.962–4.016 when the step decreases by 16, consistent with square-root truncation (Figure 18). Original-node model-column errors increase from $1.60 \times 10^{-12}$ to $6.58 \times 10^{-10}$ s as the perturbation decreases from $2^{-16}$ to $2^{-20}$, indicating a numerical floor. These controls are saved in `data/development/loma_endpoint_shooting_controls_001/`. A separate four-baseline replay saves initially omitted mixed-model trajectories and exactly reproduces the stored kernel, momentum, and boundary-secant values; it is preserved in `data/development/loma_endpoint_shooting_baselines_`

001/. They provide a local action-consistent candidate, with no native-ray, full-path optimization, or production correction claimed.

The boundary limit differs from an ordinary centered finite difference. For an affine layer with bottom speed $v_b$ at $z = B$ and positive depth gradient $g$, a branch retaining a positive guide has inward-displacement derivative $[(v_b - gh)^{-2} - v_b^{-2}]^{1/2}$ after an inward displacement $h$. Integrating this momentum gives

$$T(B-h) - T(B) = ch^{3/2} + O(h^{5/2}), \qquad c = \frac{2\sqrt{2g}}{3v_b^{3/2}}.$$

Thus $D(h) = [T(B) - T(B-h)]/h$ approaches its zero limiting normal derivative with a leading square-root term. For a smooth nonuniform field the analogous expansion is conditional on a regular, stable branch, positive normal reaction, and a remaining positive guide; it is not an exact mixed-field oracle.

A saved-output calculation cancels that leading term using $E(h) = [4D(h/16) - D(h)]/3$. Across eight ladders, the largest discrepancy from limiting momentum is $3.47 \times 10^{-9}$ s/km for the first step pair and $4.98 \times 10^{-8}$ s/km for the finer pair. The finer result worsens, consistent with a numerical floor. These 16 extrapolates support the truncation interpretation but supply no certified error bound, global branch test, or replacement for the 24 failed raw comparisons. A fixed straight element's nonzero linear term, $Ls'_z/2$, survives this cancellation. The derivation, complete saved ladders and unchanged raw failures are retained in data/development/loma_endpoint_shooting_asymptotics_002/.

### 7.8 Rejecting unsupported forward evaluations during inversion

The preceding inversion code recorded local ray failures but could still use those rays in a Jacobian, a line-search objective, or final output. An initial preflight therefore did not protect later model updates. A new guard on the integration branch applies the unchanged local, finite-value, domain, endpoint, initialization, and derivative-dimension checks before each such use. A failed evaluation stops the inverse run and saves its model, identified ray statuses and available paths separately from the last accepted state. Failure snapshots retain 17 significant digits. Unevaluated derivatives remain blank, including caught worker exceptions.

Invented component runs exercise failures at the current model, a trial step, and the final model. They reproduce the parent code's use of unconverged rays; all seven outputs of a successful fixed-model location inversion remain byte-identical. The matrix-only diagnostic retains its original behavior. Independent review and the optimized application build pass. These checks, saved in data/development/inversion_ray_gate_checks/, establish failure handling rather than global first-arrival accuracy or nonlinear recovery. No live executable was replaced, and no corrected inverse result is claimed.

A separate candidate now retries only a pure local-stopping rejection during the inverse line search. It retains the old ray kernel used for the full initial Hayward accuracy checks, the original solve increment, fixed weights and twelve halving fractions. Current/final-model failures, mixed reasons, unexpected exceptions and failed evidence writes remain fatal. Each rejected trial retains its own native states and paths, and the first rejection saves all source/model increments, residuals, fixed weights and LSQR status. The old-kernel component suite passes 1,102 recorded assertions across small invented fixtures; 28 all-valid files remain byte-identical. Separate bounded integration checks pass all 48 core checks and complete a tiny joint-DD fixture after nine and eleven local rejections, accepting fractions 1/512 and 1/2048. Its original zero ray-iteration cap is retained; this is not a new scientific stopping rule. The evidence is in data/development/hayward_old_kernel_trial_backtracking_controls_001/ and data/development/hayward_old_kernel_trial_backtracking_integration_checks_002/.

The prospective completion reviewer passes 190 saved-table controls after three preserved false acceptances were corrected: excess iterations, continuation after stage-2 no-update termination, and a shifted initial accepted state. Independent readback reconstructs all 20 joint-fixture rejected states and

both accepted updates. The launcher controls exercise its actual child/source prerequisite only, stopping before final tables; complete positive launcher integration remains separate. An all-success final gate emits no standalone full ray-status table, so its enforcement is tied to the exact audited binary and successful return. These checks, saved in `data/development/hayward_trial_completion_checks_002/`, establish evidence closure, not accepted field recovery or updated-model first-arrival accuracy.

A separately reviewed launcher binds both main-program changes to the original initial-model evidence while retaining identical mathematical headers. Its 62 checks include the actual saved eligibility assessment of 58,980 observations and 14,262 native rays, and five process-lifecycle cases. The latter exercise normal and failed exits, surviving writers and interruption; no compiler, ray or inverse calculation runs in this suite. The original unexecuted test draft remains preserved. This launcher validation preceded the separate field attempt. A successful execution would still await final ray, derivative and scientific assessment. The launcher evidence is in `data/development/hayward_old_kernel_backtracking_launcher_checks_002/`; the lossless archive and restoration instructions are in `data/development/trial_backtracking_evidence_001/README.md`. The catalog-only attempt began at 10:52 UTC on 10 October 2026. Its first update is accepted at inverse step fraction 1/8, after three larger trials fail local ray stopping. The fixed-weight inversion objective, including data residuals and log-slowness smoothing, falls from 1812.15659 to 1409.16490. The closed iteration-1 starting-state snapshot contains the result of this iteration-0 update, retaining all 1,092 event identities and their station, channel, epoch and observation mappings.

Figure 19 compares this saved state with the original identified inputs. At the actual grid depth of 8 km, the 77 nodal velocity changes range from −0.25806 to +0.07275 km/s. Horizontal event movements range from 0.00181 to 0.33425 km; the largest three-dimensional movement is 0.55600 km. No spatial interpolation is used in this comparison. The run remains incomplete. Its recorded history RMS describes the preupdate state, and updated-model first-arrival and source-derivative accuracy remain unestablished. The saved state also lacks the complete Jacobian export needed for full-DD resolution. These results do not change the manuscript panel classifications. Exact saved values, full identities, source and restoration instructions are in `data/development/hayward_accepted_update_delivery_001/README.md`.

### 7.9 Historical waveform metadata and identified correlations

The existing Hayward catalog pair at differential row 15108 links events `NC/238035` and `NC/30083101` at `NC.JPR.02.EHZ`. EarthScope routing identifies NCEDC as the archive for the two requested windows. The preserved miniSEED records each contain 500 contiguous samples at 100 Hz over a half-open five-second interval. Full response metadata cover the distinct 1991 and 1995 epochs; their three-stage response contents agree. Event, origin, pick, association, channel, and epoch identities accompany the raw files and derived records. Missing provider response resource IDs remain explicitly missing.

The written precursor method uses a tapered 2.56 s window and requires squared coherence above 0.9 at at least half the frequency bins between 2 and 10 Hz (Waldhauser and Ellsworth, 2002, section 2.2, paragraph 16). For the primary pilot calculation, the unspecified choices are declared: a window beginning 0.5 s before the catalog P pick, demeaning, a Hann taper, and five-bin spectral averaging. No additional filter, response removal, or resampling is applied. Only two of twenty frequency bins pass, where ten are required. All nine declared window/smoothing variants fail, with zero to four passing bins. Consequently, no waveform difference is adopted. A line through the two coherent phase bins has two free parameters; its near-zero residual does not measure timing precision. The source, spectra, controls, and failed result are in `data/development/waveform_correlation_pilot_001/`.

An offline audit retains all 14,262 picks and 44,718 catalog differences. Historical operation records identify `NC.CMW1`, but do not provide calibration: the saved EarthScope and NCEDC StationXML responses start in 2000, whereas 117 retained picks and 199 differences involve its 1995–1998 epochs. No later response is backdated. Among 398 picks whose channel names do not end in Z, provider

orientation metadata identify 227 as vertical, 54 as nonvertical, and 117 as unknown. Channel names alone therefore cannot define the vertical waveform selection. A further 89 picks and 542 differences precede the archive's documented March 1984 start; this date is a coverage lead, not proof that all earlier data are absent. These audit categories remove no catalog observations and establish no study exclusion. Full native rows and the original unsubmitted requests are saved in `data/authoritative/waveforms/cohort_plan_001/`.

Normalized waveform correlation supplies differential times without calibration corrections, as clarified by the user. Missing historical responses remain metadata gaps rather than retrieval prerequisites. EQcorrscan documents local-window normalization, pick-relative extraction, and `dt.cc` identifier and sign conventions.[1] Differential time is travel time for event 1 minus that for event 2. The failed spectral pilot remains preserved as a separate calculation.

NCEDC reports covering spans for 464 pick windows; 13,798 have no reported span, which does not establish historical absence. Five sample batches provide 459 complete windows of 500 native samples at 100 Hz and five identified zero-sample outcomes, all at `NC.JPR.02.EHZ`. The unchanged first response is reused; no omitted window is re-requested, and partial or overlapping support is not accepted. The complete identified inventory, raw replies, and sample audit are in `data/authoritative/waveforms/cohort_cc_002/` and `data/authoritative/waveforms/cohort_samples_002/`. The first decoding failure remains in `data/authoritative/waveforms/cohort_samples_001/`. No catalog observation or station is removed.

The declared method correlates 256 native samples over the complete $\pm 0.25$ s search. The template starts nearest to 0.5 s before its pick, with half-sample ties resolved earlier. Direct local centering and scaling evaluates signed Pearson correlation; 41 controls include large-amplitude cancellation cases. No response removal, extra filter, resampling, or fractional interpolation is used. Nonpositive, tied, and boundary maxima are rejected. After discrete alignment, the declared Hann-window and five-bin spectral screen requires at least ten of twenty 2–10 Hz bins to have squared coherence strictly above 0.9. Actual native sample offsets determine the sign:

$$\Delta t_{ij}^{\mathrm{cc}} = \Delta t_{ij}^{\mathrm{catalog}} - \mathrm{delay}_{j-i}.$$

Native-input controls check both epochs, origins, sample hashes, and timing.

Of the original 44,718 pairs, 1,195 have eligible samples on both sides. The calculation accepts 98; 1,086 fail coherence, ten peak at the boundary, and one has a nonpositive peak. The other 43,523 remain unattempted. The 98 accepted event pairs each have one accepted channel and span 103 events: 75 are at `NC.JPR.02.EHZ`, 19 at `NC.CBR.10.EHZ`, three at `NC.JSB.10.EHZ`, and one at `NC.JSF.10.EHZ`. The complete identified roster and all attempted lag/coherence arrays remain in `data/development/native_waveform_cc_001/`. These new subset observations are not identified copies of the manuscript's 17,955 waveform differences and have not entered an inversion. Neither squared correlation nor the 10 ms sample spacing is a formal timing uncertainty. The checked `dt.cc` export in `data/development/native_dt_cc_001/` retains a complete 44,718-row identity sidecar, event integer map, and receiver epochs. It preserves binary64 values, rejects ambiguous mappings, and leaves the separate catalog-only attempt unchanged.

A bounded test of NCEDC's documented event-gather route requested the original five-second windows for catalog pair 21796, events `NC12434` and `NC12437` at `NC.CMC..EHZ`. Both exact requests returned HTTP 204 with empty bodies. The requests, responses, event/pick/channel identities and authoritative epoch are retained in `data/authoritative/waveforms/eventdata_pilot_001/`. This adds no samples or differential times and does not establish historical absence. No alias, calibration correction, or catalog selection change follows.

A separate prepared dataset in `data/hayward_catalog_cc_001/` preserves all 58,980 original catalog rows and appends the 98 waveform measurements, giving 59,078 rows. Thirty-eight added pairs span

[1] EQcorrscan documentation: normalized correlation; correlation-derived differential times.

different native metadata epochs at the same receiver. Both receiver indices and full identities are retained. The combined input requires exactly the original 14,262 native source–receiver rays, with no additional pair. The CC graph has 32 connected components and a largest component of seven events; these counts do not determine the rank of the combined inverse problem.

The adopted empirical prior is literal $CC^2$ as a multiplier of both residual and Jacobian row. It therefore contributes $CC^4$ to the squared data objective before the other squared weighting factors. This explicit convention is not a calibrated inverse variance or a claim to recover the paper's exact weights. Derived timing uncertainties remain blank and marked unestimated. Seventy Python and 36 C++ controls check loading, prediction, signed Jacobian entries, weighting and identity retention; an independent full-file audit checks the actual preparation. No run configuration or inversion was created. A future calculation must declare all nine stage factors explicitly; the existing catalog covariance analysis continues to reject waveform rows until their joint noise model is justified.

Catalog differences require a different covariance calculation from independent differential measurements because their constituent picks recur. Let $C$ map original pick errors into absolute and catalog-difference rows, and assume independent original errors with covariance $\Sigma$. A difference row of $C$ contains $+1$ and $-1$. With actual assembled coefficients $G$, frozen row multipliers $W$, velocity smoothing $L$, and the exported column-scaled damping penalty $D$, define $H = G^T W^2 G + L^T L + D$. If $P$ selects model parameters, then

$$K = PH^{-1}G^T W^2 C, \qquad \mathrm{Cov}(q) = K\Sigma K^T.$$

This propagates shared errors without inverting the singular observation covariance $C\Sigma C^T$. Repeating a datum cannot create an independent measurement. Physical local model resolution is

$$R_{\mathrm{model}} = PH^{-1}G^T W^2 J_{\mathrm{model}},$$

where $J_{\mathrm{model}}$ retains prediction derivatives suppressed by the estimator's DD path-length mask. This response can differ from the model block of $H^{-1}G^T W^2 G$.

Native pick identities and coupled source elimination are retained by `resolution_full_hayward.py`. A separate dense-source implementation passes 35 invented controls, including repeated picks, masked coefficients, and the solver's ordered-entry column norms. Its largest agreement error is $1.78 \times 10^{-15}$ against a declared $10^{-10}$ algebra tolerance. The controls are saved in `data/development/catalog_dd_resolution_controls_003/`. Six direct comparisons with the maintained sparse implementation agree within $9.44 \times 10^{-16}$. The maintained calculation also rejects a nonfinite unmasked response or conditioning diagnostic; 39 controls include finite-input overflow cases and preserve the original 34 results. The reader also checks the full exported ray inventory against every original observation before matrix assembly. It verifies native source/channel identities, both receiver epochs, serialized coordinates, and the recorded local and initializer flags, including rays used only by observations omitted by weights. It refuses incomplete or failed inputs and preserves existing resolution files. All 56 invented input controls pass; the preceding 39 covariance-control records remain exactly unchanged. These checks are saved in `data/development/full_dd_resolution_ray_input_checks_001/`. The original 12-digit status table lacks path-domain, endpoint and source-partial fields; unattempted initialization remains explicitly allowed. Passing this input check therefore does not certify first arrivals or final-model derivative accuracy.

A separate geometry export now writes identified 17-digit status and complete paths from the same ray array used for the full matrix, including pairs used only by zero-weight observations. Its reader checks native support and both receiver epochs, path order, endpoints, box membership, configured initialization and signed source-matrix entries. All 82 invented controls pass on the frozen old-kernel source, with the original eight export files byte-identical. The evaluation point is the parsed saved 12-digit model, not an asserted copy of the earlier internal state. Integration adop-

tion compiled successfully after adding the reviewed checked-I/O helpers; the first compile-only failure is preserved. Neither adoption attempt evaluated rays. Source, controls and independent readback are retained in `data/development/hayward_full_dd_geometry_adoption_002/` and the lossless `data/development/geometry_export_evidence_001/README.md`. These diagnostics establish no field accuracy or covariance acceptance. No new field resolution has been calculated. Fixed weights, independent one-sigma pick errors, and unconstrained sources remain assumptions; active bounds, model and ray errors, nonlinear effects, and waveform noise are outside this calculation.

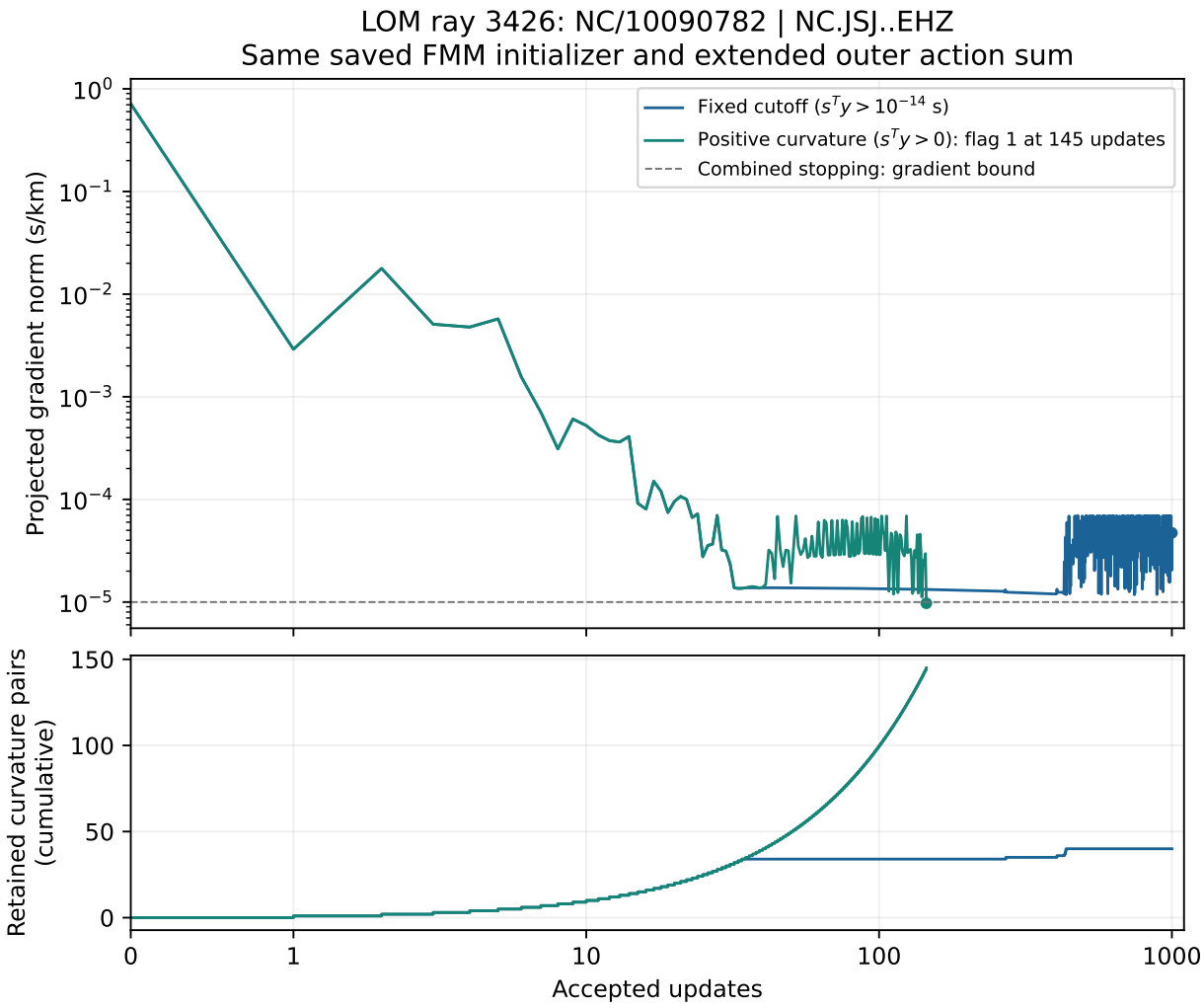



Figure 17: Isolated curvature-retention comparison for LOM initial ray 3426, event `NC/10090782` at `NC.JSJ..EHZ`. Both runs use the same saved FMM initializer and extended outer action sum. Top: projected gradient norm; the dashed line is the gradient bound of the combined stopping rule, which also requires a small action change. Bottom: cumulative retained pairs. The original cutoff exhausts 1,000 updates; positive retention satisfies the unchanged combined rule at 145. The horizontal axis is logarithmic beyond one update. This single-ray result does not establish a safe general retention rule or validate an inversion.

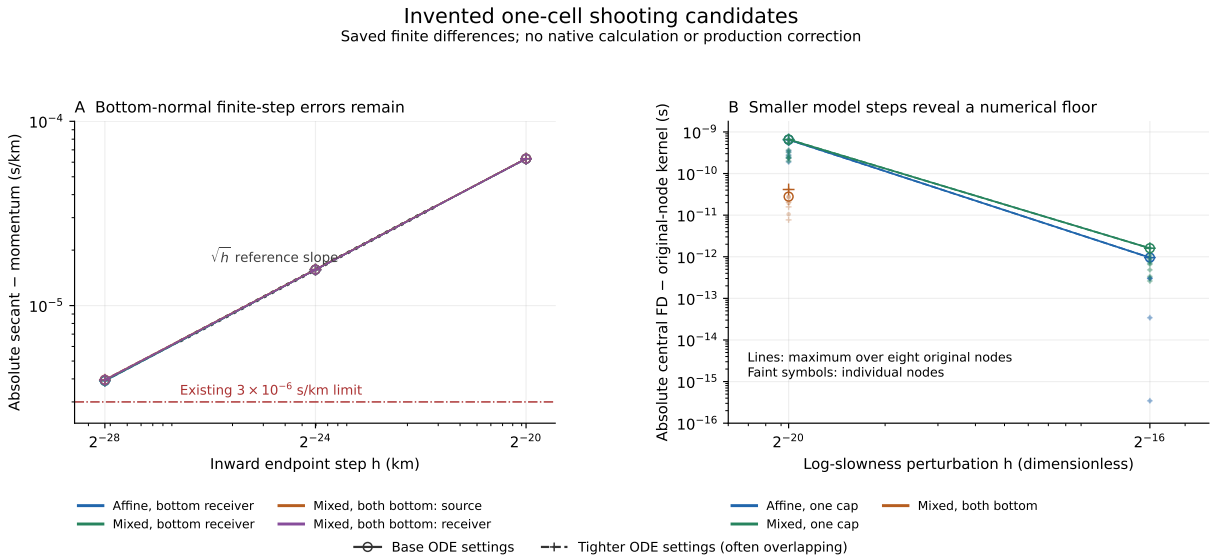



Figure 18: Invented trilinear shooting controls from saved numerical output. Left: inward boundary secant errors relative to limiting endpoint momentum, with a square-root reference slope and the unchanged $3 \times 10^{-6}$ s/km comparison limit. Every sampled inward secant remains above that limit. Right: central finite-difference errors in original-node log-slowness kernels. The smaller model perturbation increases the largest discrepancy, so further step reduction alone is not evidence of derivative accuracy. Both integration tolerances and all corresponding comparisons are retained. These are local candidate tests, not reproduced manuscript panels.

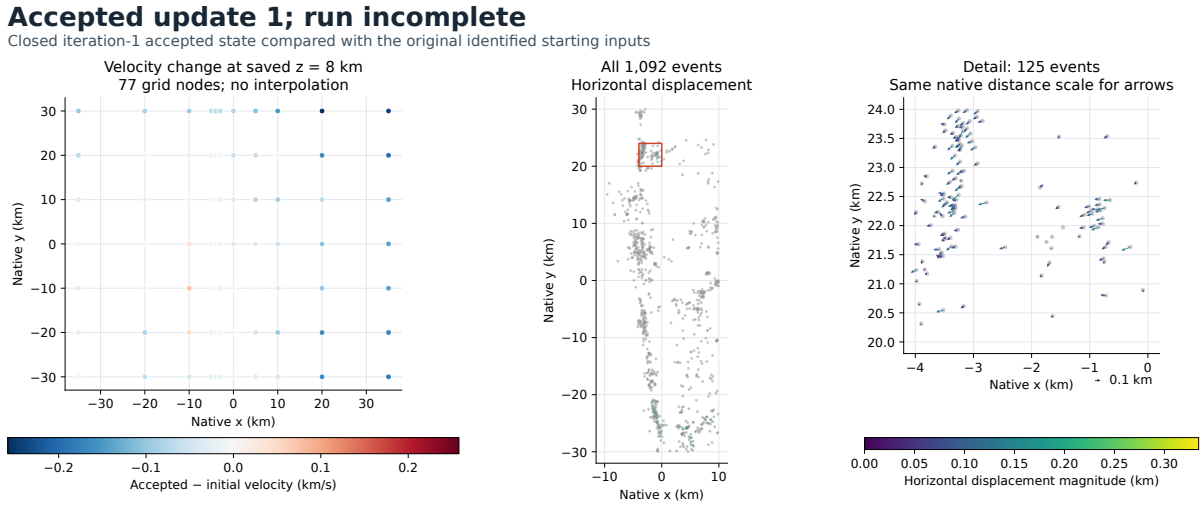



Figure 19: First accepted Hayward update from the separate catalog-only backtracking attempt; the run is incomplete. Left: accepted minus initial velocity at the 77 saved grid nodes at 8 km depth, in km/s. Centre: native horizontal positions of all 1,092 events and their displacements. Right: the indicated 4 km by 4 km window contains 125 events. Grey points denote initial positions; arrows show horizontal movement without amplification, and their colours give its magnitude in km. This saved-state comparison does not report post-update RMS, final convergence or manuscript reproduction.

# 8 Authoritative Hayward subset results

The following calculations use 1,092 catalog sources, 37 physical stations, 14,262 absolute P observations, and 44,718 catalog differences. They contain no waveform-correlation observations. The same identified subset is used for absolute-data tomography, catalog DD tomography, and relocation in the fixed initial one-dimensional velocity model. Figures 20–26 are partial counterparts to the manuscript figures because the observations, pairing, initial model, and numerical implementation differ. The ray branch-selection defect in Section 7.1 remains unresolved in these runs; its effect on these field estimates has not been quantified.

The complete original workflow exits successfully after 33,712 s, including the final checks and report build. This completion record does not establish scientific convergence. Fixed-velocity relocation increases catalog-difference RMS from 0.09340 to 0.25423 s, while the last DD update moves a source coordinate by 2.761 km. The final absolute-data resolution uses two rays with false local flags. These results therefore remain diagnostic comparisons. Independent readback verifies the complete native identities, geographic and zoom joins, and saved resolution interpolation in `data/development/hayward_completed_baseline_review_001/`.

These results use the independently selected authoritative Hayward cohort. Residuals below are unweighted final residuals; standard tomography fits absolute picks only, so its catalog-difference RMS is an additional diagnostic. Catalog displacement measures relocation, not error against known source positions.

The selected pair graph has 35 connected components; the largest contains 1058 events, and 34 events have no differential links. Unlinked events remain in the inversion with absolute observations. 28576 catalog differences span metadata epochs at identical NSLC and receiver coordinates; this geometry match does not establish unchanged timing calibration.

| Method | $R_{\mathrm{abs}}$ (s) | $R_{\mathrm{cat}}$ (s) | Updates | LSQR | Min. rays (%) |
|---|---|---|---|---|---|
| Initial | 0.39163 | 0.09340 | – | – | – |
| standard | 0.07869 | 0.07162 | 12/13 | 13/13 | 95.75 |
| dd | 0.16064 | 0.06160 | 18/18 | 18/18 | 95.76 |
| loc1d | 0.29188 | 0.25423 | 10/13 | 13/13 | 97.93 |

The initial row uses the first pre-update forward prediction. Updates counts accepted steps; LSQR counts solves meeting the existing relative normal-residual tolerance of $10^{-8}$. The last column is the minimum converged fraction over all used rays in any saved pre-update history row. These history entries precede their update and do not represent the final model. Frozen-weight objectives are compared within each update; weights can change between iterations. The solver stops at its iteration limit or when no line-search step is accepted; it does not test nonlinear stationarity or a small-source-update criterion.

No waveform-correlation observations are included. The history column for precise-difference RMS is therefore empty (NaN), not a nonfinite modeled residual.

For standard, median and maximum catalog displacements are 1.100 and 5.956 km; 3 sources lie within $10^{-6}$ km of a grid boundary. Nodal velocities range from 3.326 to 6.288 km/s. There are 1 zero-update iterations and 0 accepted objective increases above the existing $10^{-8}$ audit tolerance; the worst LSQR relative normal residual is 5.14e-09. The final recorded iteration changes a source coordinate by at most 0.000 km and nodal log slowness by at most 0. Geoid depths span -0.960 to 16.494 km, with 1 negative-depth estimates and 0 estimates outside the grid beyond the boundary tolerance. Boundary counts by face are 1 at $y = -30$ km; 2 at $y = 30$ km; a source on intersecting faces contributes to each count. Across all 14262–14262 rays per recorded iteration, the largest projected/raw gradient norms are 0.00577/0.00577 s/km. At the final model, production/refined convergence counts are 31/31 of 32 sampled rays, with maximum projected gradients 0.00049/0.000399 s/km and raw

gradients 0.00049/0.000399 s/km. Halving the segment spacing changes sampled travel times by at most 0.101 ms. For 24 source-coordinate derivatives, maximum production/refined errors against centered travel-time differences are 2.88e-05/2.16e-05 s/km; 0/0 components exceed the existing 0.003 s/km component tolerance. 0 derivative components are omitted at domain boundaries.

For dd, median and maximum catalog displacements are 1.178 and 8.799 km; 15 sources lie within $10^{-6}$ km of a grid boundary. Nodal velocities range from 4.427 to 5.618 km/s. There are 0 zero-update iterations and 0 accepted objective increases above the existing $10^{-8}$ audit tolerance; the worst LSQR relative normal residual is 6.97e-09. The final recorded iteration changes a source coordinate by at most 2.761 km and nodal log slowness by at most 0.00541. Geoid depths span -1.000 to 17.244 km, with 19 negative-depth estimates and 0 estimates outside the grid beyond the boundary tolerance. Boundary counts by face are 2 at $y = 30$ km; 13 at $z = -1$ km; a source on intersecting faces contributes to each count. Across all 14262–14262 rays per recorded iteration, the largest projected/raw gradient norms are 0.00544/0.00544 s/km. At the final model, production/refined convergence counts are 32/32 of 32 sampled rays, with maximum projected gradients 8.08e-07/2.77e-07 s/km and raw gradients 8.08e-07/2.77e-07 s/km. Halving the segment spacing changes sampled travel times by at most 0.008 ms. For 24 source-coordinate derivatives, maximum production/refined errors against centered travel-time differences are 1.52e-06/1.24e-07 s/km; 0/0 components exceed the existing 0.003 s/km component tolerance. 0 derivative components are omitted at domain boundaries.

For loc1d, median and maximum catalog displacements are 2.638 and 8.350 km; 8 sources lie within $10^{-6}$ km of a grid boundary. Nodal velocities range from 3.770 to 7.180 km/s. There are 3 zero-update iterations and 0 accepted objective increases above the existing $10^{-8}$ audit tolerance; the worst LSQR relative normal residual is 9.4e-09. The final recorded iteration changes a source coordinate by at most 0.000 km and nodal log slowness by at most 0. Geoid depths span -1.000 to 15.282 km, with 23 negative-depth estimates and 0 estimates outside the grid beyond the boundary tolerance. Boundary counts by face are 1 at $y = 30$ km; 7 at $z = -1$ km; a source on intersecting faces contributes to each count. Across all 14262–14262 rays per recorded iteration, the largest projected/raw gradient norms are 0.00795/0.00795 s/km. At the final model, production/refined convergence counts are 32/32 of 32 sampled rays, with maximum projected gradients 9.32e-06/9.84e-06 s/km and raw gradients 9.32e-06/9.84e-06 s/km. Halving the segment spacing changes sampled travel times by at most 0.164 ms. For 24 source-coordinate derivatives, maximum production/refined errors against centered travel-time differences are 0.000121/8.83e-05 s/km; 0/0 components exceed the existing 0.003 s/km component tolerance. 0 derivative components are omitted at domain boundaries.

Some source estimates reach the computational top, added to contain elevated receivers. These constrained estimates do not establish source-depth recovery. Negative geoid depth means elevation above the geoid, not necessarily above local terrain. The velocity sections show the complete computational depth range so these estimates remain visible.

Over the saved model nodes, DD minus standard velocity has mean -0.0213 km/s, RMS 0.4164 km/s, and maximum absolute difference 1.3754 km/s. The median separation between their source estimates is 0.623 km. These differences do not establish which field model is more accurate.

The absolute-data resolution calculation is linearized at the final DD model with fixed uncertainty weights, first-difference log-slowness smoothing, and column-scaled damping. The model Schur-complement condition number is 2.81e+03, its inverse relative residual is 4.01e-15, and the smallest damped source-block eigenvalue is 3.39e-05. Resolution diagonals range from -0.00011 to 0.838; conditional velocity-noise standard deviations range from 0.00407 to 0.033 km/s. For this final DD linearization, 14260/14262 unique absolute-data rays meet the ray stopping criterion; maximum projected/raw gradient norms are 2.95e-05/0.000162 s/km.

This is a conditional absolute-data calculation, not full DD resolution. Reported symmetric pick uncertainties are interpreted as independent one-standard-deviation errors; provider confidence levels are absent. Propagation through the damped estimator excludes correlated station/model errors, reg-

ularization bias, and nonlinear or ray errors. It eliminates unconstrained source parameters and does not represent the active domain bounds on some inferred source coordinates. The plotted sections interpolate nodal resolution and nodal noise standard deviations; they do not propagate covariance to interpolated velocity. Small propagated noise therefore does not establish small total velocity error. Ray convergence means a projected gradient below $10^{-7}$ s/km, or a travel-time change below $10^{-11}$ s with gradient below $10^{-5}$ s/km. The source finite-difference half-step is 0.01 km. Small spacing changes alone do not establish derivative convergence or globally earliest arrivals, and the deterministic final samples do not certify every production ray.

Rebuild this calculation from the saved authoritative responses with

```
./rebuild_hayward.sh
```

For the recorded baseline, use the frozen solver and inputs preserved at commit `3b674ba`; the integration branch contains later forward changes. Source, executable, configuration, and input hashes are recorded in `data/hayward/run_manifest.json`. The method-specific history and residual files retain numerical diagnostics and observational identities. Synthetic baseline results earlier in this report are a separate experiment and have not been replaced by these field inversions.

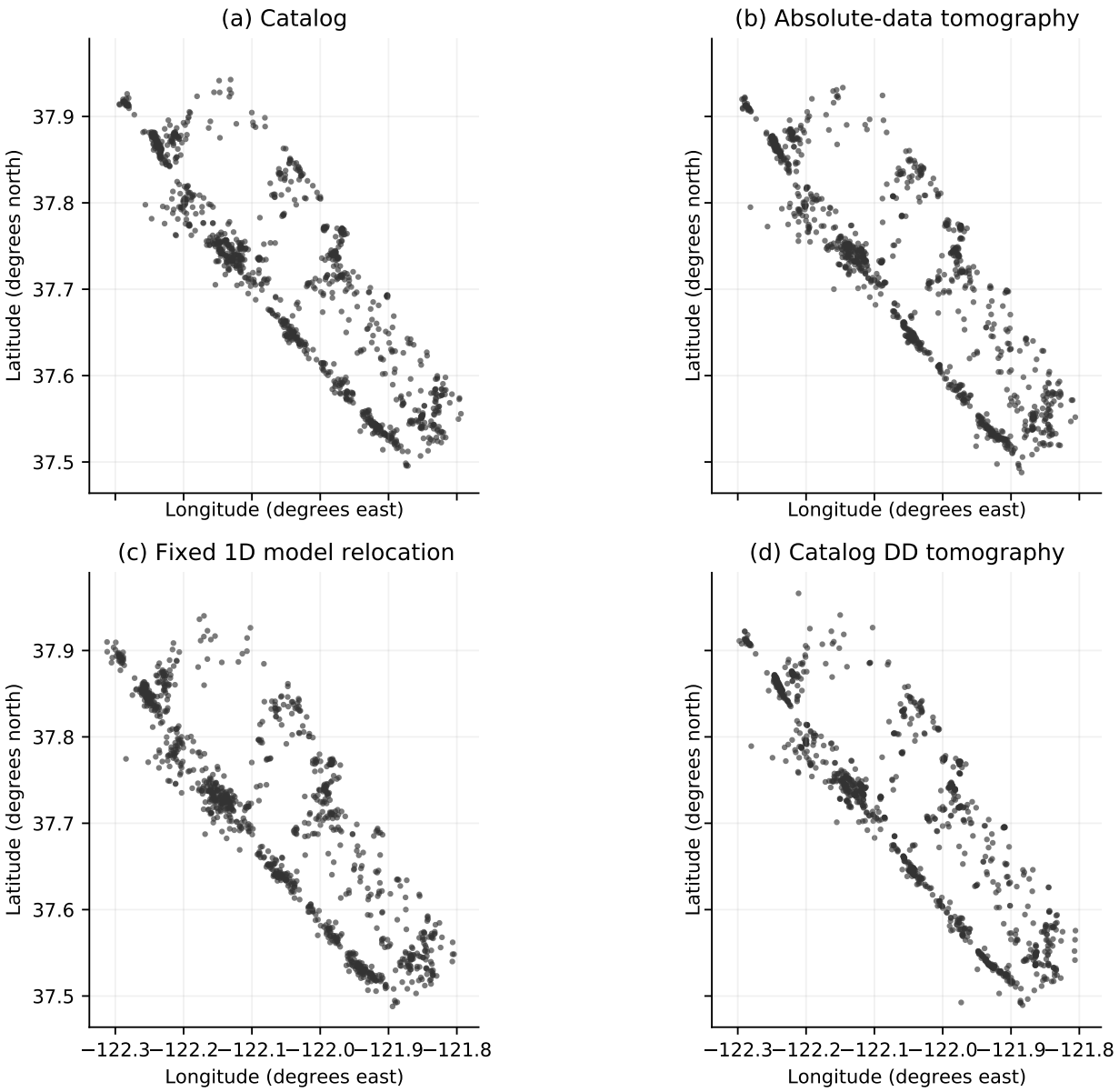



Figure 20: Independent Figure 5 counterpart. All four maps retain the same catalog-selected events. Source movement measures change relative to the reference catalog, not location error against known truth. The fixed-model relocation uses absolute arrivals for anchoring as well as catalog differences.

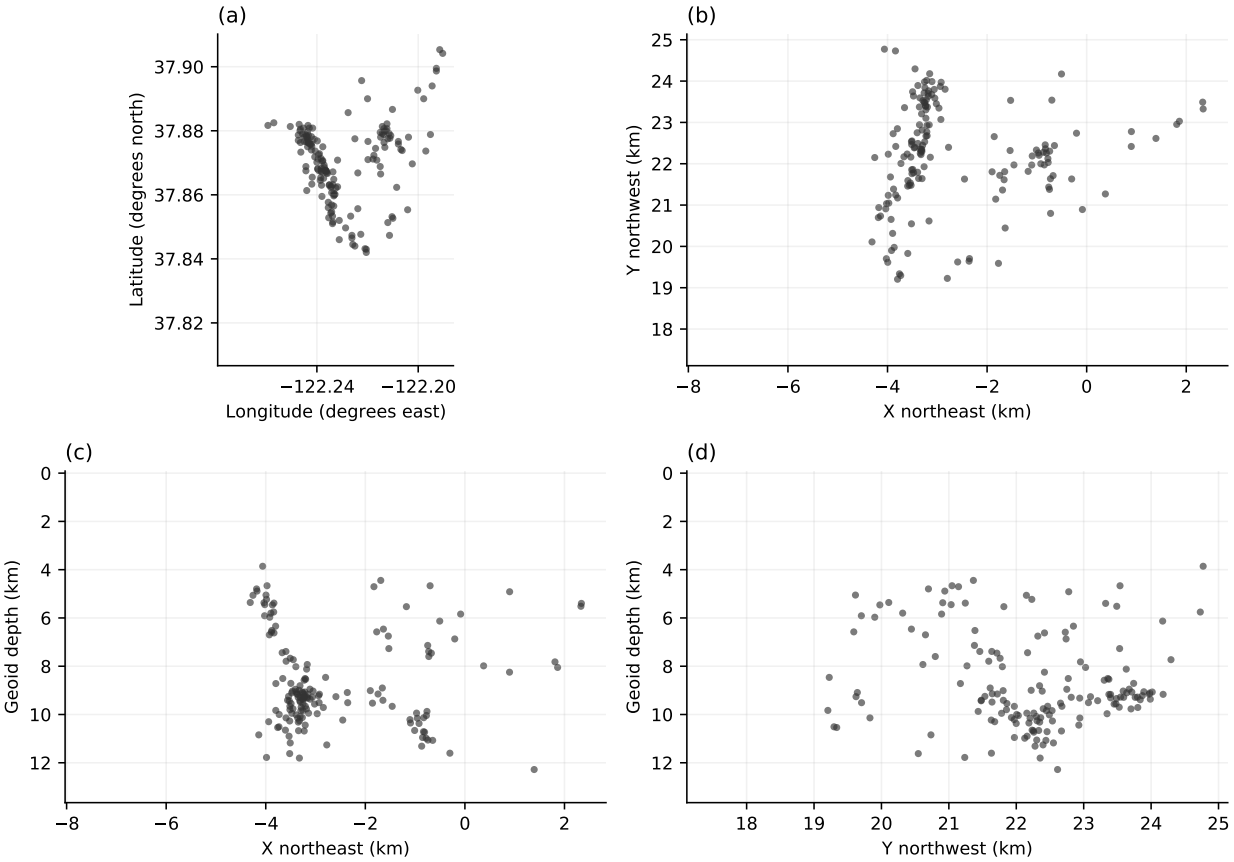



Figure 21: Independent Figure 6 counterpart: catalog events within the documented geographic zoom window. Both depth views show all selected zoom events; the manuscript's undocumented AA′ and BB′ section-box memberships are not reconstructed. This same event subset and common coordinate limits are retained in the following three figures.

Figure 7 counterpart: Absolute-data tomography; 159 common zoom events

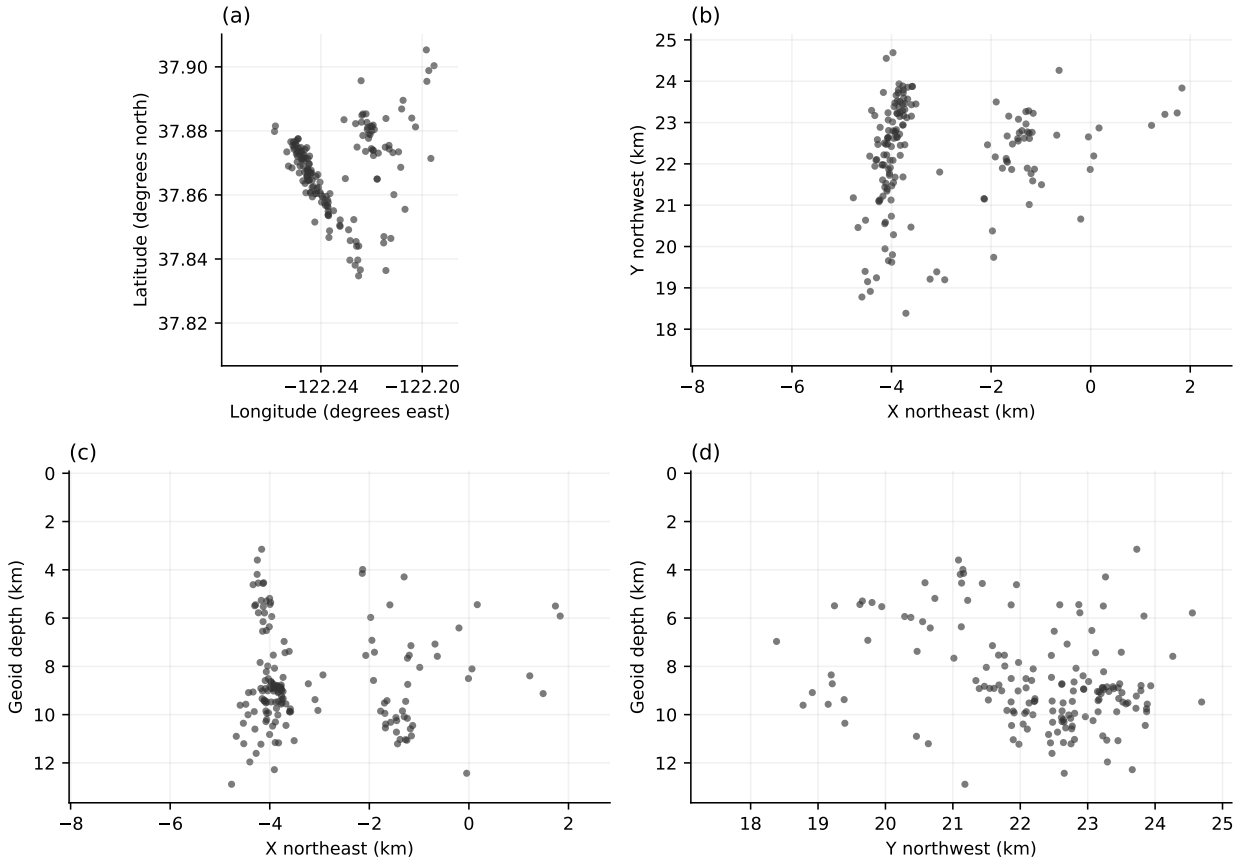



Figure 22: Independent Figure 7 counterpart: the common zoom events after absolute-data tomography. Reference event and origin identifiers remain attached to the independently estimated coordinates.

Figure 8 counterpart: Fixed 1D model relocation; 159 common zoom events

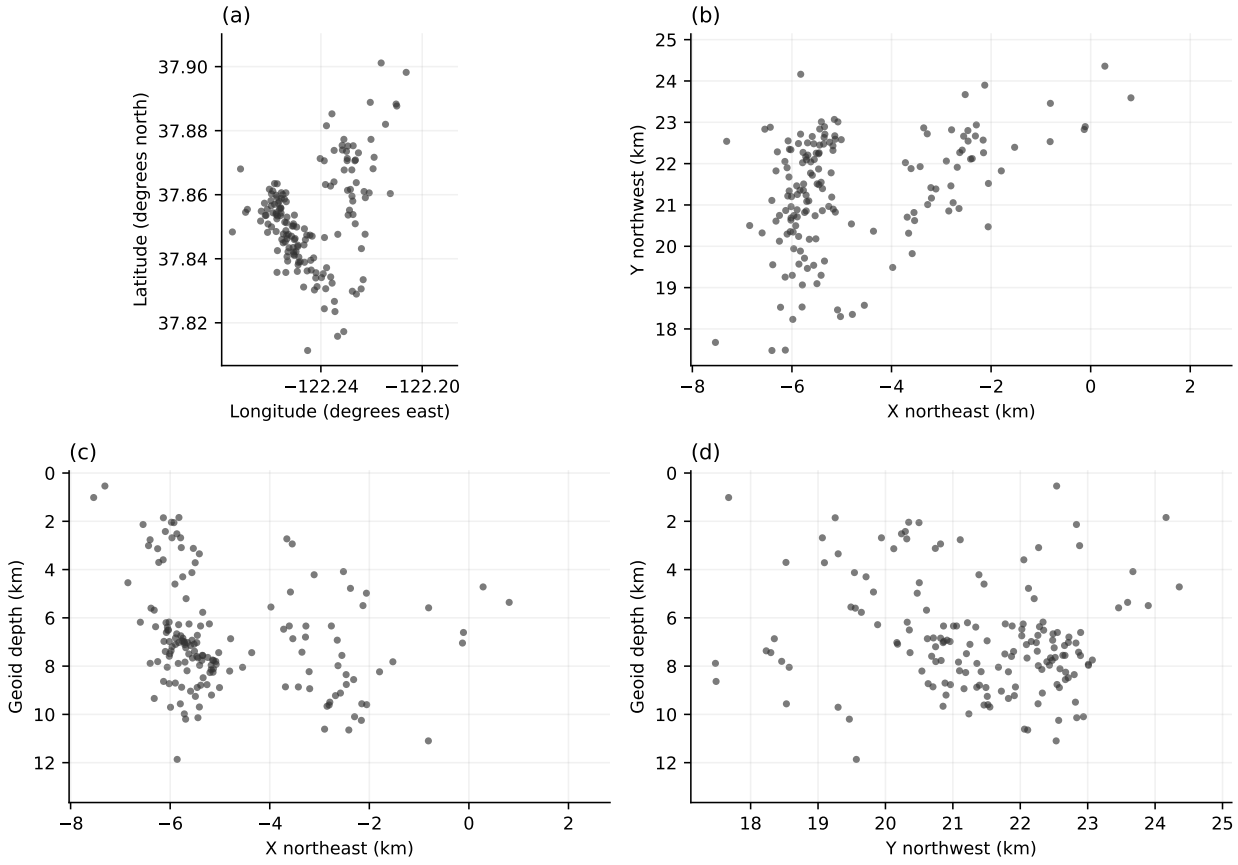



Figure 23: Independent Figure 8 counterpart: relocation in the fixed initial one-dimensional velocity model using catalog differences and absolute anchoring. The point set is fixed by the original catalog selection.

Figure 9 counterpart: Catalog DD tomography; 159 common zoom events

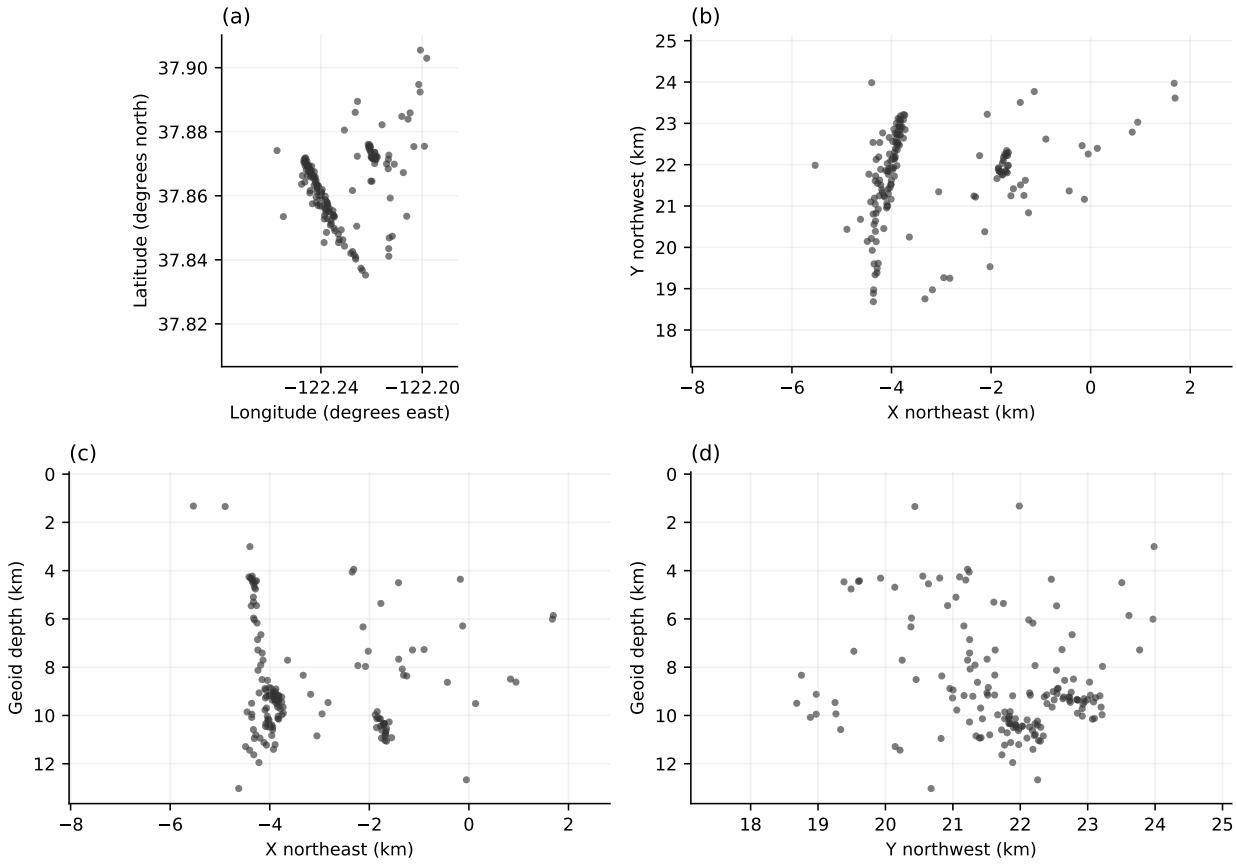



Figure 24: Independent Figure 9 counterpart: catalog DD tomography for the common zoom events. No waveform-correlation observations enter this calculation.

Figure 10 counterpart: independent subset; points within 5 km of each section

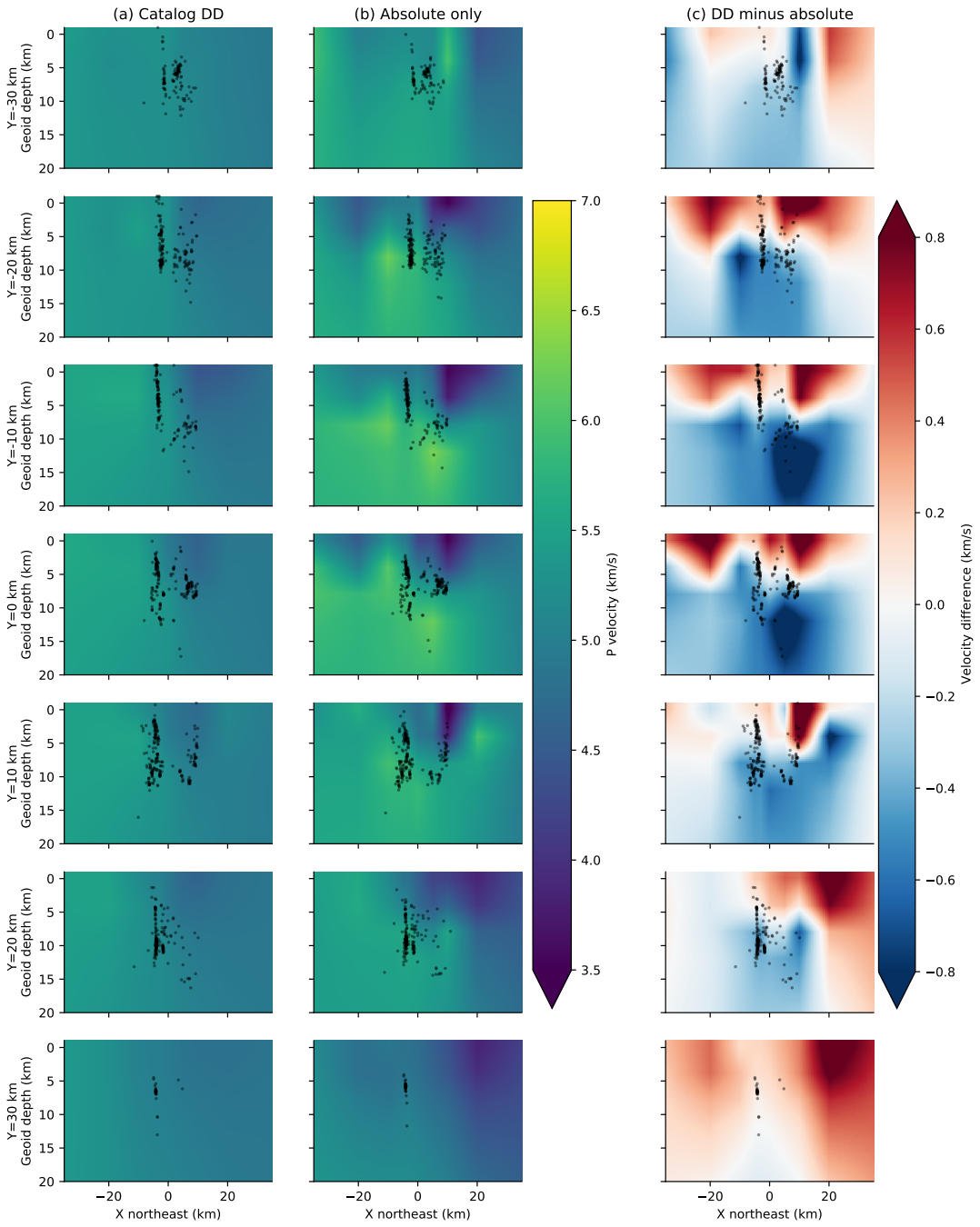



Figure 25: Independent Figure 10 counterpart: seven across-strike sections of catalog DD velocity, absolute-data velocity, and their signed difference. Points lie within 5 km of each section. Interpolation displays the saved nodal model. The depth range includes the added elevated-receiver layer and shows source estimates at the computational top; ray coverage and smoothing do not by themselves establish resolution.

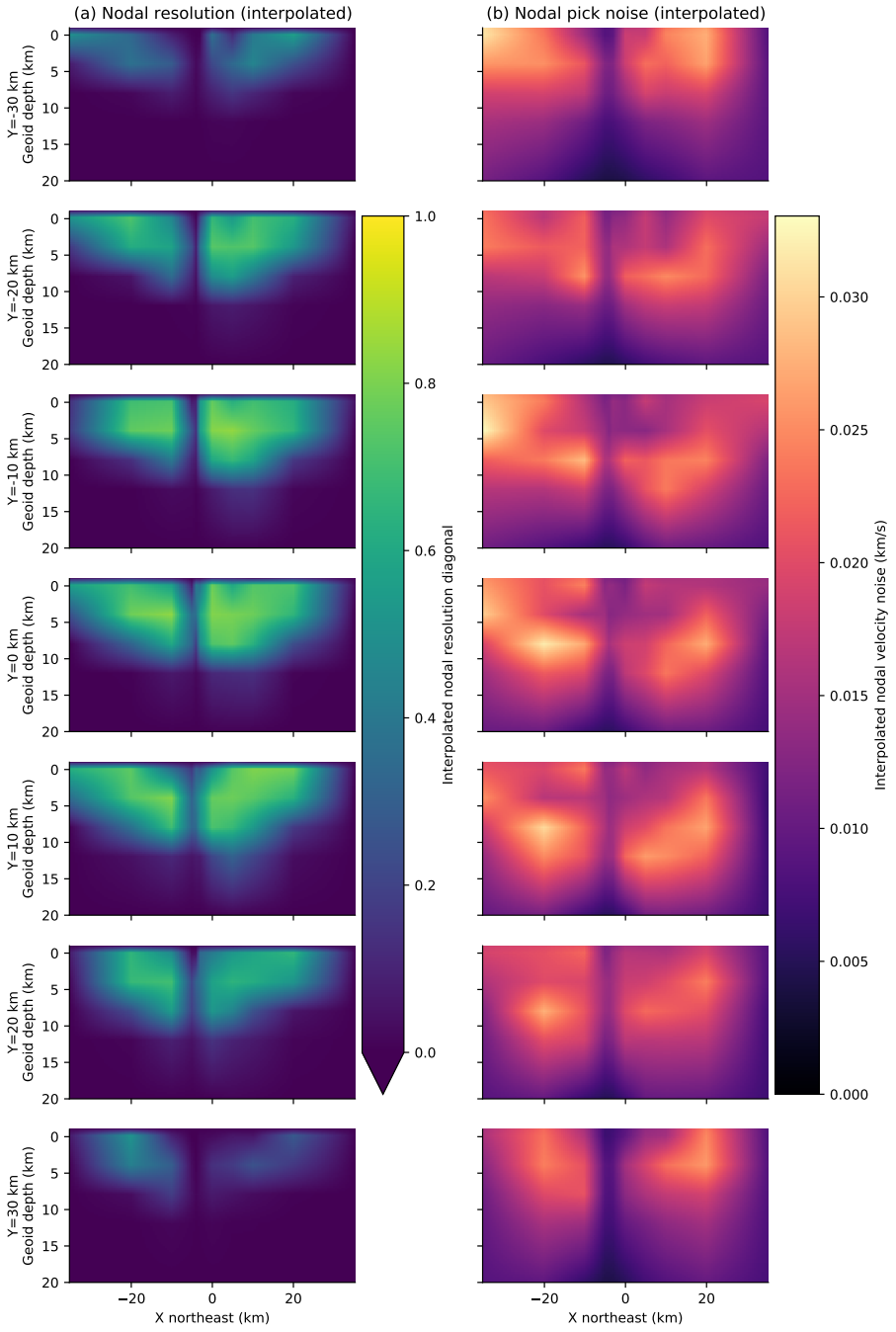



Figure 26: Independent Figure 11 counterpart: absolute-arrival model-resolution diagonal and propagated picking noise, linearized at the catalog DD solution. Reported symmetric uncertainties are assumed to represent independent one-standard-deviation errors. Values between grid nodes interpolate nodal diagnostics; they do not propagate covariance to interpolated velocity. The linear calculation treats source coordinates as unconstrained and omits the effect of active domain bounds. These conditional noise estimates also omit model, metadata, correlation, nonlinear, and regularization-bias contributions.

# A Numerical validation record

This appendix retains the baseline 32-check record supporting the baseline figures.

| Check | Error | Tolerance |
|---|---|---|
| homogeneous_travel_time | 4.44e-15 | 1e-10 |
| homogeneous_source_derivative | 5.19e-17 | 1e-09 |
| homogeneous_velocity_derivative | 4.44e-16 | 1e-09 |
| partition_of_unity | 1.42e-14 | 1e-09 |
| ray_reciprocity | 6.58e-07 | 2e-05 |
| bending_not_slower_than_straight | 0 | 1e-10 |
| ray_refinement_seconds | 4.14e-05 | 0.015 |
| bent_source_finite_difference | 9.22e-05 | 0.003 |
| velocity_finite_difference | 3.71e-13 | 1e-07 |
| identical_event_dd_cancellation | 0 | 1e-12 |
| station_bias_cancellation | 0 | 1e-12 |
| lsqr_known_solution | 7.11e-16 | 1e-09 |
| matrix_adjoint | 0 | 1e-12 |
| interpolation_constant_x_symmetry | 0 | 0 |
| interpolation_constant_y_symmetry | 0 | 0 |
| interpolation_constant_z_symmetry | 0 | 0 |
| surface_ray_plane_symmetry | 0 | 0 |
| surface_ray_source_symmetry | 0 | 0 |
| snell_reference_travel_time | 6.9e-05 | 0.001 |
| snell_reference_source_derivative | 8e-06 | 0.0001 |
| trilinear_gradient_finite_difference | 3.52e-12 | 2e-10 |
| boundary_ray_feasible_descent | 0 | 1e-10 |
| boundary_ray_projected_gradient | 1.84e-06 | 1e-05 |
| boundary_ray_domain | 0 | 0 |
| collapsed_vertices_preserve_endpoints | 0 | 0 |
| collapsed_ray_minimum_segment | 0 | 0 |
| collapsed_ray_gradient_regression | 4.53e-05 | 0.001 |
| scaled_damped_lsqr_solution | 2.48e-16 | 1e-10 |
| scaled_damped_lsqr_normal_residual | 5.55e-17 | 1e-08 |
| scaled_damped_lsqr_converged | 0 | 0 |
| zero_rhs_lsqr | 0 | 0 |
| truncated_lsqr_not_converged | 0 | 0 |

Error units follow each test: seconds for time comparisons, seconds per km for source derivatives, seconds per (km/s) for velocity derivatives, and algebraic units for the sparse-system checks. These tolerances were chosen for numerical consistency tests, not to enforce agreement with the paper.

# B Figure-by-figure record

The complete machine-readable record is in `data/panel_audit.csv` and `data/panel_audit.json`; its readable counterpart is `FIGURE_AUDIT.md`. "Independently reproduced" means the quantity is reconstructed from the written specification using our model output. "Partially reproduced" means an independent numerical counterpart exists but one or more specified experimental inputs or algorithms differ. "Not reproduced" means no corresponding result is claimed.

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 1 | Horizontal true P-velocity slice | independently reproduced. Physical velocity profile reconstructed from prose; plotting extent and depth chosen from model domain. |
| 2 | Synthetic-test event/station geometry and inversion grid | partially reproduced. The saved baseline uses random geometry. A documented authoritative Loma Prieta subset is authorized; it has not yet replaced this experiment. No digitization used. |
| 3a | Standard tomographic P velocity; all generated depth slices | partially reproduced. Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences. |
| 3b | DD tomographic P velocity; all generated depth slices | partially reproduced. Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences. |
| 3c | True minus DD P velocity; all generated depth slices | partially reproduced. Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences. |
| 3d | True minus standard P velocity; all generated depth slices | partially reproduced. Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences. |
| 4 | Hayward network, catalog seismicity, and inversion grid | partially reproduced. Generated geometry uses authoritative observations and an independent selection; the original 1489-event/52-station cohort remains unidentified. CMW1 historical elevation interpretation and conflicting borehole wellhead tables are documented. This input-geometry panel does not establish an inferred velocity model. |
| 5a | Hayward event map: NCSN catalog | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Catalog coordinates are provider preferred-origin estimates; this panel contains no inverse estimate. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 5b | Hayward event map: standard tomography | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |
| 5c | Hayward event map: DD location | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Fixed-model relocation uses catalog differences with absolute-data anchoring; it is an independent counterpart of the published DD-location experiment. Saved history records 10 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 5d | Hayward event map: DD tomography | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |
| 6a | catalog: geographical map | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Catalog coordinates are provider preferred-origin estimates; this panel contains no inverse estimate. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |
| 6b | catalog: rotated Cartesian map | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Catalog coordinates are provider preferred-origin estimates; this panel contains no inverse estimate. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 6c | catalog: across-strike depth section AA-prime | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Catalog coordinates are provider preferred-origin estimates; this panel contains no inverse estimate. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |
| 6d | catalog: along-strike depth section BB-prime | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Catalog coordinates are provider preferred-origin estimates; this panel contains no inverse estimate. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 7a | standard tomography: geographical map | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |
| 7b | standard tomography: rotated Cartesian map | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 7c | standard tomography: across-strike depth section AA-prime | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |
| 7d | standard tomography: along-strike depth section BB-prime | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 8a | DD location: geographical map | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Fixed-model relocation uses catalog differences with absolute-data anchoring; it is an independent counterpart of the published DD-location experiment. Saved history records 10 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 8b | DD location: rotated Cartesian map | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Fixed-model relocation uses catalog differences with absolute-data anchoring; it is an independent counterpart of the published DD-location experiment. Saved history records 10 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 8c | DD location: across-strike depth section AA-prime | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Fixed-model relocation uses catalog differences with absolute-data anchoring; it is an independent counterpart of the published DD-location experiment. Saved history records 10 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 8d | DD location: along-strike depth section BB-prime | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Fixed-model relocation uses catalog differences with absolute-data anchoring; it is an independent counterpart of the published DD-location experiment. Saved history records 10 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 9a | DD tomography: geographical map | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |
| 9b | DD tomography: rotated Cartesian map | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 9c | DD tomography: across-strike depth section AA-prime | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |
| 9d | DD tomography: along-strike depth section BB-prime | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 10a (y=-30) | Hayward across-strike DD velocity | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Seven written y sections are generated from the saved standard and catalog-DD models, with hypocenters within 5 km. Sections include the full saved depth range, including the above-geoid layer added to contain receivers and any source estimates at its top. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 10a (y=-20) | Hayward across-strike DD velocity | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Seven written y sections are generated from the saved standard and catalog-DD models, with hypocenters within 5 km. Sections include the full saved depth range, including the above-geoid layer added to contain receivers and any source estimates at its top. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 10a (y=-10) | Hayward across-strike DD velocity | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Seven written y sections are generated from the saved standard and catalog-DD models, with hypocenters within 5 km. Sections include the full saved depth range, including the above-geoid layer added to contain receivers and any source estimates at its top. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 10a (y=0) | Hayward across-strike DD velocity | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Seven written y sections are generated from the saved standard and catalog-DD models, with hypocenters within 5 km. Sections include the full saved depth range, including the above-geoid layer added to contain receivers and any source estimates at its top. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

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| 10a (y=10) | Hayward across-strike DD velocity | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Seven written y sections are generated from the saved standard and catalog-DD models, with hypocenters within 5 km. Sections include the full saved depth range, including the above-geoid layer added to contain receivers and any source estimates at its top. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

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| 10a (y=20) | Hayward across-strike DD velocity | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Seven written y sections are generated from the saved standard and catalog-DD models, with hypocenters within 5 km. Sections include the full saved depth range, including the above-geoid layer added to contain receivers and any source estimates at its top. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

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| 10b (y=-30) | Hayward across-strike standard velocity | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Seven written y sections are generated from the saved standard and catalog-DD models, with hypocenters within 5 km. Sections include the full saved depth range, including the above-geoid layer added to contain receivers and any source estimates at its top. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

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| 10c (y=-30) | Hayward across-strike DD minus standard velocity | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Seven written y sections are generated from the saved standard and catalog-DD models, with hypocenters within 5 km. Sections include the full saved depth range, including the above-geoid layer added to contain receivers and any source estimates at its top. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

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| 11a (y=-30) | Hayward resolution-matrix diagonal | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. The generated resolution is the absolute-data velocity block at the catalog-DD estimate, with source parameters included in the joint algebra. It is not a full DD resolution matrix or the original simul2000 calculation. Noise propagation assumes reported symmetric pick uncertainties are independent 1-sigma errors; provider confidence levels are absent. Sections interpolate nodal diagnostics; they do not propagate covariance to interpolated velocity. Unconstrained source elimination omits the effect of active source-coordinate bounds. It omits correlated station/model errors, ray and nonlinear errors, selection effects and regularization bias. Small conditional noise does not establish small total velocity error. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

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| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 11b (y=-30) | Hayward interpolated nodal conditional velocity noise from independent picks | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. The generated resolution is the absolute-data velocity block at the catalog-DD estimate, with source parameters included in the joint algebra. It is not a full DD resolution matrix or the original simul2000 calculation. Noise propagation assumes reported symmetric pick uncertainties are independent 1-sigma errors; provider confidence levels are absent. Sections interpolate nodal diagnostics; they do not propagate covariance to interpolated velocity. Unconstrained source elimination omits the effect of active source-coordinate bounds. It omits correlated station/model errors, ray and nonlinear errors, selection effects and regularization bias. Small conditional noise does not establish small total velocity error. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 11b (y=-20) | Hayward interpolated nodal conditional velocity noise from independent picks | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. The generated resolution is the absolute-data velocity block at the catalog-DD estimate, with source parameters included in the joint algebra. It is not a full DD resolution matrix or the original simul2000 calculation. Noise propagation assumes reported symmetric pick uncertainties are independent 1-sigma errors; provider confidence levels are absent. Sections interpolate nodal diagnostics; they do not propagate covariance to interpolated velocity. Unconstrained source elimination omits the effect of active source-coordinate bounds. It omits correlated station/model errors, ray and nonlinear errors, selection effects and regularization bias. Small conditional noise does not establish small total velocity error. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 11b (y=-10) | Hayward interpolated nodal conditional velocity noise from independent picks | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. The generated resolution is the absolute-data velocity block at the catalog-DD estimate, with source parameters included in the joint algebra. It is not a full DD resolution matrix or the original simul2000 calculation. Noise propagation assumes reported symmetric pick uncertainties are independent 1-sigma errors; provider confidence levels are absent. Sections interpolate nodal diagnostics; they do not propagate covariance to interpolated velocity. Unconstrained source elimination omits the effect of active source-coordinate bounds. It omits correlated station/model errors, ray and nonlinear errors, selection effects and regularization bias. Small conditional noise does not establish small total velocity error. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 11b (y=0) | Hayward interpolated nodal conditional velocity noise from independent picks | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. The generated resolution is the absolute-data velocity block at the catalog-DD estimate, with source parameters included in the joint algebra. It is not a full DD resolution matrix or the original simul2000 calculation. Noise propagation assumes reported symmetric pick uncertainties are independent 1-sigma errors; provider confidence levels are absent. Sections interpolate nodal diagnostics; they do not propagate covariance to interpolated velocity. Unconstrained source elimination omits the effect of active source-coordinate bounds. It omits correlated station/model errors, ray and nonlinear errors, selection effects and regularization bias. Small conditional noise does not establish small total velocity error. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 11b (y=10) | Hayward interpolated nodal conditional velocity noise from independent picks | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. The generated resolution is the absolute-data velocity block at the catalog-DD estimate, with source parameters included in the joint algebra. It is not a full DD resolution matrix or the original simul2000 calculation. Noise propagation assumes reported symmetric pick uncertainties are independent 1-sigma errors; provider confidence levels are absent. Sections interpolate nodal diagnostics; they do not propagate covariance to interpolated velocity. Unconstrained source elimination omits the effect of active source-coordinate bounds. It omits correlated station/model errors, ray and nonlinear errors, selection effects and regularization bias. Small conditional noise does not establish small total velocity error. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 11b (y=20) | Hayward interpolated nodal conditional velocity noise from independent picks | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. The generated resolution is the absolute-data velocity block at the catalog-DD estimate, with source parameters included in the joint algebra. It is not a full DD resolution matrix or the original simul2000 calculation. Noise propagation assumes reported symmetric pick uncertainties are independent 1-sigma errors; provider confidence levels are absent. Sections interpolate nodal diagnostics; they do not propagate covariance to interpolated velocity. Unconstrained source elimination omits the effect of active source-coordinate bounds. It omits correlated station/model errors, ray and nonlinear errors, selection effects and regularization bias. Small conditional noise does not establish small total velocity error. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

| Panel | Quantity | Status and limiting dependency |
|---|---|---|
| 11b (y=30) | Hayward interpolated nodal conditional velocity noise from independent picks | partially reproduced. Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. The generated resolution is the absolute-data velocity block at the catalog-DD estimate, with source parameters included in the joint algebra. It is not a full DD resolution matrix or the original simul2000 calculation. Noise propagation assumes reported symmetric pick uncertainties are independent 1-sigma errors; provider confidence levels are absent. Sections interpolate nodal diagnostics; they do not propagate covariance to interpolated velocity. Unconstrained source elimination omits the effect of active source-coordinate bounds. It omits correlated station/model errors, ray and nonlinear errors, selection effects and regularization bias. Small conditional noise does not establish small total velocity error. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. |

## B.1 Specifications, commands, and attempted dependencies

The following records specify the source description, parameters, generation command, and plot command for each panel. Seven sections of a common Figure 10 or 11 panel share the explicitly listed settings. Unavailable commands are reported as unavailable rather than replaced by placeholders that would fabricate field observations.

### Panel 1

**Quantity:** Horizontal true P-velocity slice
**Written specification:** Synthetic Test, p.1877; caption 1
**Parameters:** Written sandwich profile, x=-35..35 km, y=-60..40 km; constant with depth
**Status:** independently reproduced

**Attempt and limitation:** Physical velocity profile reconstructed from prose; plotting extent and depth chosen from model domain.
**Generate data:**

```
./build/dd generate configs/synthetic.cfg data/synthetic
```

**Create plot:**

```
python3 plot.py --figure 1
```

### Panel 2

**Quantity:** Synthetic-test event/station geometry and inversion grid
**Written specification:** Caption 2, p.1879
**Parameters:** Published 8x6x5 inversion grid; 120 independent sources and 32 independent stations, seed 2003
**Status:** partially reproduced
**Attempt and limitation:** The saved baseline uses random geometry. A documented authoritative Loma Prieta subset is authorized; it has not yet replaced this experiment. No digitization used.
**Generate data:**

```
./build/dd generate configs/synthetic.cfg data/synthetic
```

**Create plot:**

```
python3 plot.py --figure 2
```

### Panel 3a

**Quantity:** Standard tomographic P velocity; all generated depth slices
**Written specification:** Synthetic Test pp.1877-1879; captions 3 pp.1880-1881
**Parameters:** Depths 0,3,7,11,16 km; grid from caption 2; noise 0.04 s and station bias +/-0.3 s; configs/synthetic.cfg
**Status:** partially reproduced
**Attempt and limitation:** Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
**Generate data:**

```
./build/dd generate configs/synthetic.cfg data/synthetic && ./build/dd invert configs/synthetic.cfg data/synthetic
    standard
```

**Create plot:**

```
python3 plot.py --figure 3a
```

### Panel 3b

**Quantity:** DD tomographic P velocity; all generated depth slices
**Written specification:** Synthetic Test pp.1877-1879; captions 3 pp.1880-1881
**Parameters:** Depths 0,3,7,11,16 km; grid from caption 2; noise 0.04 s and station bias +/-0.3 s; configs/synthetic.cfg
**Status:** partially reproduced
**Attempt and limitation:** Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
**Generate data:**

```
./build/dd generate configs/synthetic.cfg data/synthetic && ./build/dd invert configs/synthetic.cfg data/synthetic
    dd
```

**Create plot:**

```
python3 plot.py --figure 3b
```

### Panel 3c

**Quantity:** True minus DD P velocity; all generated depth slices
**Written specification:** Synthetic Test pp.1877-1879; captions 3 pp.1880-1881
**Parameters:** Depths 0,3,7,11,16 km; grid from caption 2; noise 0.04 s and station bias +/-0.3 s; configs/synthetic.cfg
**Status:** partially reproduced
**Attempt and limitation:** Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
**Generate data:**

```
./build/dd generate configs/synthetic.cfg data/synthetic && ./build/dd invert configs/synthetic.cfg data/synthetic
    dd
```

**Create plot:**

```
python3 plot.py --figure 3c
```

### Panel 3d

**Quantity:** True minus standard P velocity; all generated depth slices
**Written specification:** Synthetic Test pp.1877-1879; captions 3 pp.1880-1881
**Parameters:** Depths 0,3,7,11,16 km; grid from caption 2; noise 0.04 s and station bias +/-0.3 s; configs/synthetic.cfg
**Status:** partially reproduced
**Attempt and limitation:** Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
**Generate data:**

```
./build/dd generate configs/synthetic.cfg data/synthetic && ./build/dd invert configs/synthetic.cfg data/synthetic
    standard
```

**Create plot:**

```
python3 plot.py --figure 3d
```

### Panel 4

**Quantity:** Hayward network, catalog seismicity, and inversion grid
**Written specification:** Caption 4, p.1882
**Parameters:** 1092 independently selected current catalog earthquakes, 37 physical stations, 77 horizontal grid nodes; identified source and receiver inputs in data/hayward
**Status:** partially reproduced
**Attempt and limitation:** Generated geometry uses authoritative observations and an independent selection; the original 1489-event/52-station cohort remains unidentified. CMW1 historical elevation interpretation and conflicting borehole wellhead tables are documented. This input-geometry panel does not establish an inferred velocity model.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction
```

**Create plot:**

```
python3 plot_hayward.py --geometry-only
```

### Panel 5a

**Quantity:** Hayward event map: NCSN catalog
**Written specification:** Event Locations pp.1881-1883; caption 5
**Parameters:** 1489 earthquakes, 52 stations; 20257 absolute picks, 767127 catalog differences, 17955 precise differences
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Catalog coordinates are provider preferred-origin estimates; this panel contains no inverse estimate.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 5b

**Quantity:** Hayward event map: standard tomography
**Written specification:** Event Locations pp.1881-1883; caption 5
**Parameters:** 1489 earthquakes, 52 stations; 20257 absolute picks, 767127 catalog differences, 17955 precise differences
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward standard;
    ./build/dd raycheck configs/hayward.cfg data/hayward standard
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 5c

**Quantity:** Hayward event map: DD location
**Written specification:** Event Locations pp.1881-1883; caption 5
**Parameters:** 1489 earthquakes, 52 stations; 20257 absolute picks, 767127 catalog differences, 17955 precise differences

**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Fixed-model relocation uses catalog differences with absolute-data anchoring; it is an independent counterpart of the published DD-location experiment. Saved history records 10 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward loc1d; ./
    build/dd raycheck configs/hayward.cfg data/hayward loc1d
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 5d

**Quantity:** Hayward event map: DD tomography
**Written specification:** Event Locations pp.1881-1883; caption 5
**Parameters:** 1489 earthquakes, 52 stations; 20257 absolute picks, 767127 catalog differences, 17955 precise differences
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward dd; ./build
    /dd raycheck configs/hayward.cfg data/hayward dd
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 6a

**Quantity:** catalog: geographical map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Catalog coordinates are provider preferred-origin estimates; this panel contains no inverse estimate. Zoom

membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 6b

**Quantity:** catalog: rotated Cartesian map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Catalog coordinates are provider preferred-origin estimates; this panel contains no inverse estimate. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 6c

**Quantity:** catalog: across-strike depth section AA-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Catalog coordinates are provider preferred-origin estimates; this panel contains no inverse estimate. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 6d

**Quantity:** catalog: along-strike depth section BB-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Catalog coordinates are provider preferred-origin estimates; this panel contains no inverse estimate. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 7a

**Quantity:** standard tomography: geographical map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward standard;
    ./build/dd raycheck configs/hayward.cfg data/hayward standard
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 7b

**Quantity:** standard tomography: rotated Cartesian map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E

**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward standard;
    ./build/dd raycheck configs/hayward.cfg data/hayward standard
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 7c

**Quantity:** standard tomography: across-strike depth section AA-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward standard;
    ./build/dd raycheck configs/hayward.cfg data/hayward standard
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 7d

**Quantity:** standard tomography: along-strike depth section BB-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E

**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward standard;
    ./build/dd raycheck configs/hayward.cfg data/hayward standard
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 8a

**Quantity:** DD location: geographical map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Fixed-model relocation uses catalog differences with absolute-data anchoring; it is an independent counterpart of the published DD-location experiment. Saved history records 10 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward loc1d; ./
    build/dd raycheck configs/hayward.cfg data/hayward loc1d
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 8b

**Quantity:** DD location: rotated Cartesian map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E

**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Fixed-model relocation uses catalog differences with absolute-data anchoring; it is an independent counterpart of the published DD-location experiment. Saved history records 10 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward loc1d; ./
    build/dd raycheck configs/hayward.cfg data/hayward loc1d
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 8c

**Quantity:** DD location: across-strike depth section AA-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Fixed-model relocation uses catalog differences with absolute-data anchoring; it is an independent counterpart of the published DD-location experiment. Saved history records 10 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward loc1d; ./
    build/dd raycheck configs/hayward.cfg data/hayward loc1d
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 8d

**Quantity:** DD location: along-strike depth section BB-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9

**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Fixed-model relocation uses catalog differences with absolute-data anchoring; it is an independent counterpart of the published DD-location experiment. Saved history records 10 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward loc1d; ./
    build/dd raycheck configs/hayward.cfg data/hayward loc1d
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 9a

**Quantity:** DD tomography: geographical map
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward dd; ./build
    /dd raycheck configs/hayward.cfg data/hayward dd
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 9b

**Quantity:** DD tomography: rotated Cartesian map

**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward dd; ./build
    /dd raycheck configs/hayward.cfg data/hayward dd
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 9c

**Quantity:** DD tomography: across-strike depth section AA-prime
**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward dd; ./build
    /dd raycheck configs/hayward.cfg data/hayward dd
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 9d

**Quantity:** DD tomography: along-strike depth section BB-prime

**Written specification:** Event Locations pp.1882-1884; captions 6-9
**Parameters:** Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Zoom membership is fixed by the provider catalog coordinates within the written geographic window. AA/BB section-box boundaries and fault traces were not recovered or digitized; both depth views include every selected zoom event, with common coordinate limits across methods.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward dd; ./build
    /dd raycheck configs/hayward.cfg data/hayward dd
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 10a: all seven sections at y=-30,-20,-10,0,10,20,30 km

**Quantity:** Hayward across-strike DD velocity
**Written specification:** Velocity Structure p.1884; caption 10 p.1887
**Parameters:** Published Hayward grid; smoothing weight 10; all seven y sections listed above; hypocenters within 5 km
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Seven written y sections are generated from the saved standard and catalog-DD models, with hypocenters within 5 km. Sections include the full saved depth range, including the above-geoid layer added to contain receivers and any source estimates at its top. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward standard;
    ./build/dd raycheck configs/hayward.cfg data/hayward standard; ./build/dd invert configs/hayward.cfg data/
    hayward dd; ./build/dd raycheck configs/hayward.cfg data/hayward dd
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

**Panel 10b: all seven sections at y=-30,-20,-10,0,10,20,30 km**

**Quantity:** Hayward across-strike standard velocity
**Written specification:** Velocity Structure p.1884; caption 10 p.1887
**Parameters:** Published Hayward grid; smoothing weight 10; all seven y sections listed above; hypocenters within 5 km
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Seven written y sections are generated from the saved standard and catalog-DD models, with hypocenters within 5 km. Sections include the full saved depth range, including the above-geoid layer added to contain receivers and any source estimates at its top. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward standard;
    ./build/dd raycheck configs/hayward.cfg data/hayward standard; ./build/dd invert configs/hayward.cfg data/
    hayward dd; ./build/dd raycheck configs/hayward.cfg data/hayward dd
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

**Panel 10c: all seven sections at y=-30,-20,-10,0,10,20,30 km**

**Quantity:** Hayward across-strike DD minus standard velocity
**Written specification:** Velocity Structure p.1884; caption 10 p.1887
**Parameters:** Published Hayward grid; smoothing weight 10; all seven y sections listed above; hypocenters within 5 km
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. Seven written y sections are generated from the saved standard and catalog-DD models, with hypocenters within 5 km. Sections include the full saved depth range, including the above-geoid layer added to contain receivers and any source estimates at its top. Absolute-data tomography uses the independently adapted continuous starting profile and numerical settings. Saved history records 12 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or

nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward standard;
    ./build/dd raycheck configs/hayward.cfg data/hayward standard; ./build/dd invert configs/hayward.cfg data/
    hayward dd; ./build/dd raycheck configs/hayward.cfg data/hayward dd
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 11a: all seven sections at y=-30,-20,-10,0,10,20,30 km

**Quantity:** Hayward resolution-matrix diagonal
**Written specification:** Velocity Structure pp.1884,1886; caption 11 p.1888
**Parameters:** Absolute catalog-P Jacobian at saved DD sources/velocity; source parameters eliminated; independent log-slowness smoothing and column-scaled damping; y=-30
**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. The generated resolution is the absolute-data velocity block at the catalog-DD estimate, with source parameters included in the joint algebra. It is not a full DD resolution matrix or the original simul2000 calculation. Noise propagation assumes reported symmetric pick uncertainties are independent 1-sigma errors; provider confidence levels are absent. Sections interpolate nodal diagnostics; they do not propagate covariance to interpolated velocity. Unconstrained source elimination omits the effect of active source-coordinate bounds. It omits correlated station/model errors, ray and nonlinear errors, selection effects and regularization bias. Small conditional noise does not establish small total velocity error. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward dd; ./build
    /dd raycheck configs/hayward.cfg data/hayward dd; ./build/dd linearize configs/hayward.cfg data/hayward dd;
    OPENBLAS_NUM_THREADS=1 python3 resolution_hayward.py
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

### Panel 11b: all seven sections at y=-30,-20,-10,0,10,20,30 km

**Quantity:** Hayward interpolated nodal conditional velocity noise from independent picks
**Written specification:** Velocity Structure pp.1884,1886; caption 11 p.1888
**Parameters:** Absolute catalog-P Jacobian at saved DD sources/velocity; source parameters eliminated; independent log-slowness smoothing and column-scaled damping; y=-30

**Status:** partially reproduced
**Attempt and limitation:** Independent subset: 1092 events, 37 physical stations, 14262 absolute P picks and 44718 catalog differences; the manuscript reports 1489, 52, 20257 and 767127, respectively. Its exact cohort remains unidentified, and its 17955 waveform differences are absent here. The generated resolution is the absolute-data velocity block at the catalog-DD estimate, with source parameters included in the joint algebra. It is not a full DD resolution matrix or the original simul2000 calculation. Noise propagation assumes reported symmetric pick uncertainties are independent 1-sigma errors; provider confidence levels are absent. Sections interpolate nodal diagnostics; they do not propagate covariance to interpolated velocity. Unconstrained source elimination omits the effect of active source-coordinate bounds. It omits correlated station/model errors, ray and nonlinear errors, selection effects and regularization bias. Small conditional noise does not establish small total velocity error. Joint catalog-DD tomography uses catalog differences with absolute-data anchoring and preserves both native channel epochs. Saved history records 18 accepted updates and 0 failed LSQR stopping checks. Catalog displacement is not location error against independent truth. Source-derivative and ray-refinement checks sample the saved model; they do not establish all-ray or nonlinear inverse convergence. Residual reduction alone does not establish recovery or unchanged timing calibration across metadata epochs.
**Generate data:**

```
python3 prepare_hayward.py --selection reproduction; ./build/dd invert configs/hayward.cfg data/hayward dd; ./build
    /dd raycheck configs/hayward.cfg data/hayward dd; ./build/dd linearize configs/hayward.cfg data/hayward dd;
    OPENBLAS_NUM_THREADS=1 python3 resolution_hayward.py
```

**Create plot:**

```
python3 plot_hayward.py # complete field pipeline: ./rebuild_hayward.sh
```

# C References and guidance

Zhang, H., and C. H. Thurber (2003). Double-Difference Tomography: The Method and Its Application to the Hayward Fault, California. *BSSA*, 93, 1875–1889. doi:10.1785/0120020190.

Oppenheimer, D. H., F. W. Klein, J. P. Eaton, and F. W. Lester (1993). The Northern California Seismic Network bulletin, January–December 1992. USGS Open-File Report 93-578, Table 4. doi:10.3133/ofr93578.

Waldhauser, F., and W. L. Ellsworth (2002). Fault structure and mechanics of the Hayward Fault, California, from double-difference earthquake locations. *Journal of Geophysical Research*, 107(B3), ESE 3-1–ESE 3-15. doi:10.1029/2000JB000084.

The two user-permitted guidance files are repository guidelines and academic style guide. They govern research-code simplicity and prose. They were not used as permission to inspect any original model implementation or output.